

Generative Image and Video Coding: Principles, Methods, and Emerging Directions

Jianhui Chang¹, Giuseppe Valenzise², Shiqi Wang³

¹ Cloud Computing Institute, China Telecom

² Univ. Paris-Saclay, CNRS, CentraleSupélec, L2S, Gif-sur-Yvette, France

³ Department of Computer Science, City University of Hong Kong, Hong Kong, China



Speakers



Jianhui Chang



Giuseppe Valenzise



Shiqi Wang



Tutorial Schedule

Session	Topic	Speaker
Part 1	Foundations of Generative Visual Coding Coding paradigms and rate–distortion–perception Diffusion priors and posterior reconstruction	Giuseppe Valenzise Université Paris-Saclay CNRS, CentraleSupélec, L2S
Part 2	Diffusion Methods for Image and Video Coding Bitstream representations and generation control Efficient decoding and temporal modeling	Jianhui Chang China Telecom Cloud Computing Institute
Part 3	Generative Face Video Coding and Beyond Compact facial representations and motion modeling Scalable coding, standardization and future directions	Shiqi Wang City University of Hong Kong Department of Computer Science

PART 01

Foundations of Generative Visual Coding

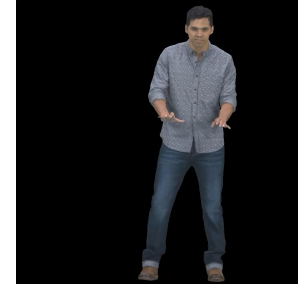
Giuseppe Valenzise

Université Paris-Saclay
CNRS, CentraleSupélec, L2S

Video compression matters

- Video streaming represents over **80%** of the total Internet traffic today

more, more, **more**....



- Higher spatial and temporal resolutions

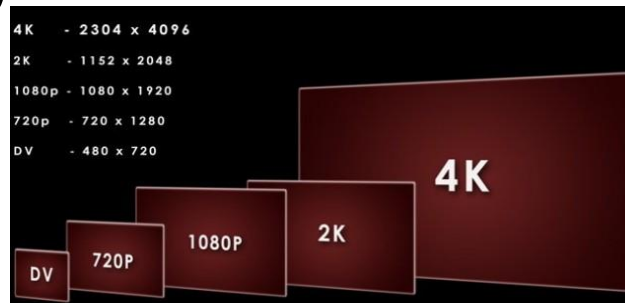
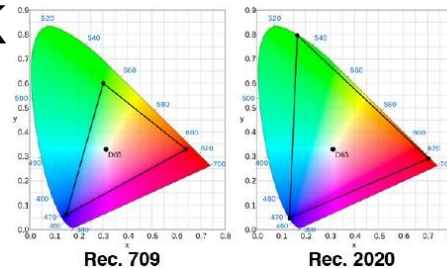
- Ultra High Definition (UHD), 4K, 8K
- High Frame Rate (HFR)

- Higher pixel depth

- High Dynamic Range (HDR)

- More colors

- Wide color gamut



- More views

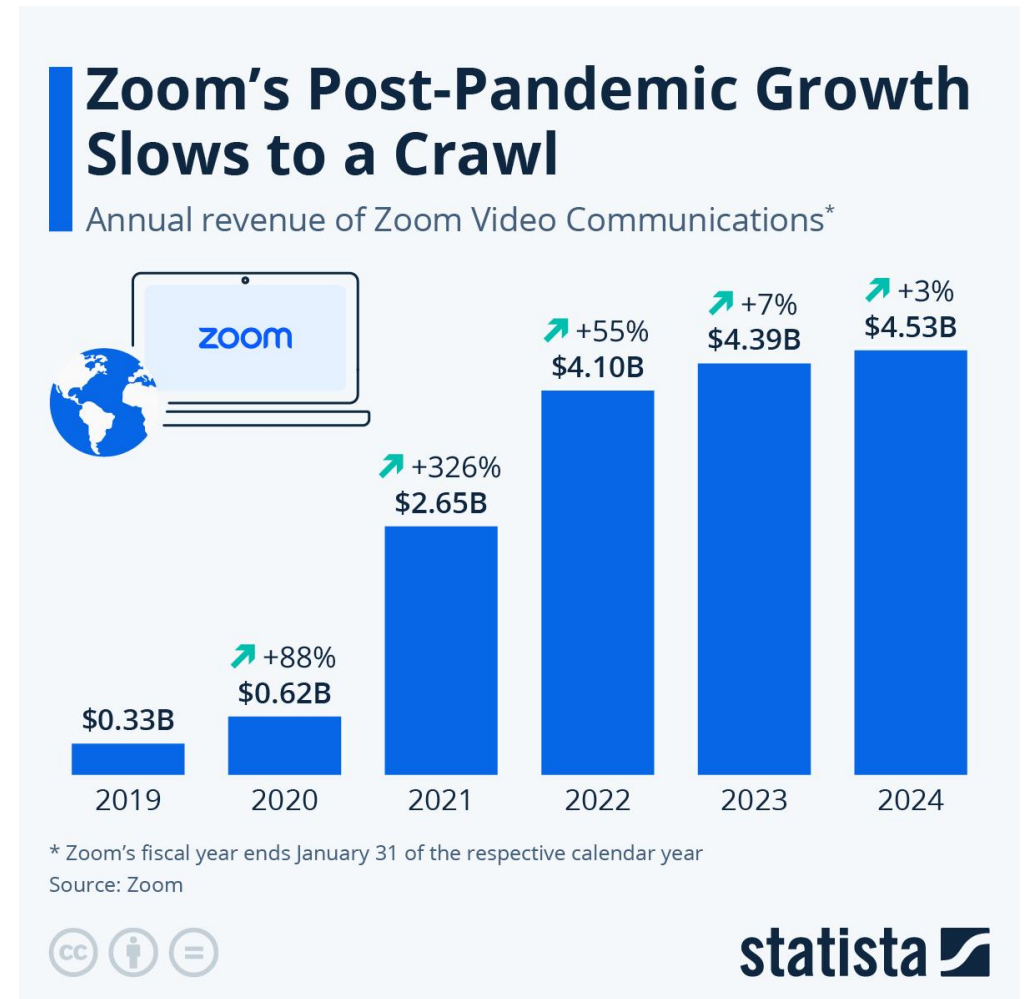
- Stereo 3D, multi-view

- Immersive video

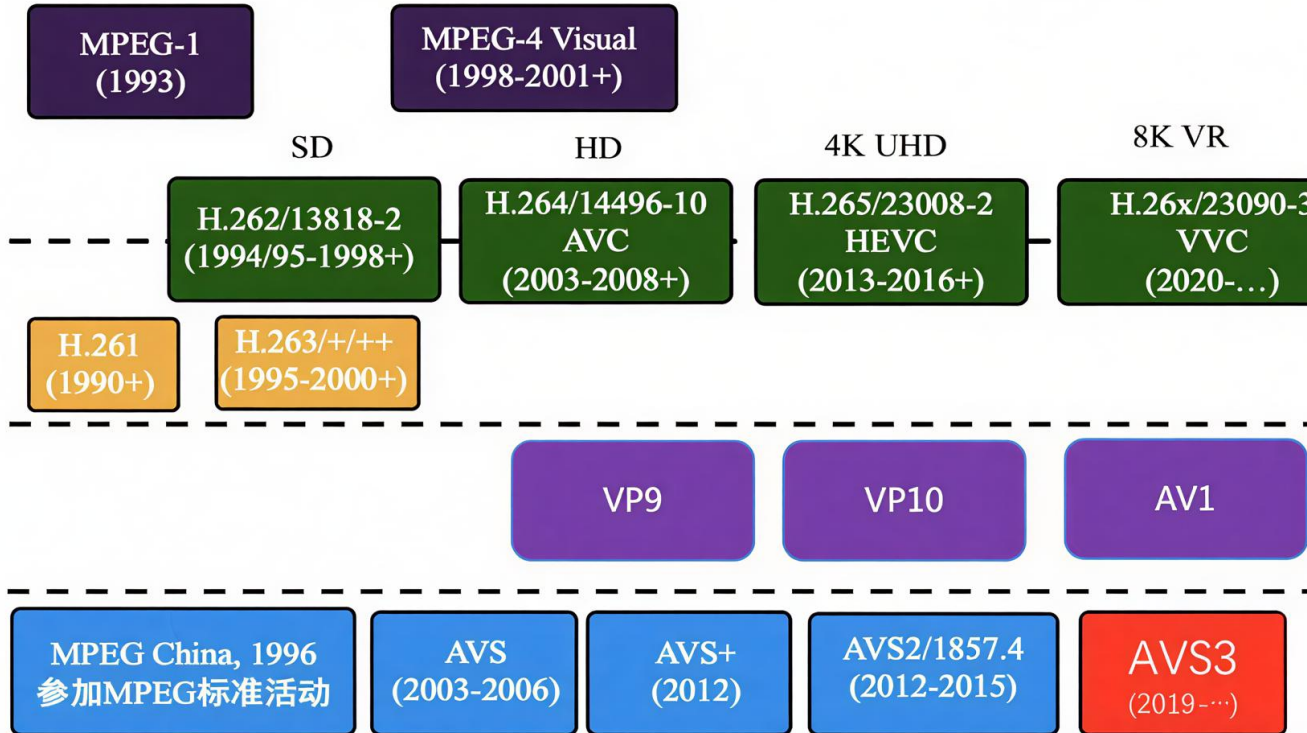
- Light fields
- 3D point clouds
- 3D meshes
- Neural Radiance Fields
- Gaussian Splats

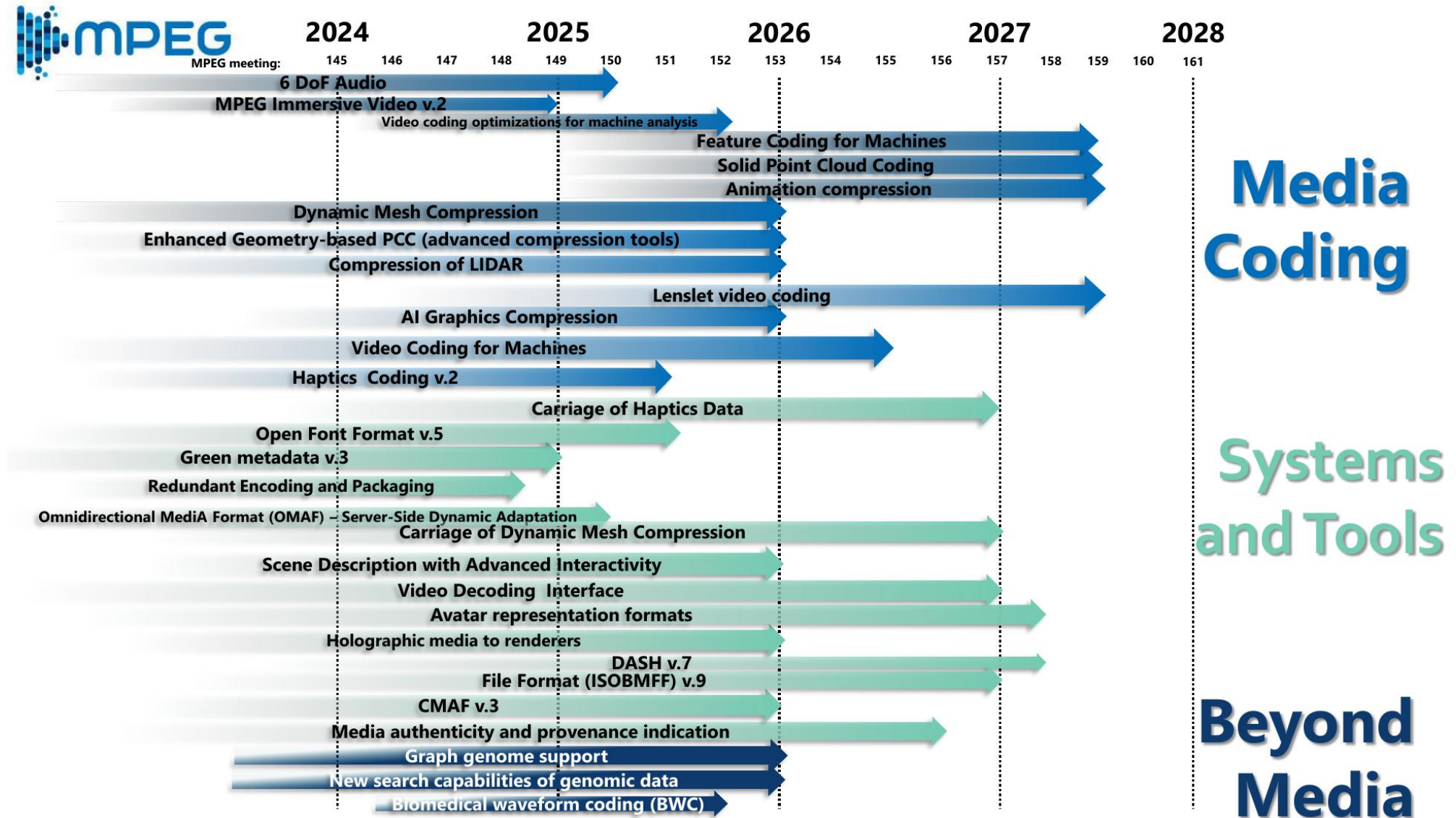
Video compression matters

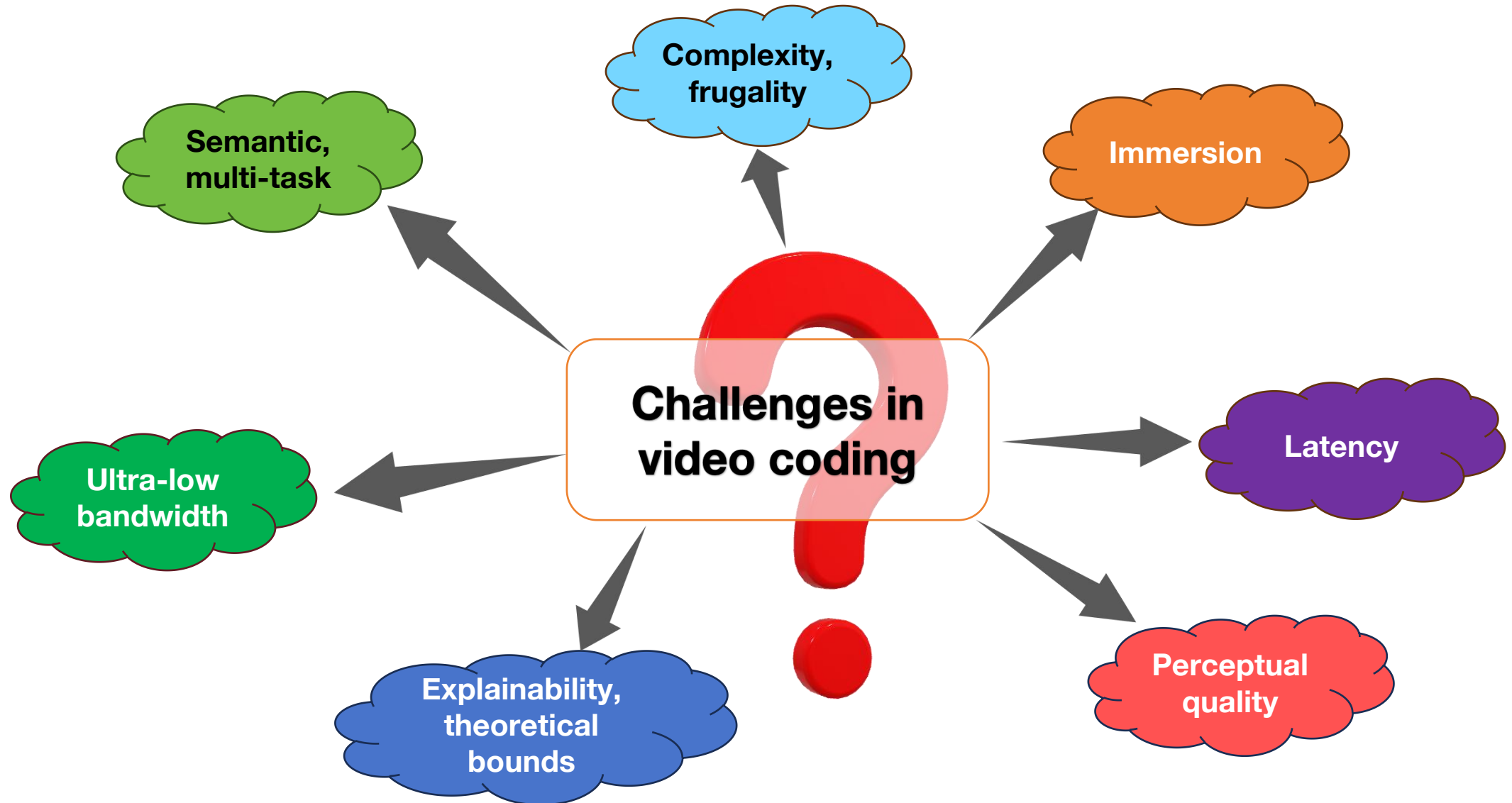
- Video streaming represents over **80% of the total Internet traffic** today
- Video conferencing applications currently represent a substantial share of the video traffic:
 - Zoom
 - Google Meet/classroom
 - Microsoft Teams
 - Cisco Webex meetings
 - ...



Visual coding standards



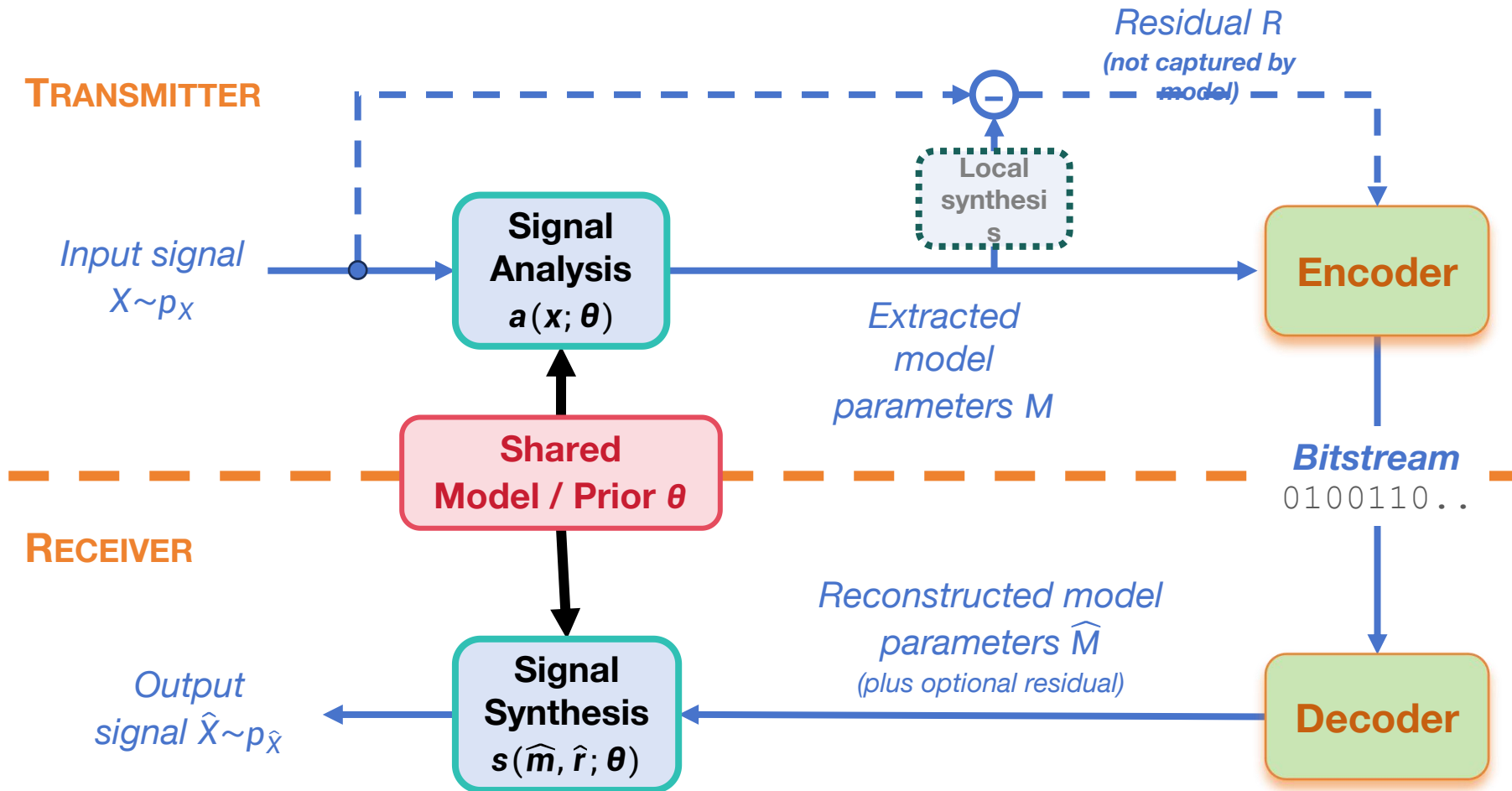




Fundamentals

- The general compression paradigm
- Traditional and neural codecs

Typical Compression Paradigm



Objective:

Minimize average transmitted bitrate, subject to a fidelity constraint

Information-theoretic rate-fidelity tradeoff:

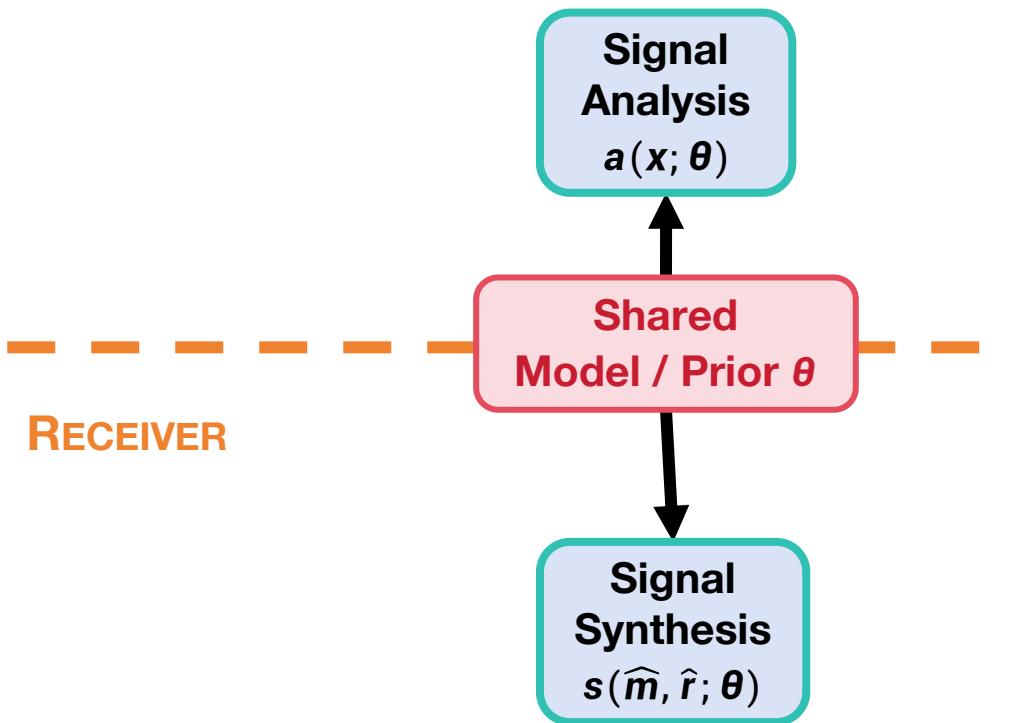
$$R(D) = \inf_{p_{\hat{X}|X}} I(X; \hat{X})$$

$$\text{s. t. } f(p_{X, \hat{X}}) \leq D$$

Typical Compression Paradigm

Data modeling

TRANSMITTER



RECEIVER

- **Analysis function:** extract the relevant representation/message to be transmitted
 - Make the message to be transmitted low dimensional
 - Discard irrelevant information
 - E.g., linear/nonlinear transforms
- **Synthesis function:** reconstruct an output signal of interest
 - Pixels
 - Detected objects/classes
 - ...
- **Shared model/prior:** what the transmitter already knows and need not be transmitted
 - A probability distribution of typical/relevant signals;
 - Some smoothness properties of the input;
 - A foundation model;
 - A random generator (common randomness)
 - ...

Typical Compression Paradigm

Coding

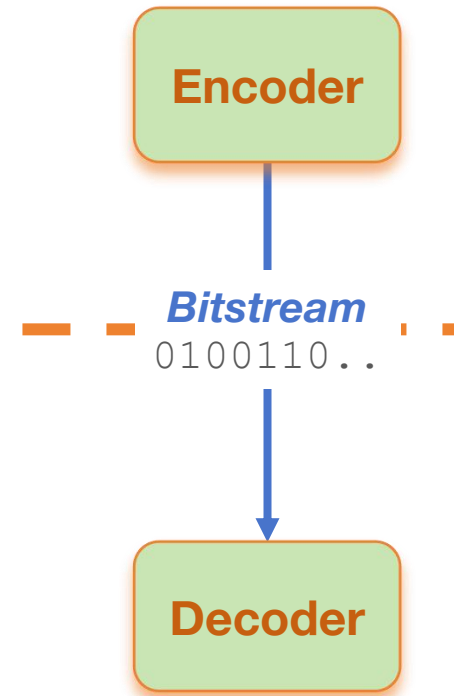
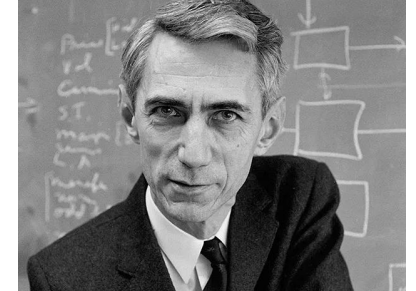
- **Noiseless transmission channel:**

- Remove redundancy/Transmit the minimum number of bits to perfectly reconstruct the message at the decoder
- **Entropy coding:** use $H(M)$ bits/symbol for long sequences of symbols
- **Quantization** to handle continuous sources/reduce precision when needed

- **Noisy transmission channel:**

- Add redundancy to achieve a transmission rate asymptotically approaching the channel capacity
- **Channel coding** to protect against transmission errors

C. E. Shannon, "A Mathematical Theory of Communication," Bell System Technical Journal, vol. 27, pp. 379–423 and 623–656, 1948.



Typical Compression Paradigm

The rate-distortion optimization

- Distortion/fidelity function f
 - E.g.: MSE, feature distance (LPIPS), “world intelligibility”¹, etc.
- Rate: minimum number of bits to resolve uncertainty given a certain distortion are given by $I(X; \hat{X}) = H(X) - H(X|\hat{X})$
- The system is optimized end-to-end, over all possible conditional distributions $p_{\hat{X}|X}$ (“test channels”)
- The decoder can be **stochastic**
 - If the fidelity criterion is an *average distortion constraint*, deterministic block codes suffice to approach $R(D)$ as the block length increases

Objective:

Minimize average transmitted bitrate, subject to a fidelity constraint

Information-theoretic rate-fidelity tradeoff:

$$R(D) = \inf_{p_{\hat{X}|X}} I(X; \hat{X})$$

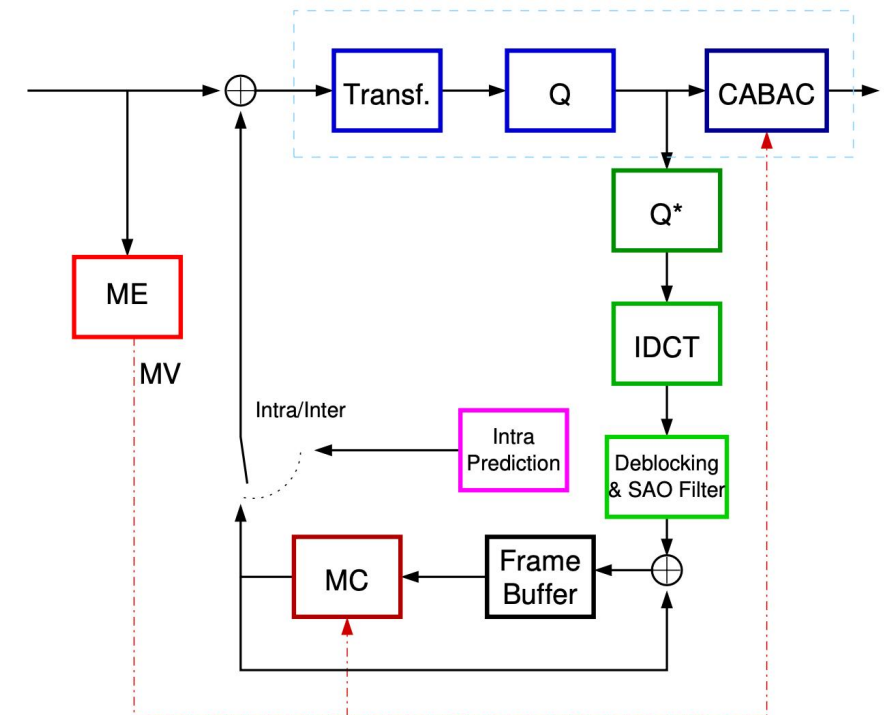
$$\text{s. t. } f(p_{X, \hat{X}}) \leq D$$

From the basic model to practical video codecs

- Traditional hybrid video codecs
- Neural video codecs

Traditional video codecs (non learning-based)

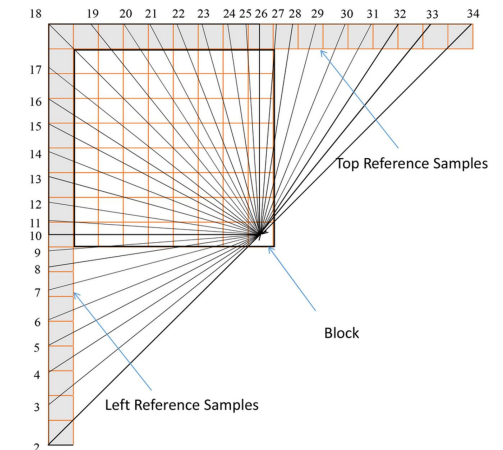
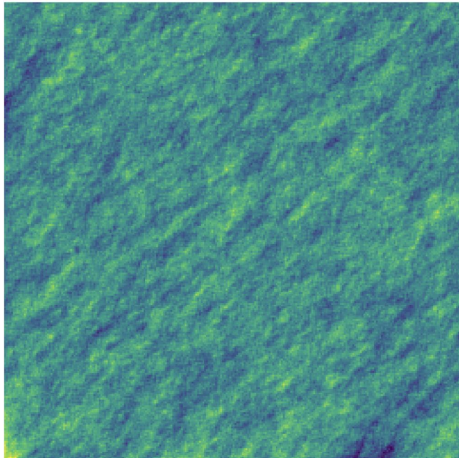
- Goal: **remove redundancies in video**
- Spatial redundancy
 - Correlations between pixels inside a frame
 - **Spatial transforms (e.g., DCT)**
 - **Spatial prediction**
- Time redundancy
 - Correlations between pixels across consecutive frames
 - **Motion estimation and compensation**
- Perceptual and statistical redundancy
 - **Quantization**
 - **Entropy coding**



Limitations of traditional video codecs

- Limited **shared/prior knowledge** and analysis/synthesis tools
 - Very simple pixel and symbol dependency models, limited contexts
- Examples:
 - **DCT**: optimal for AR(1) signals with correlation coefficient close to 1
 - **Motion vectors**: translational motion
 - **Intra Prediction**: bilinear interpolation from limited context

AR(1) with $\rho = 0.95$



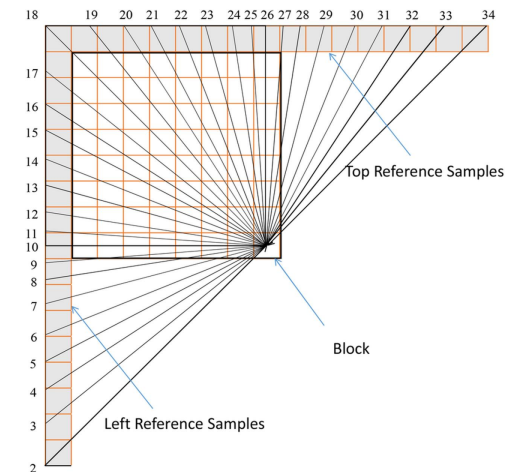
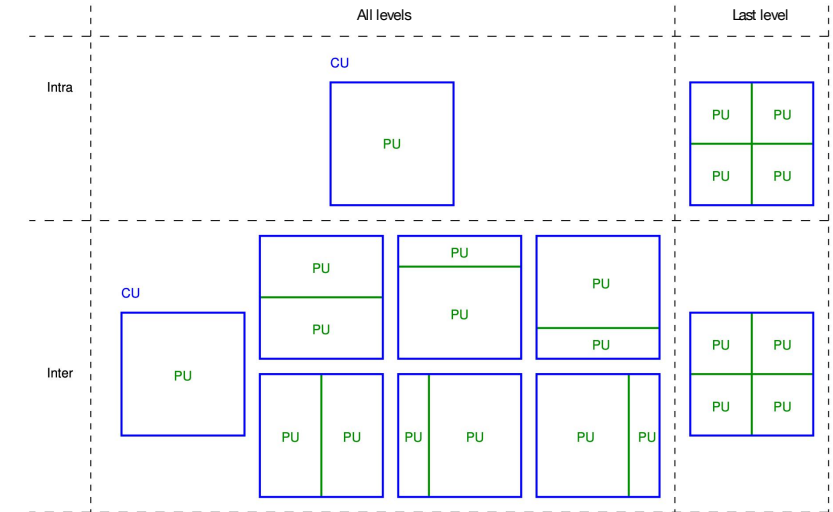
Limitations of traditional video codecs

- **Disjoint rate-distortion** optimization
 - To compress efficiently, codecs use a very large number of possible coding modes/transforms
- **Discrete** rate-distortion optimization (RDO) to choose the best modes:

$$\arg \min_{i_k} D_k(i_k, Q) + \lambda R_k(i_k, Q), \quad i_k$$

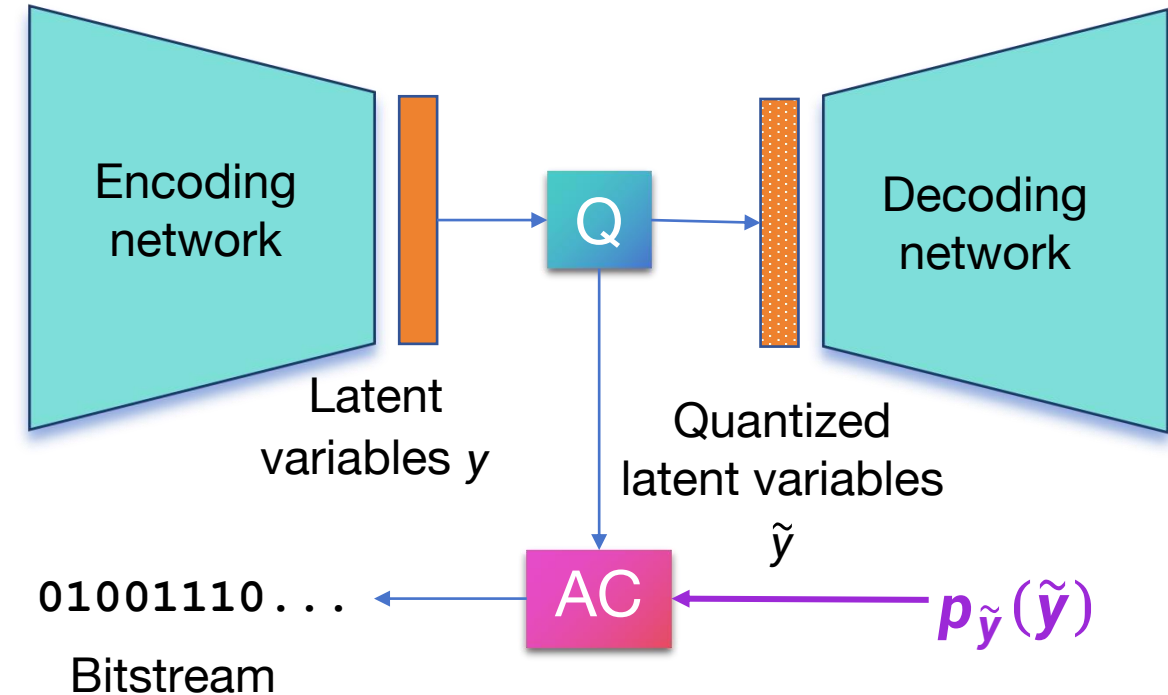
$\in \{\text{coding modes}\}$

where D is typically a **pixel distance function** (e.g., SSD or SAD)



Can we solve these problems using learned codecs?

- E.g., autoencoder-based codecs
- More **expressive learned signal models**
 - Nonlinear transform coding
- **End-to-end optimization** (gradient descent)
- Still optimized using a **distance criterion** for fidelity...



$$\min \mathbb{E}[\|x - \hat{x}\|^2] + \lambda H(\tilde{y})$$

Using a point distance for fidelity

- Traditional codecs and early neural codecs are optimized using **point-based distortion functions**
 - E.g., MSE, PSNR, MSSIM,...
- More sophisticated/learned signal models can improve RD performance
- But all models fail to provide acceptable quality at very low bitrate...
- **Why?**



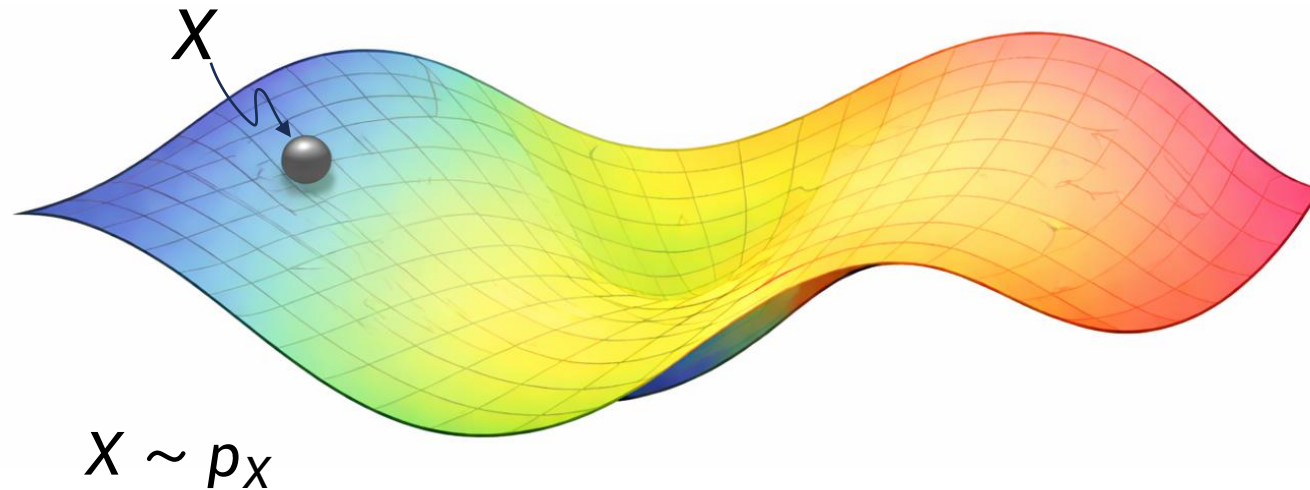
J. Ballé, V. Laparra, and E. P. Simoncelli, "End-to-end Optimized Image Compression," International Conference on Learning Representations, 2017.

The generative coding paradigm

- Realism and the rate-perception-distortion tradeoff

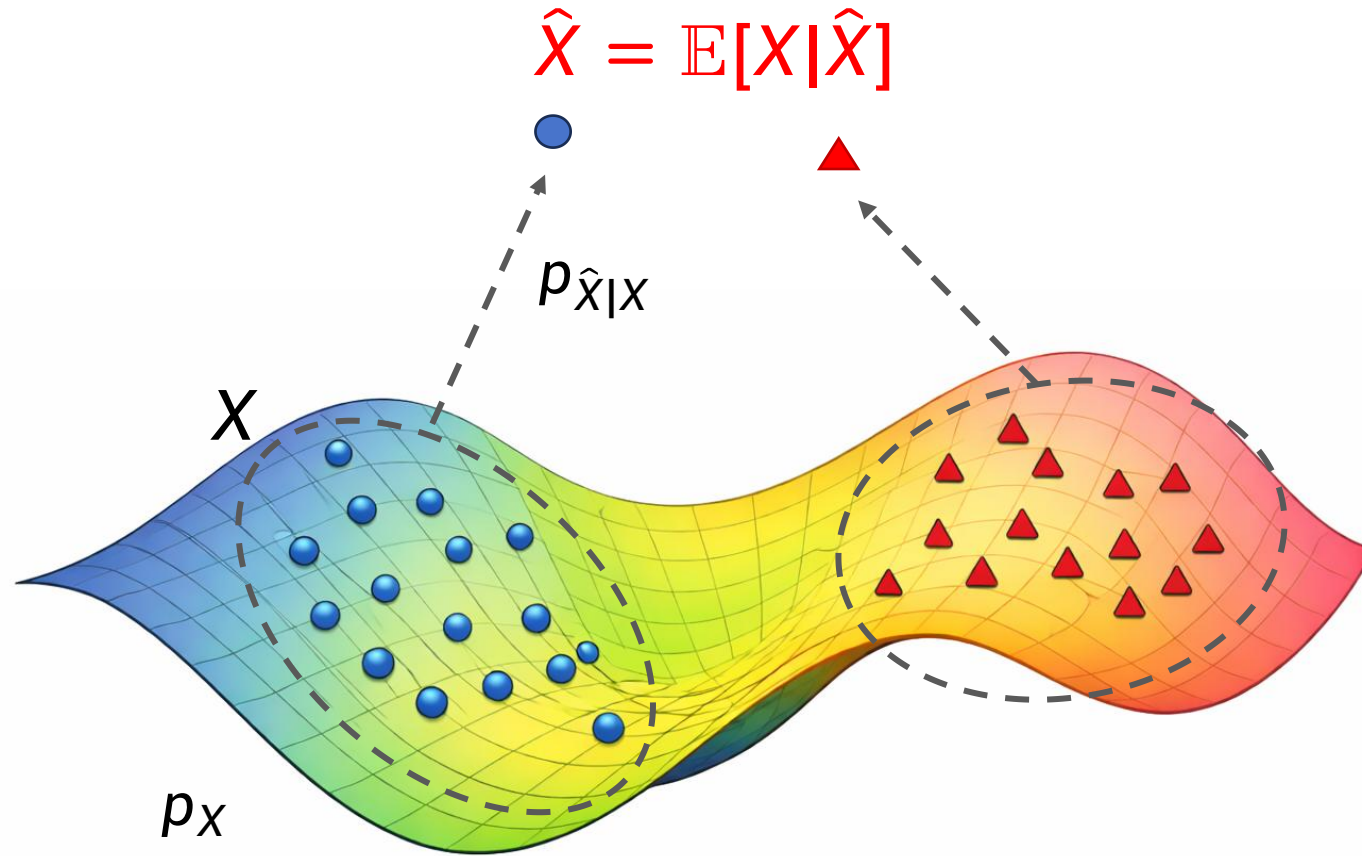
Optimizing for a point distance at low bitrate

- Manifold hypothesis



- Natural images live on the manifold
- Images outside the manifold have poor perceptual quality
- We take the example of Mean Squared Error (MSE) as distance

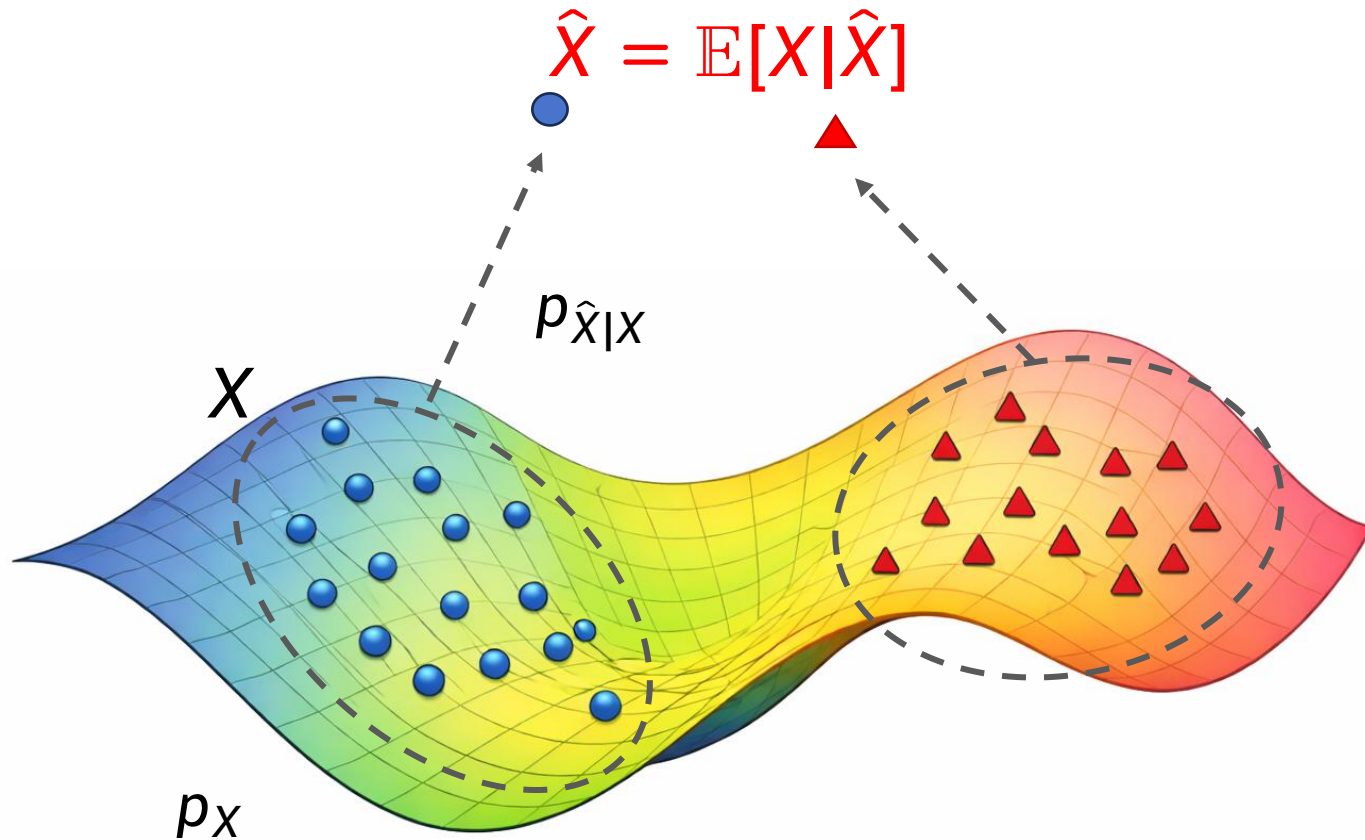
Optimizing for MSE at low bitrate



$$R(D) = \min_{p_{\hat{X}|X}} I(X; \hat{X}) \quad \text{s. t.} \quad \mathbb{E} \left[(X - \hat{X})^2 \right] \leq D$$

- For a given rate R , there are $2^{nR} < 2^{nH}$ possible reconstructions when coding blocks of n symbols
- Optimizing for MSE pushes the reconstruction away from the original data manifold

Optimizing for MSE at low bitrate



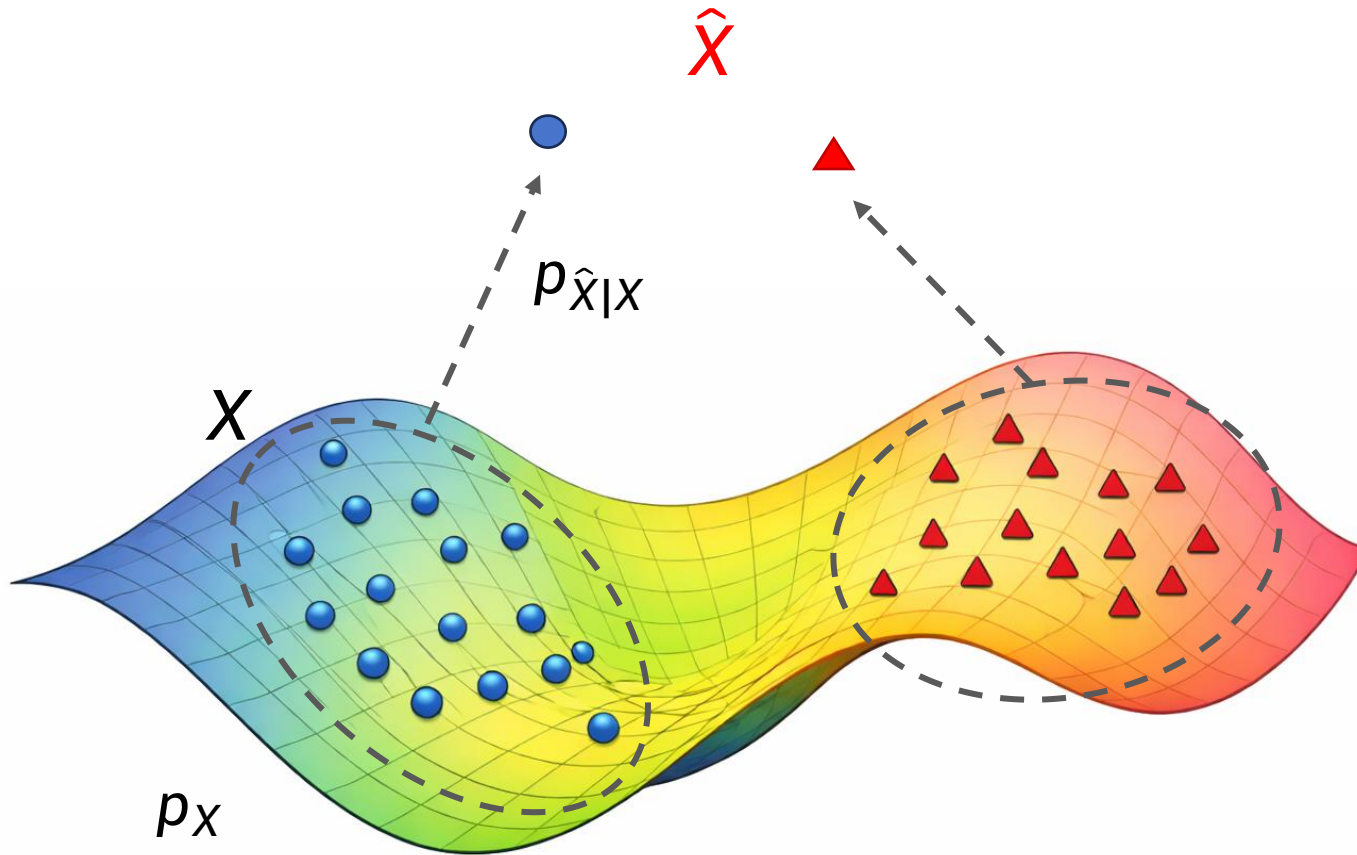
$$R(D) = \min_{p_{\hat{X}|X}} I(X; \hat{X}) \quad \text{s. t.} \quad \mathbb{E} \left[(X - \hat{X})^2 \right] \leq D$$

- At low bitrate, perceptual quality becomes very poor



Bitrate = 4.53 kbps

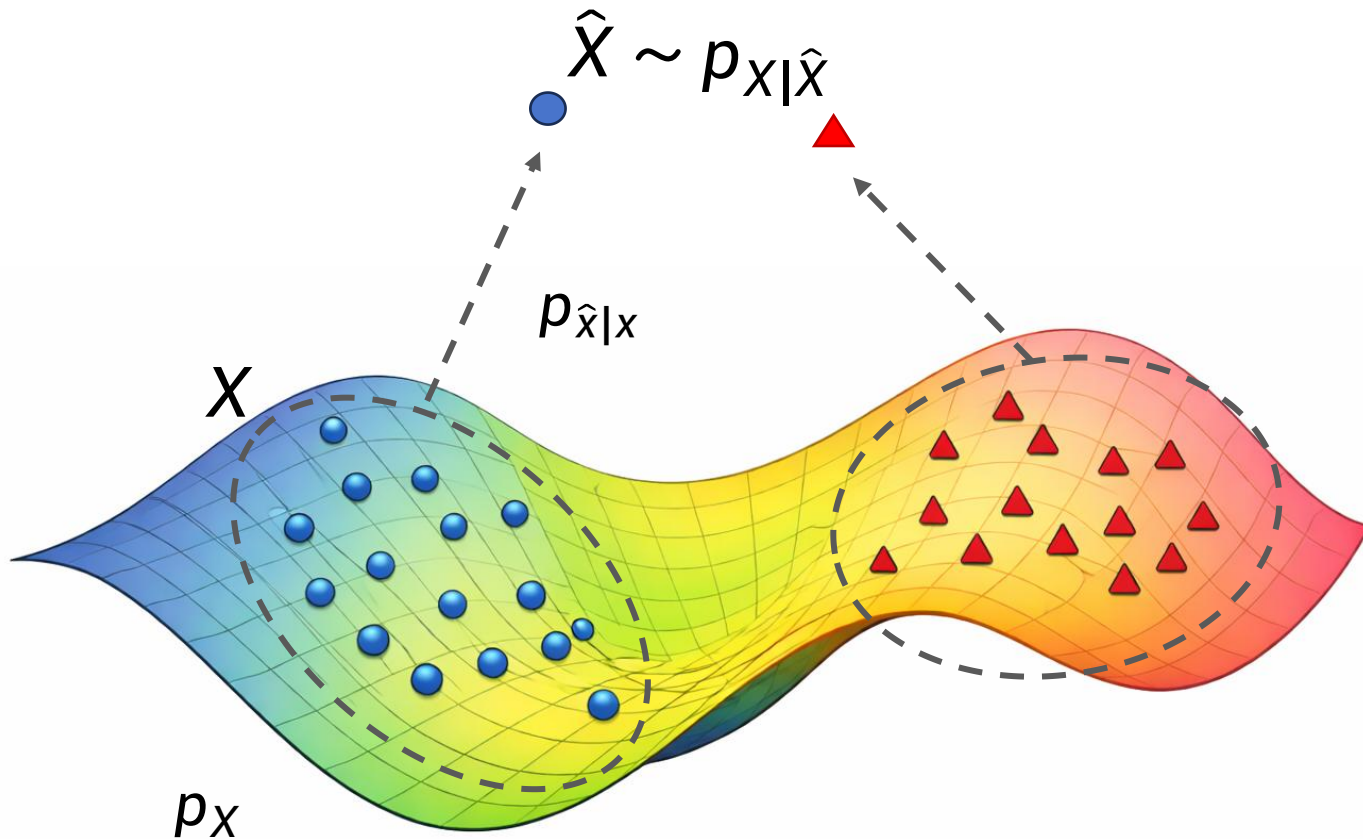
Optimizing for MSE at low bitrate



$$R(D) = \min_{p_{\hat{X}|X}} I(X; \hat{X}) \quad \text{s. t.} \quad \mathbb{E} \left[(X - \hat{X})^2 \right] \leq D$$

- Using more sophisticated distance functions between X and $\hat{X}$ (e.g., LPIPS) can improve visual quality
- But still the reconstructions can be pushed outside the manifold

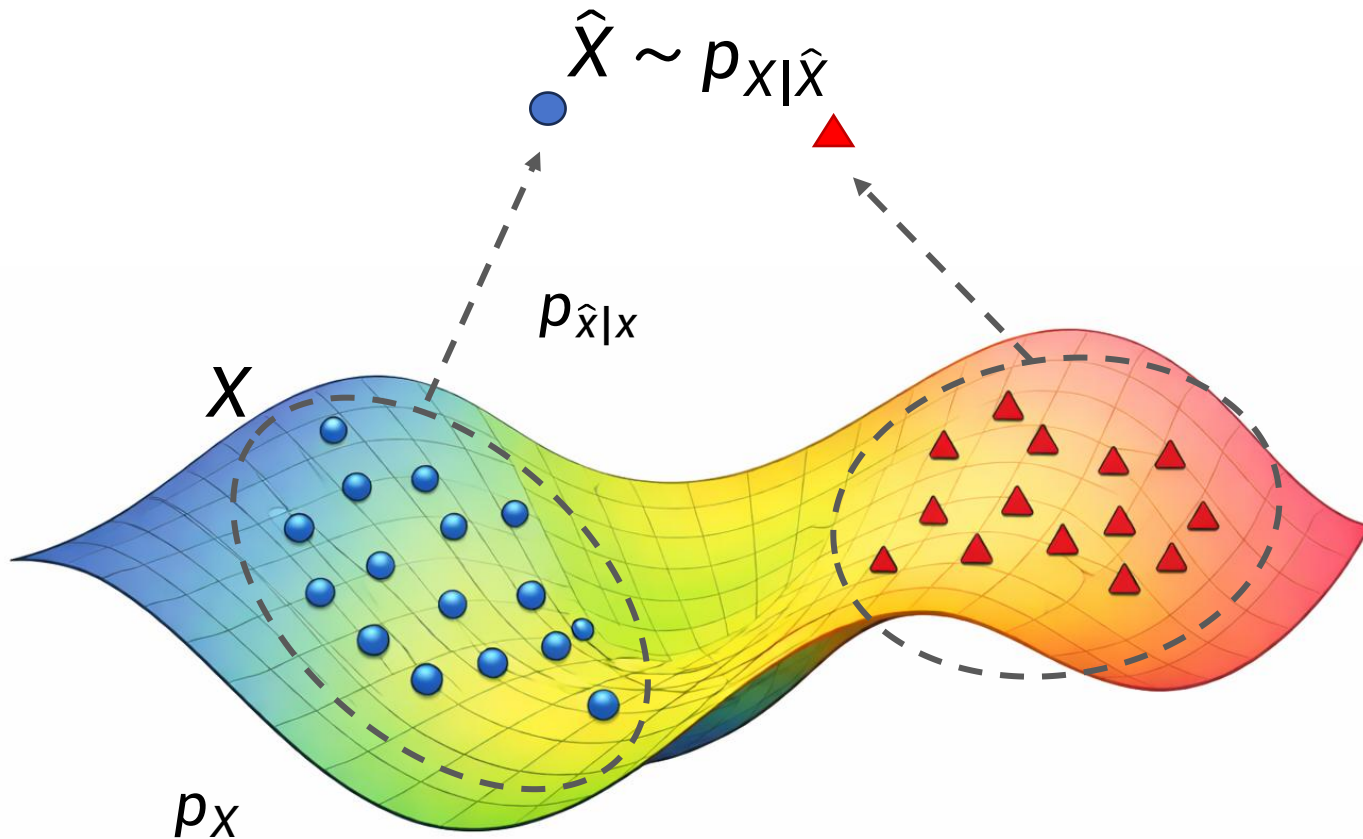
Generative coding



- To optimize perceptual quality, the reconstruction should live on the manifold
- Sample a reconstructed image from the *posterior distribution* $p_{X|\hat{X}}$
- **Generative codecs**

$$R(D) = \min_{p(\hat{X}|X)} I(X; \hat{X}) \quad \text{s. t.} \quad \mathbb{E} \left[(X - \hat{X})^2 \right] \leq D$$

Generative coding



- **Generative codecs**
- The reconstructed signals should follow the same distribution as the original data
- **Perception criterion:** *distance between distributions*

Perception/realism term

$$R(D) = \min_{p(\hat{X}|X)} I(X; \hat{X}) \quad \text{s. t.} \quad \mathbb{E} \left[(X - \hat{X})^2 \right] \leq D, \quad \overbrace{d(p_X, p_{\hat{X}})} \leq \gamma$$

Perception optimization in traditional coding

- Distribution preserving quantizers, dithering,...

1962

IRE TRANSACTIONS ON INFORMATION THEORY

145

Picture Coding Using Pseudo-Random Noise*

LAWRENCE GILMAN ROBERTS†

IEEE TRANSACTIONS ON COMMUNICATIONS, VOL. 39, NO. 11, NOVEMBER 1991

Moment Preserving Quantization

Edward J. Delp, *Senior Member, IEEE*, and Owen Robert Mitchell, *Senior Member, IEEE*

1014

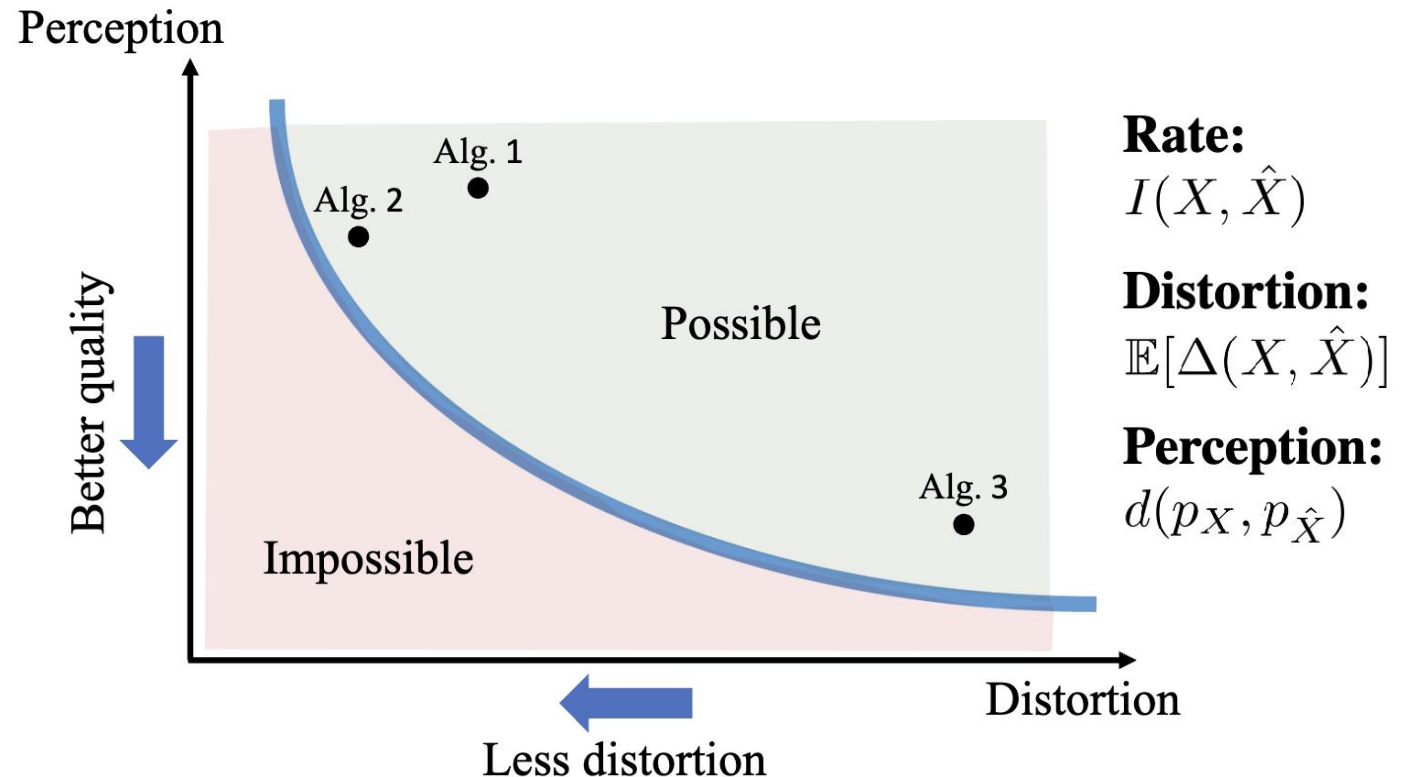
IEEE SIGNAL PROCESSING LETTERS, VOL. 17, NO. 12, DECEMBER 2010

Distribution Preserving Quantization With Dithering and Transformation

Minyue Li, *Student Member, IEEE*, Janusz Klejsa, *Student Member, IEEE*, and W. Bastiaan Kleijn, *Fellow, IEEE*

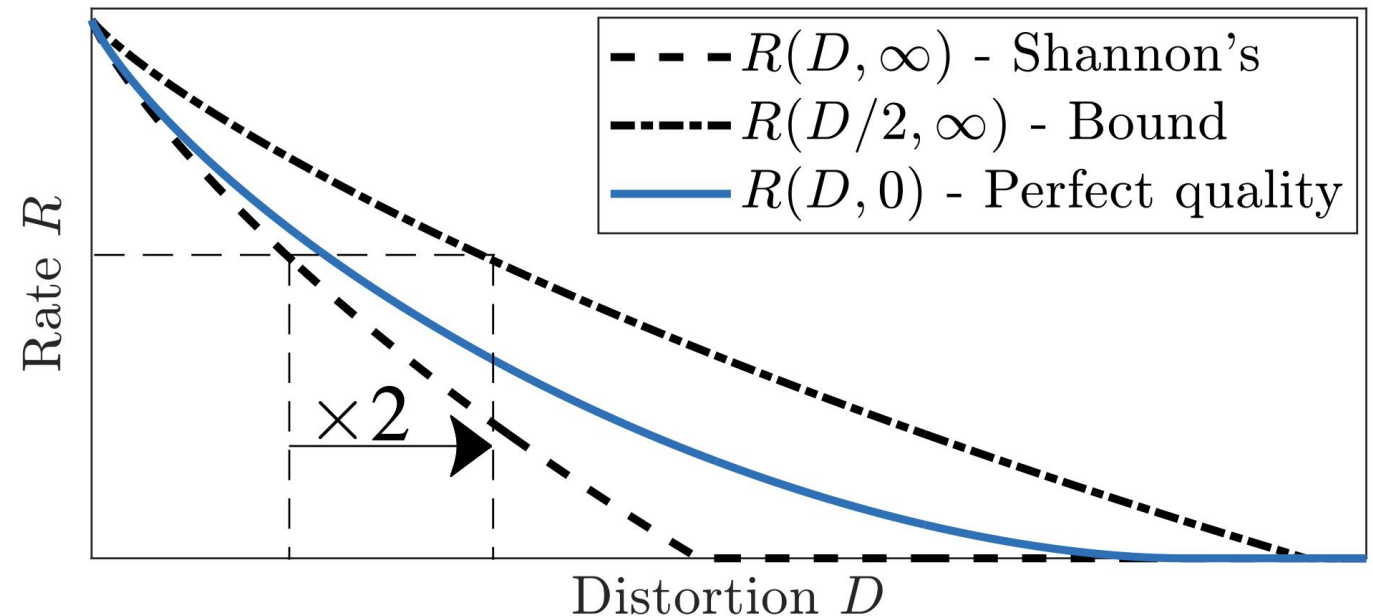
The Rate-Perception-Distortion (RPD) Trade-off

- **One cannot achieve at the same time good perceptual quality and good fidelity, with arbitrary low bitrate**
- **Distortion:** pixel- or feature-based fidelity
- **Perceptual quality/realism:** divergence between data distributions

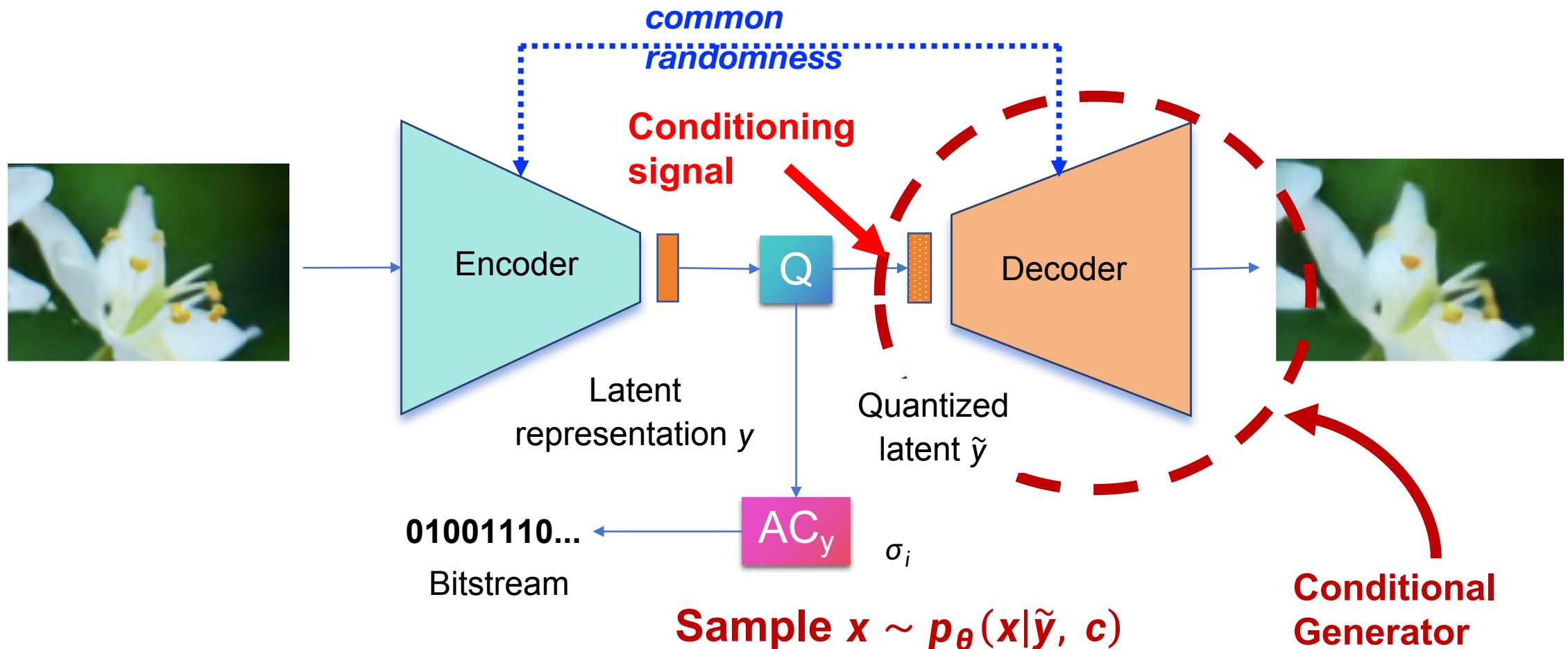


The Rate-Perception-Distortion (RPD) Trade-off

- ***One cannot achieve at the same time good perceptual quality and good fidelity, with arbitrary low bitrate***
- With MSE distortion, perfect realism in reconstruction can increase distortion by 2x, for a given bitrate



Ultra-low bitrate compression with generative codecs



How to improve the RPD tradeoff?

- **Better generative models**

- For a fixed conditioning representation, more accurate posterior sampling can improve reconstruction realism

- **More informative conditioning**

- Better representations can improve fidelity at a given bitrate
- Strong domain priors can make compact conditioning particularly effective, as in GFVC

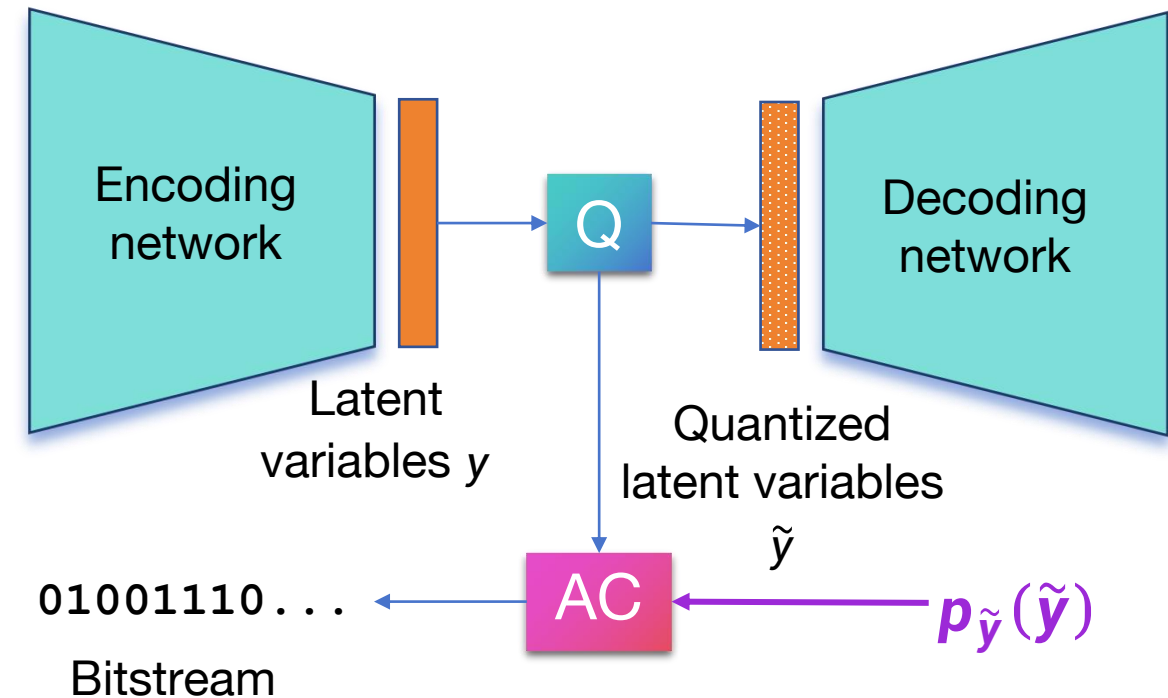
- **Common randomness**

- With sufficient shared randomness, channel simulation makes the information-theoretic RDP bound asymptotically achievable.

Generative models

• Variational AutoEncoders (VAE)

- First-generation neural codecs (e.g., Ballé et al., 2016)
- Maximize a variational lower bound (ELBO) on the data log-likelihood.
- Simple Gaussian decoder likelihoods favor conditional-mean reconstructions, which can blur uncertain details.
- Uncertain details are averaged out, blurred reconstructions, especially at low rates



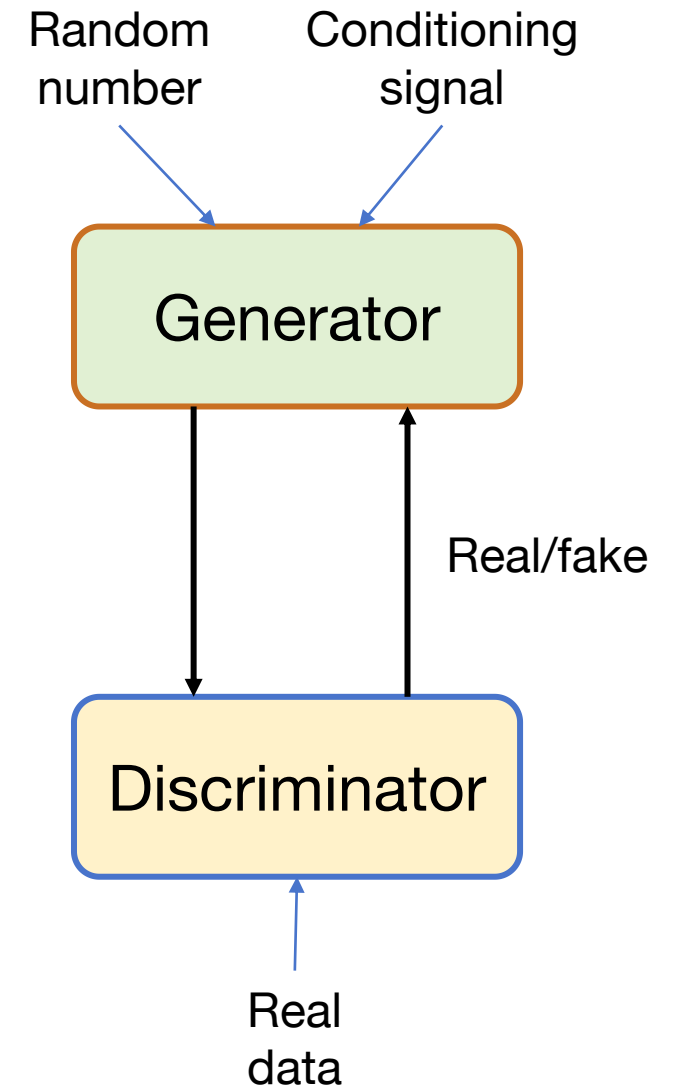
Generative models

• Generative Adversarial Networks

- Minimax game and adversarial training

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log D_{\theta_d}(x) + \mathbb{E}_{z \sim p(z)} \log(1 - D_{\theta_d}(G_{\theta_g}(z))) \right]$$

- Equivalent to minimizing the Jensen-Shannon divergence between the generated and the original data distributions
- Other variants with different divergences exist (e.g., Wasserstein GANs)
- Recent neural codecs include an adversarial loss (e.g., HiFIC) to increase realism at low bitrate



Generative models

• Diffusion models (DDPM: Denoising Diffusion Probabilistic Models)

- Forward process: gradually add noise with a known schedule
- Learn to denoise/estimate reverse conditional distributions, with optional conditioning
- The generation is divided into many smaller denoising steps

● Forward / noising process



● Reverse / denoising process

J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," Advances in Neural Information Processing Systems, vol. 33, pp. 6840–6851, 2020.

P. Dhariwal and A. Nichol, "Diffusion Models Beat GANs on Image Synthesis," Advances in Neural Information Processing Systems, vol. 34, 2021.

Generative models

• Diffusion models

- DDPMs are slow and computational expensive due to the high number of denoising steps
- Faster sampling methods can use deterministic trajectories
- **Denoising Diffusion Implicit Models (DDIM)**: reuse a DDPM-trained denoiser with deterministic sampling and fewer steps, enabling faster generation [1].
- **Flow Matching**: learn a velocity field that transports noise into images along deterministic trajectories, allowing more direct paths and efficient sampling [2].

[1] J. Song, C. Meng, and S. Ermon, “Denoising Diffusion Implicit Models,” International Conference on Learning Representations, 2021.

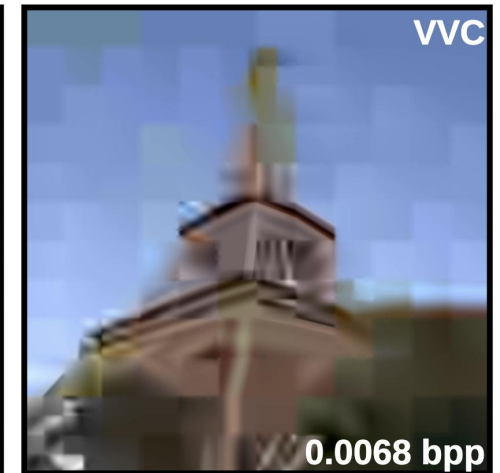
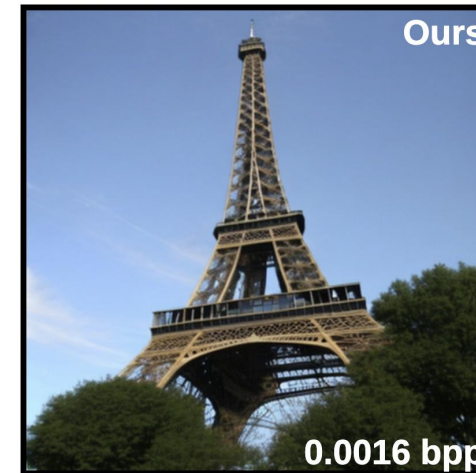
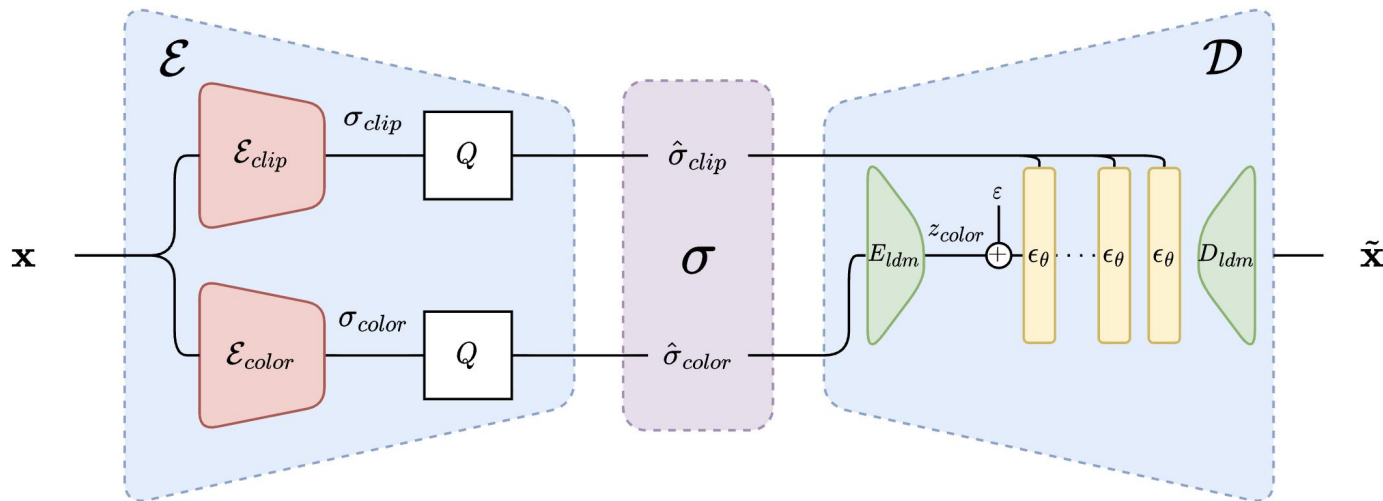
[2] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow Matching for Generative Modeling,” International Conference on Learning Representations, 2023.

Informative conditioning

- For the same transmitted bitrate, retain source-specific information that most effectively guides reconstruction
- **Information-bottleneck** principle: preserve the most relevant information within the available information budget
 - But difficult to define “relevance”
- Examples:
 - Semantic Coding
 - Generative Face Video Coding

Informative conditioning

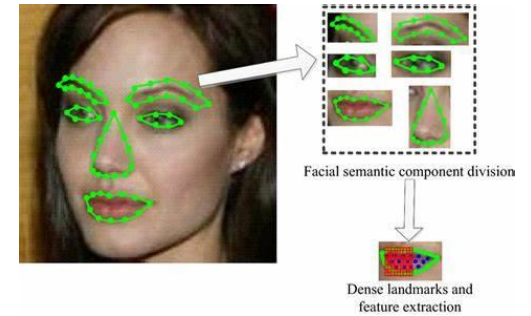
- **Semantic coding** example
- Conditioning with semantic features
 - E.g., CLIP embeddings (text-image)
 - Multimodal conditioning



Informative conditioning

• Generative Face Video Coding (GFVC)

- Physical face constraints + quasi static background
- Physical appearance conveyed by a reference image
- Face motion governed by a compact set of parameters (e.g., keypoints)
- **Model-based coding**



$$X \sim p_X(x_t | \theta, x_0)$$

Small set of
parameters

Reference
image

Common randomness

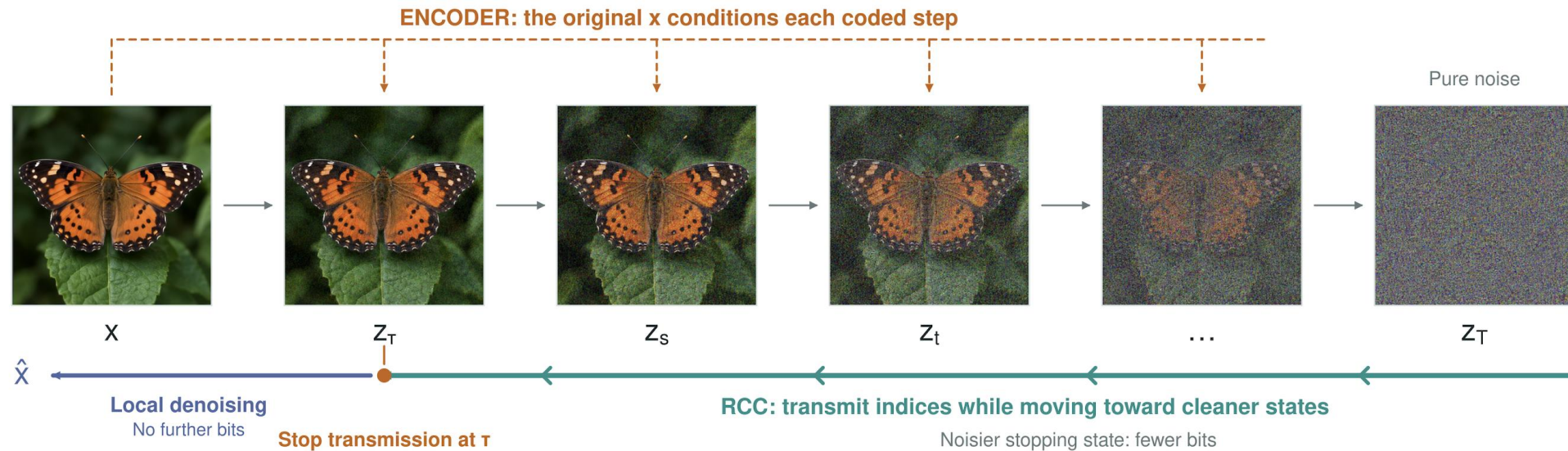
- Generative coding use randomness to sample the posterior distribution
- **Private randomness:**
 - Decoder uses a random generator to sample
 - Ensure diversity and enables minimizing the perception criterion
- **Common (shared) randomness:**
 - The encoder knows the decoder's randomness and can adapt the transmitted message (e.g., dithering)
 - But optimizing the message for each random realization can require **expensive** searches through the generative decoder

L. Theis and A. B. Wagner, "A coding theorem for the rate-distortion-perception function," arXiv preprint arXiv:2104.13662, 2021.

Y. Yang and S. Mandt, "Advances in Diffusion-Based Generative Compression," IEEE BITS the Information Theory Magazine, pp. 1–12, 2026.

Common randomness

- **Reverse Channel Coding (RCC)** with diffusion models:



- **Rate control:** stop transmitting refinements at a chosen noise level. The decoder completes denoising using unconditional sampling.

L. Theis and A. B. Wagner, "A coding theorem for the rate-distortion-perception function," arXiv preprint arXiv:2104.13662, 2021.

Y. Yang and S. Mandt, "Advances in Diffusion-Based Generative Compression," IEEE BITS the Information Theory Magazine, pp. 1–12, 2026.

Common randomness

- **Reverse Channel Coding (RCC)** with diffusion models:
 - In DDPM, the encoder knows the reverse distribution *conditioned on the original* image; the decoder only knows the learned reverse model
 - Both sides generate the same **random candidate sequence** using the model and shared randomness
 - The **encoder selects a candidate** so that it follows the target distribution and transmit its corresponding index; the decoder reproduces that sample from its index
 - **Rate control:** stop transmitting refinements at a chosen noise level. The decoder completes denoising using unconditional sampling.
- With sufficient shared randomness and optimal channel design, **rate approaches $I(X; \hat{X})$** , achieving the information-theoretic RDP bound asymptotically

L. Theis and A. B. Wagner, “A coding theorem for the rate-distortion-perception function,” arXiv preprint arXiv:2104.13662, 2021.

Y. Yang and S. Mandt, “Advances in Diffusion-Based Generative Compression,” IEEE BITS the Information Theory Magazine, pp. 1–12, 2026.

PART 02

Diffusion Methods for Image and Video Coding

Jianhui Chang

China Telecom
Cloud Computing Institute

Diffusion Methods in Generative Image and Video Coding

1 Unified Framework

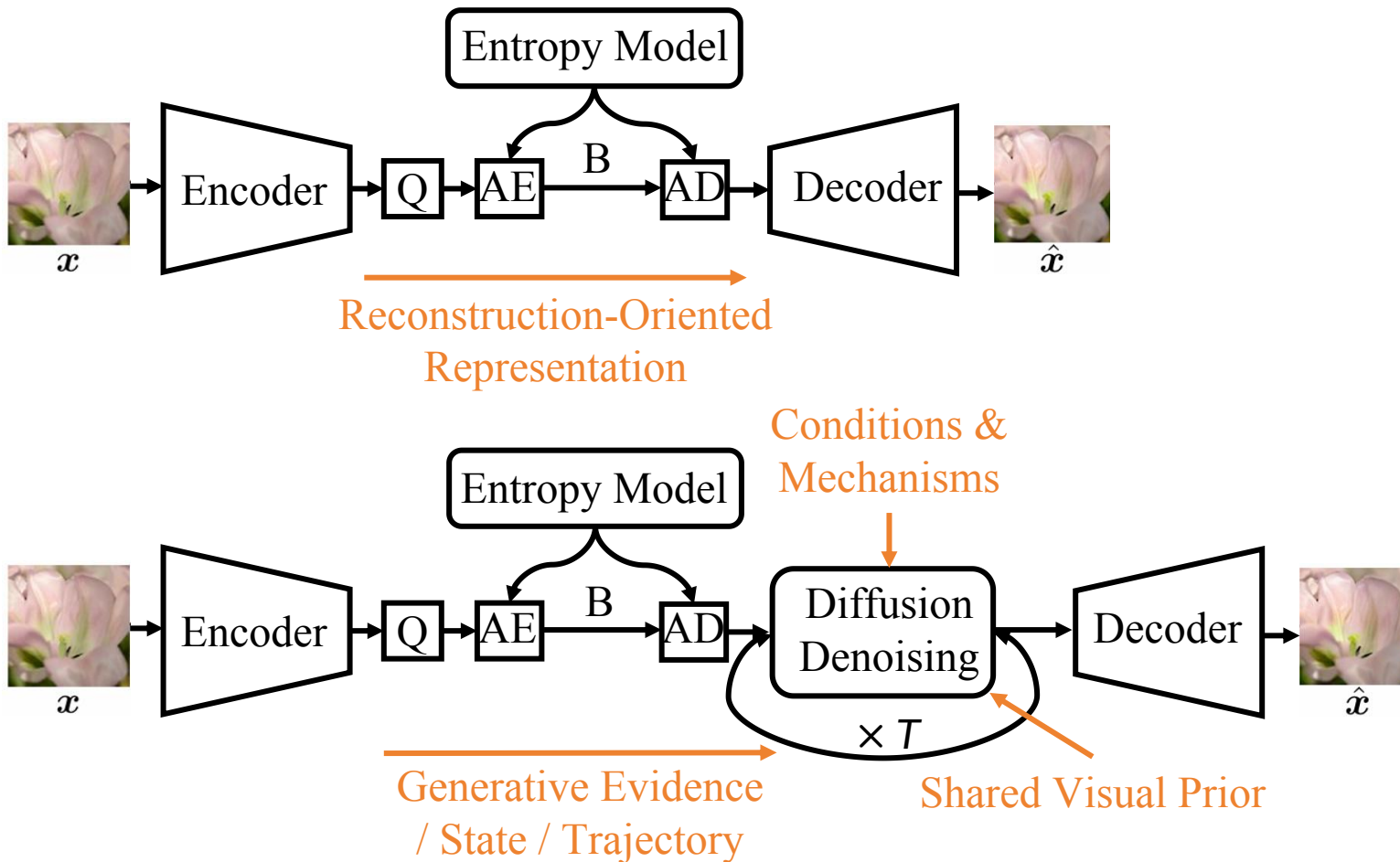
2 Image Coding

- 2.1 What Is Transmitted
- 2.2 Control Mechanisms
- 2.3 Efficient Decoding

3 Video Coding

- 3.1 Frame-Level Predictive
- 3.2 Video-Native Diffusion

A Unified View of Diffusion-Based Visual Coding



Key Shift

- What is transmitted**

Reconstruction feature $\rightarrow$
Generative evidence / state / trajectory

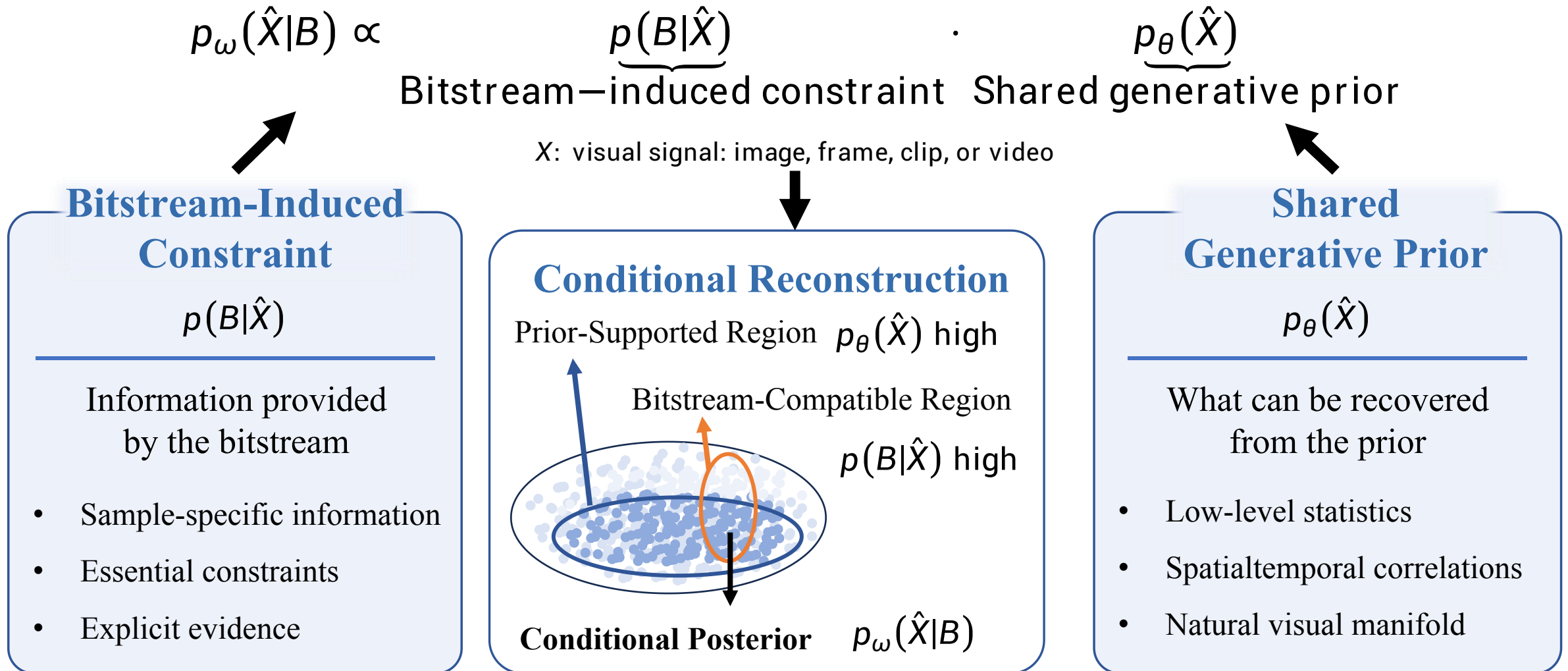
- What is shared**

Codec transforms & latent statistics $\rightarrow$
+ Generative visual prior

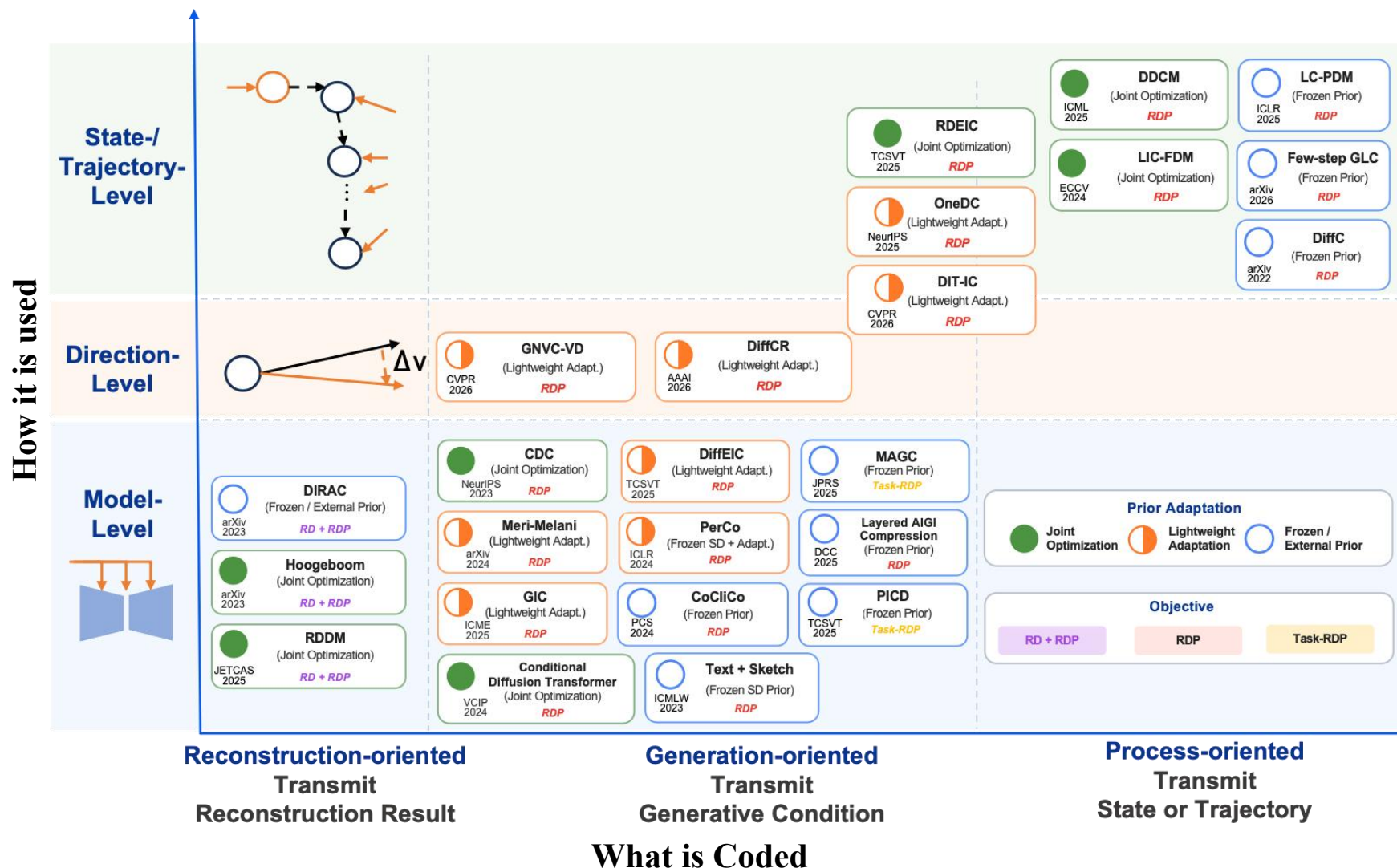
- What the decoder does**

Decode representation $\rightarrow$
Constrain generative reconstruction

- **A Bayesian Perspective of Generative Reconstruction**



• A Taxonomy of Diffusion-Based Visual Coding



What is coded?

Reconstruction, conditions, or process variables.

How is it used?

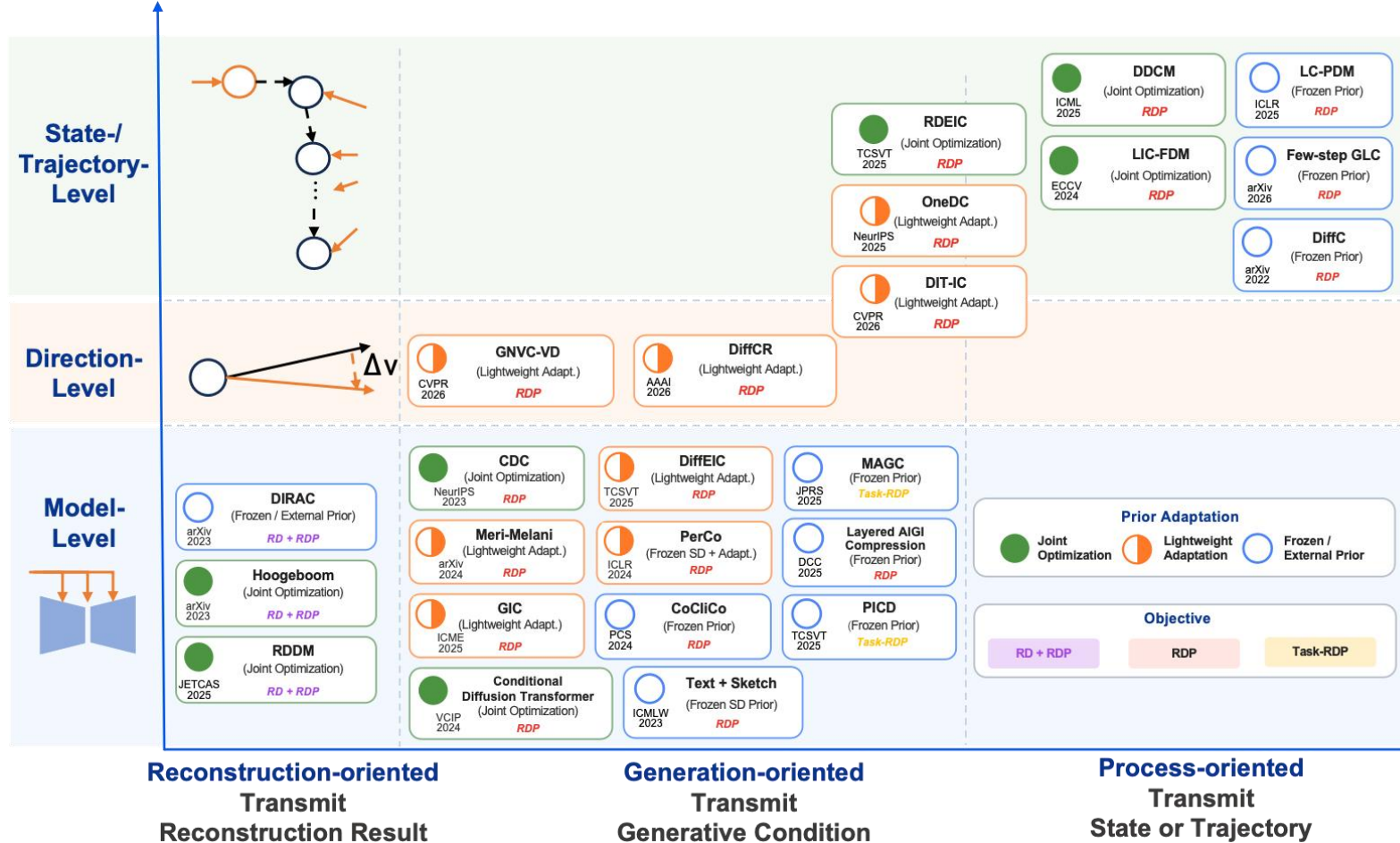
Network conditioning, direction-level guidance, or state / trajectory control.

Prior adaptation

Colors indicate how much of the prior is adapted.

Methods can combine multiple mechanisms.

X-axis: Evolution of Coded Representation



- **Reconstruction-oriented** $B \rightarrow \tilde{x}$
An independently decodable representation gives a base reconstruction.
- **Generation-oriented** $B \rightarrow c_B$
Source-specific conditions constrain generation.
- **Process-oriented** $B \rightarrow \{s_t\}_{t=1}^T$
Coded states or sampling choices specify parts of the generative process.

Diffusion Methods in Generative Image and Video Coding

OUTLINE

1 Unified Framework

2 Image Coding

2.1 What Is Transmitted

2.2 Control Mechanisms

2.3 Efficient Decoding

3 Video Coding

3.1 Frame-Level Predictive

3.2 Video-Native Diffusion

SUBTOPICS

A Reconstruction

B Conditions

B.1 Codec-Native Latents

B.2 Generative-Native Latents

B.3 Cross-Modal Cues

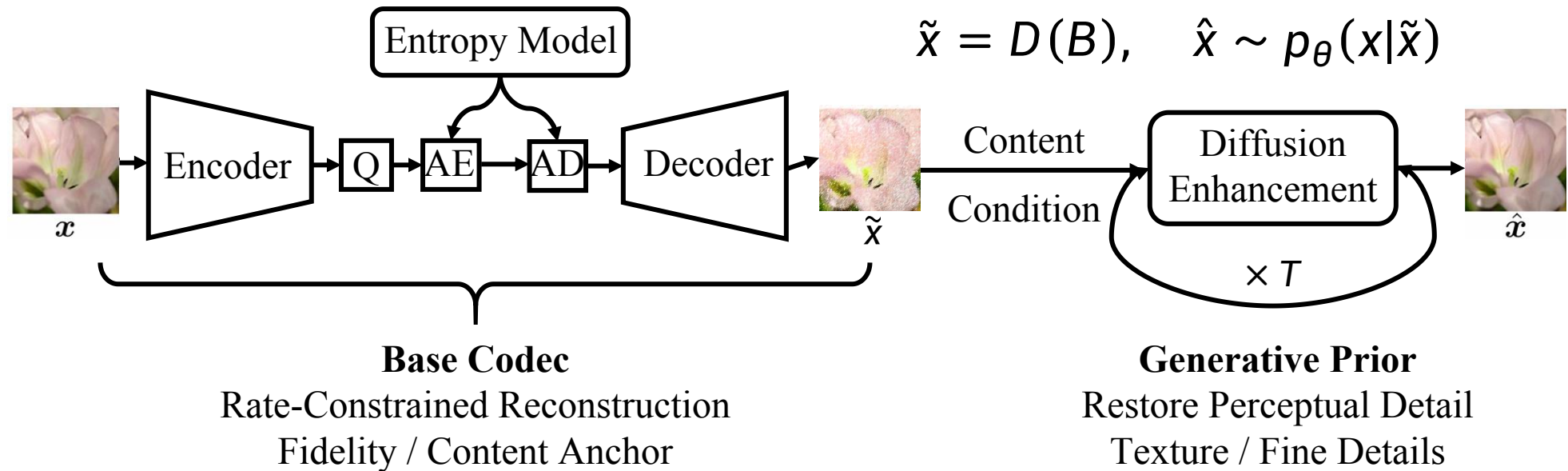
B.4 Task-Critical Evidence

C Generative Process

C.1 State

C.2 Trajectory

Diffusion Enhancement for Basic Codec Reconstruction



Residual generation and image-space refinement

Two representative approaches to codec enhancement

Representative work	What is generated	Conditioning input
DIRAC (2023)	Residual added to the base image	Base codec reconstruction
Hoogetboom et al. (2023) HFD/DDPM	Image-space reconstruction	MSE-optimized reconstruction

Both use the decoded image as a content anchor

The prior improves appearance without extra transmitted bits.

Exact recovery of the original fine detail is not guaranteed.

Base codec



Same bits + DIRAC



JPEG: less blocking, better PSNR and FID

JPEG: 12.8 kB before and after
PSNR: 24.5 to 26.0 dB
FID/256: 81.6 to 18.0



More natural texture and better FID, but lower PSNR

SwinT-ChARM: 9.3 kB before and after
PSNR: 30.2 to 28.1 dB
FID/256: 9.6 to 5.7

Better perceptual quality can come at a cost in pixel accuracy.

Diffusion Methods in Generative Image and Video Coding

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3.1 Frame-Level Predictive

3.2 Video-Native Diffusion

SUBTOPICS

A Reconstruction

B Conditions

B.1 Codec-Native Latents

B.2 Generative-Native Latents

B.3 Cross-Modal Cues

B.4 Task-Critical Evidence

C Generative Process

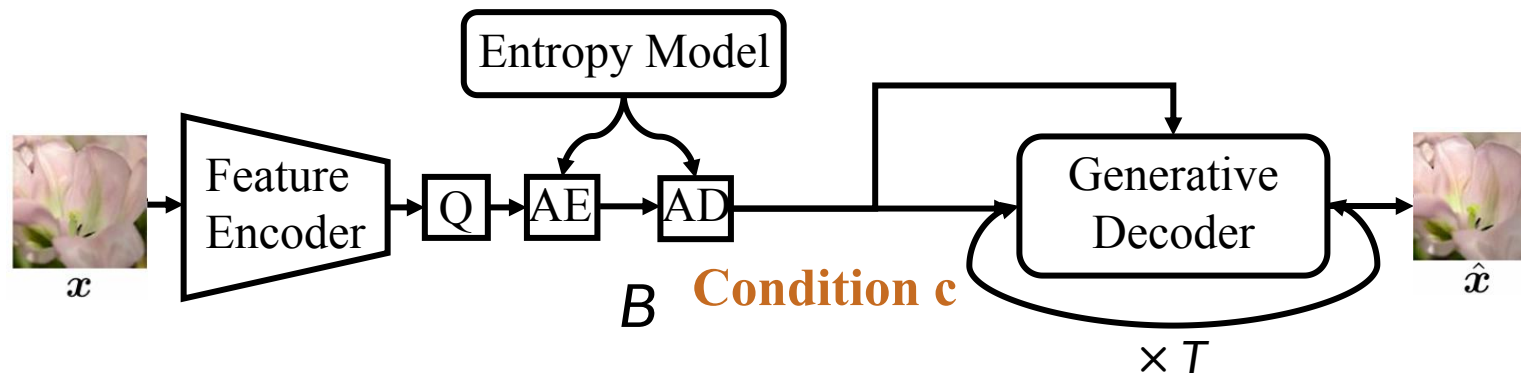
C.1 Process Initialization

C.2 Process Communication

Compressed Representations as Generative Conditions

Compressed representations become generative conditions

Diffusion uses the condition without first reconstructing pixels



TRANSMITTED CONDITION

Layout or identity may need explicit support

SHARED GENERATIVE PRIOR

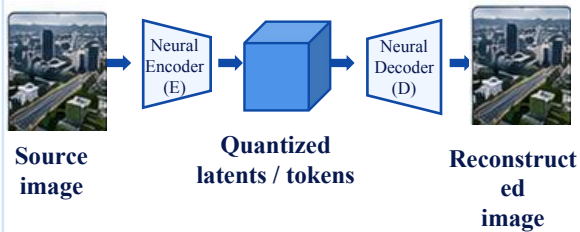
Supplies plausible visual detail

Informative, compressible and usable by the generator

Conditioning Signals for Diffusion Decoding

Codec-Native

Codec Latents

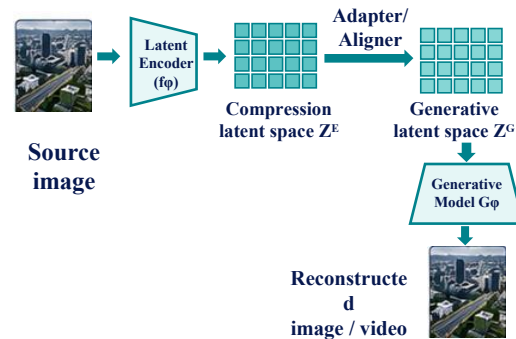


- 🎯 Leverage mature compression representations
- 💡 A good compression representation is already highly informative.

Examples: CDC, GIC

Generator-Aligned

Aligned Latents

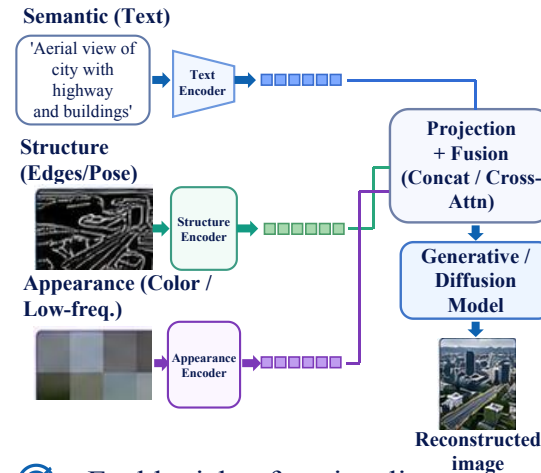


- 🎯 Make transmitted evidence directly usable for generation
- 💡 Represent source information in the latent space naturally understood by the generator.

Examples: DiffGIC, PerCo

Cross-Modal

Explicit Semantic / Structural / Appearance Cues

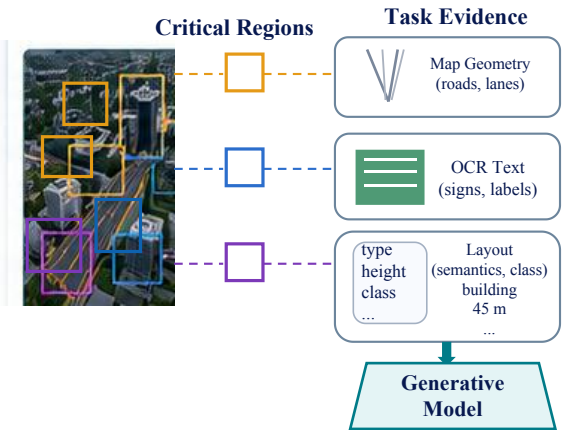


- 🎯 Enable richer functionality
- ⚡ Interpretable Controllable Editable · Scalable

Examples: Layered AIGI, MISC

Task-Critical

Task-Specific Evidence



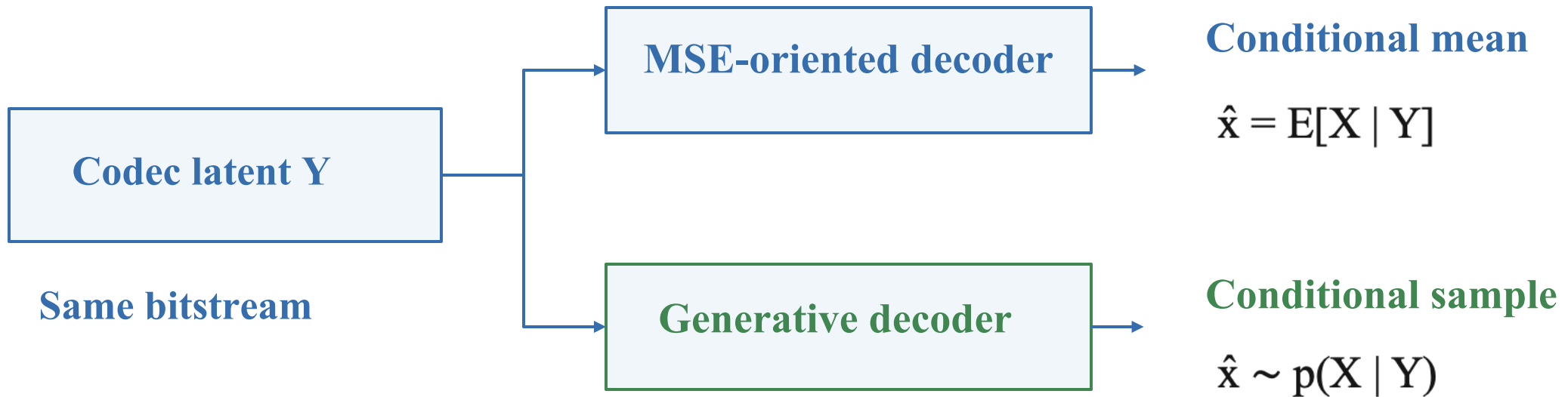
- 🎯 Focus bits on task-sensitive information
- 💡 Preserve information that demands high fidelity.

Examples: MAGC, PICD

Choose conditions for coding efficiency, generator compatibility, explicit control, or task reliability.

One Representation, Different Reconstruction Objectives

The encoder preserves source information. The decoder chooses how to reconstruct it.



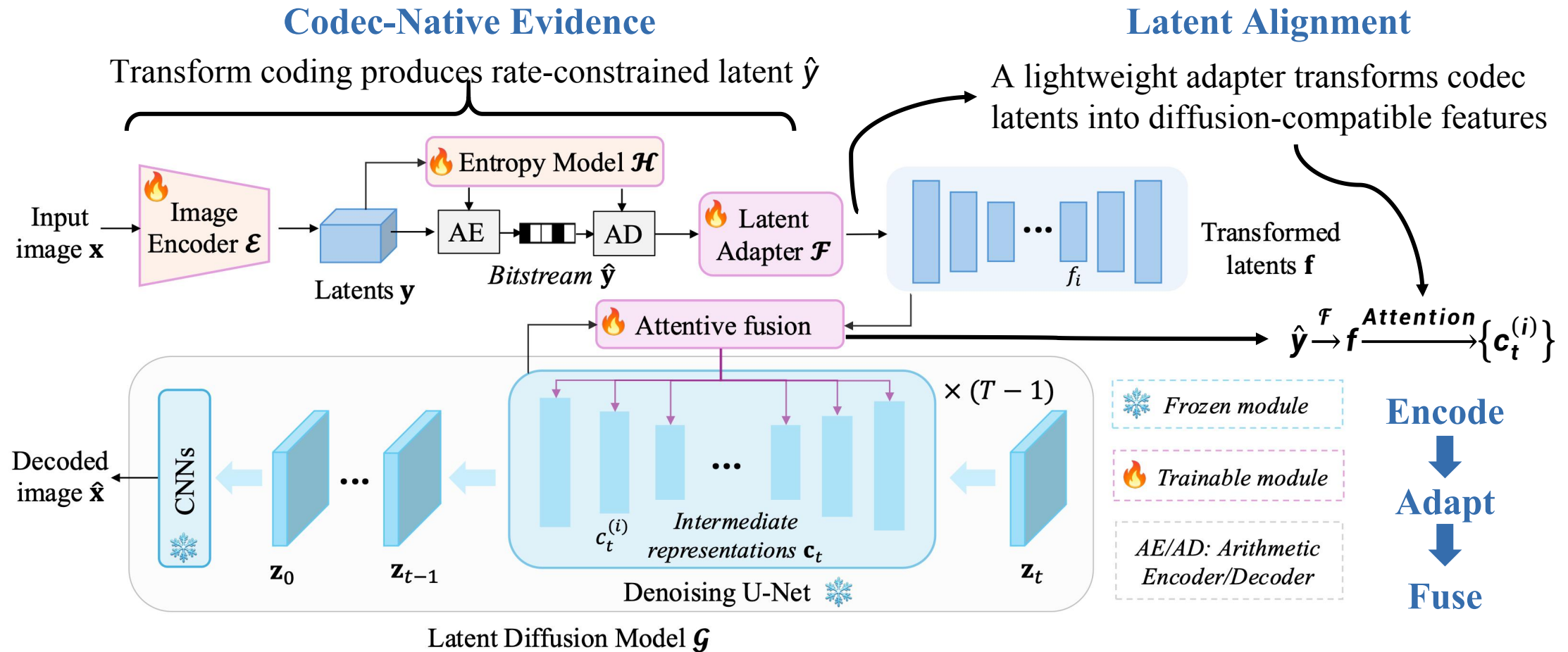
Theoretical insight

An ideal RD-optimal encoder supports the optimal distortion–perception tradeoff through changes at the decoder.

Ideal setting: fixed rate, MSE + Wasserstein-2 perception, no common randomness.

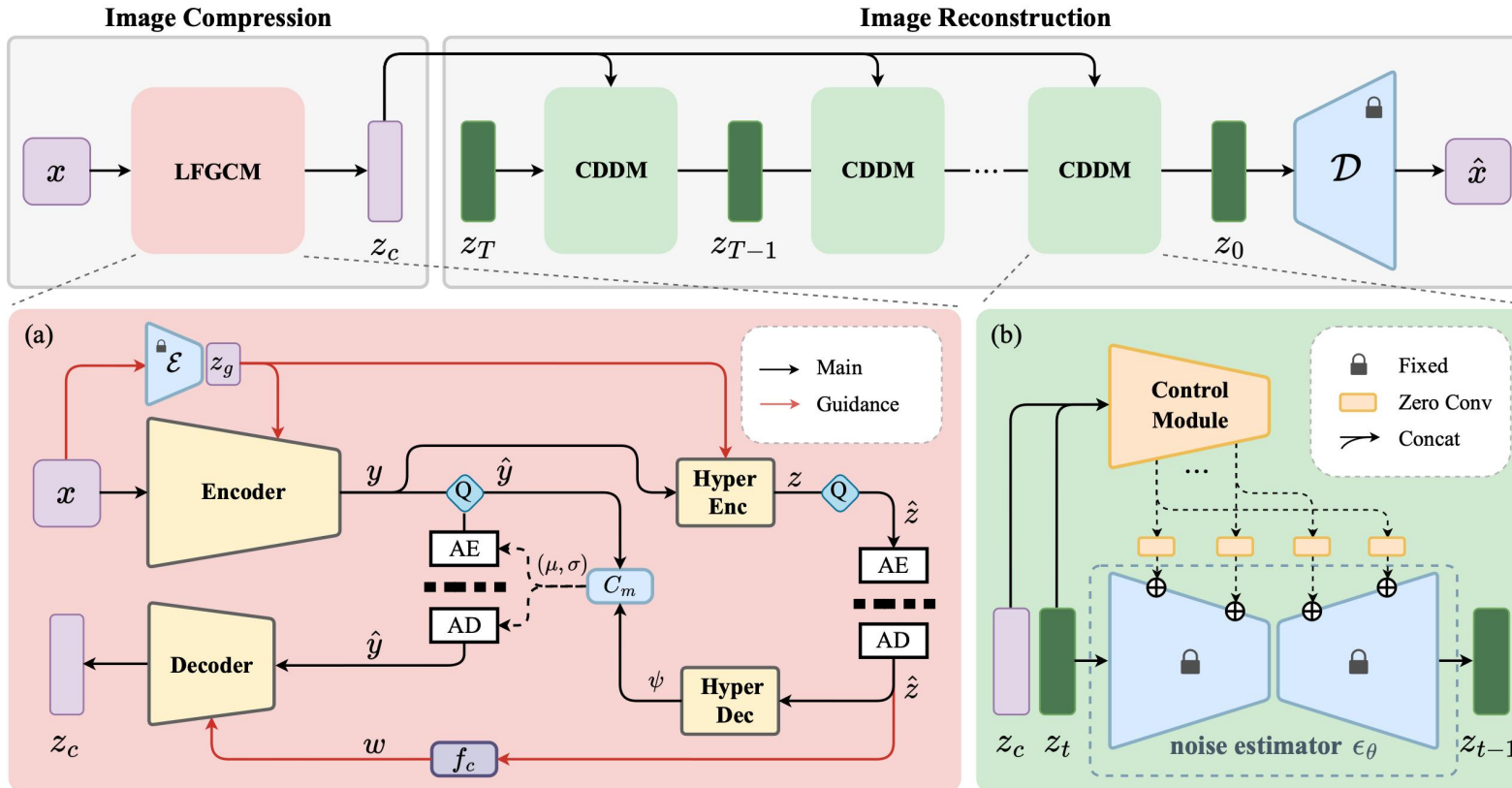
One coded representation can serve different reconstruction objectives.

- Injecting Codec Latents into a Frozen Diffusion Prior



Codec-Native Latents → Reinterpret the Latent → Enable Generative Decoding

Compact Information in a Form the Generator Can Use



Why align?

Match the feature space learned by the pretrained diffusion model.

What is learned?

A rate-constrained code that decodes into a generator-aligned condition.

DiffEIC uses this condition to guide iterative decoding.

Compact code and a decoded latent condition the generator can use.

Designing Effective Conditions

The condition should preserve source content in a form the generator can use.

Reuse generative features

Combine features from the diffusion model's pretrained VAE with features learned for compression.

Learn aligned conditions

Train the decoded condition with feature supervision, balanced against the code's bit cost.

These approaches can be combined.

Rate controls how much information is transmitted.

Alignment helps the generator use the decoded condition effectively.

Why encode explicit conditions?

Text, edges, poses and color maps expose selected source attributes, so we can inspect, guide or modify what the decoder generates.

Capability	What makes it possible	Practical benefit
Interpretable	Readable captions and spatial maps	Inspect what information is coded
Controllable	Cues specify selected source attributes	Guide content, layout or appearance
Editable	Modify cues before generating the image	Change attributes through the decoder
Scalable	Send layered cues progressively	Adapt transmitted evidence to bandwidth

Explicit attributes connect compression with control and editing

Layered representations also support scalable delivery through the same decoder.

How layered cues enable scalable reconstruction and editing

Chen et al.: extract and code cues, then reconstruct with SD 1.5 + T2I-Adapter

Text	BLIP-2 caption + Zstd	Subject and semantic content
Edges	PiDiNet + downsampling + VVC	Geometric structure
Color map	8×8 color map, 8-bit values	Coarse spatial appearance

Source



AI-generated image

Text



0.0012 bpp / 0.40 DISTS

Text + edges



0.0095 bpp / 0.33 DISTS

Text + edges + color



0.0154 bpp / 0.28 DISTS

The same cues support layered delivery and attribute editing.

These capabilities have a bit cost. Generated detail can still differ from the source.

- Different Content Types Demand Different Reconstruction Fidelity



Text-Rich Images

Character accuracy and layout matter.

Character identity · position · layout

Diffusion may alter characters and layouts

Natural Images

Plausible appearance is often sufficient.

Texture · fine details · local variations

Separate Text Coding

The exact-text requirement

Z: text content and position

Y: a code that already preserves Z exactly

$$H(Z | Y) = 0$$

Knowing Y leaves no uncertainty about Z

Separate Text Coding

The ideal total rate

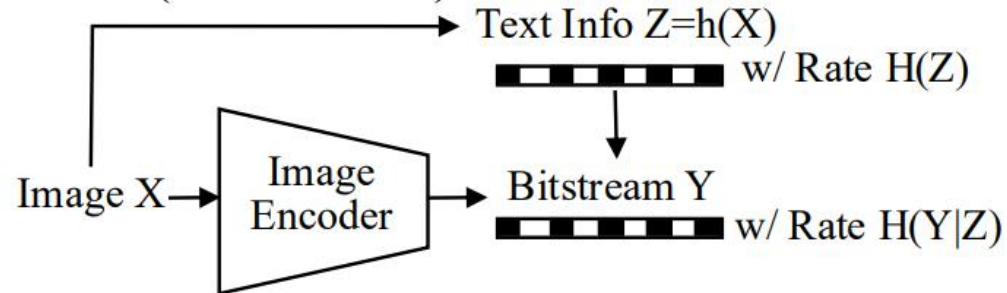
$$H(Z) + H(Y | Z) = H(Y, Z) = H(Y)$$

Text bits + remaining bits given text = original code entropy

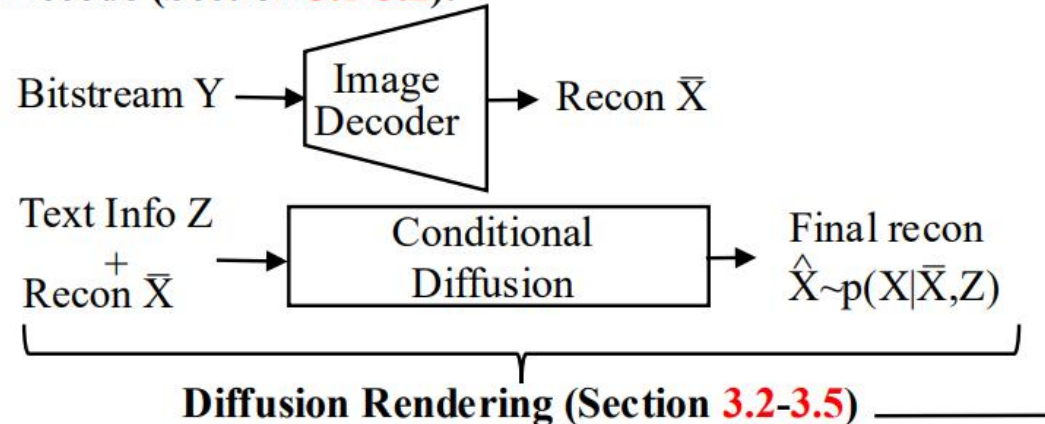
Same exact-text requirement and ideal conditional coding
Practical overhead may remain

PICD: Text-Conditioned Coding and Diffusion Rendering

Encode (Section 3.1-3.2):



Decode (Section 3.1-3.2):



Transmit text and position losslessly

Z carries the extracted text and position.

Encode the image conditioned on text

Use glyphs rendered from Z .

Reduce redundant image information.

Render with glyph and image guidance

Recreate the glyphs from decoded Z .

Combine them with the decoded image.

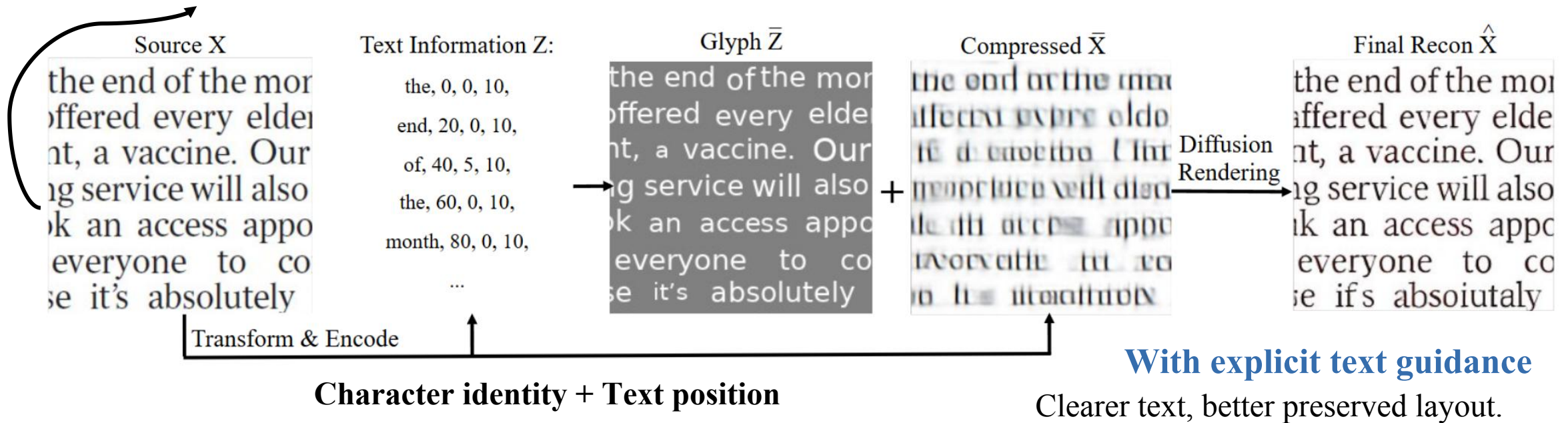
Lossless text transmission preserves the OCR payload. Rendered text can still contain errors.

- Text Conditions Improve Content Fidelity

Text must remain readable and correct.

Before refinement

Compression distorts characters.



Explicit key information improves content fidelity.
No extra ideal rate under the exact-text assumption.

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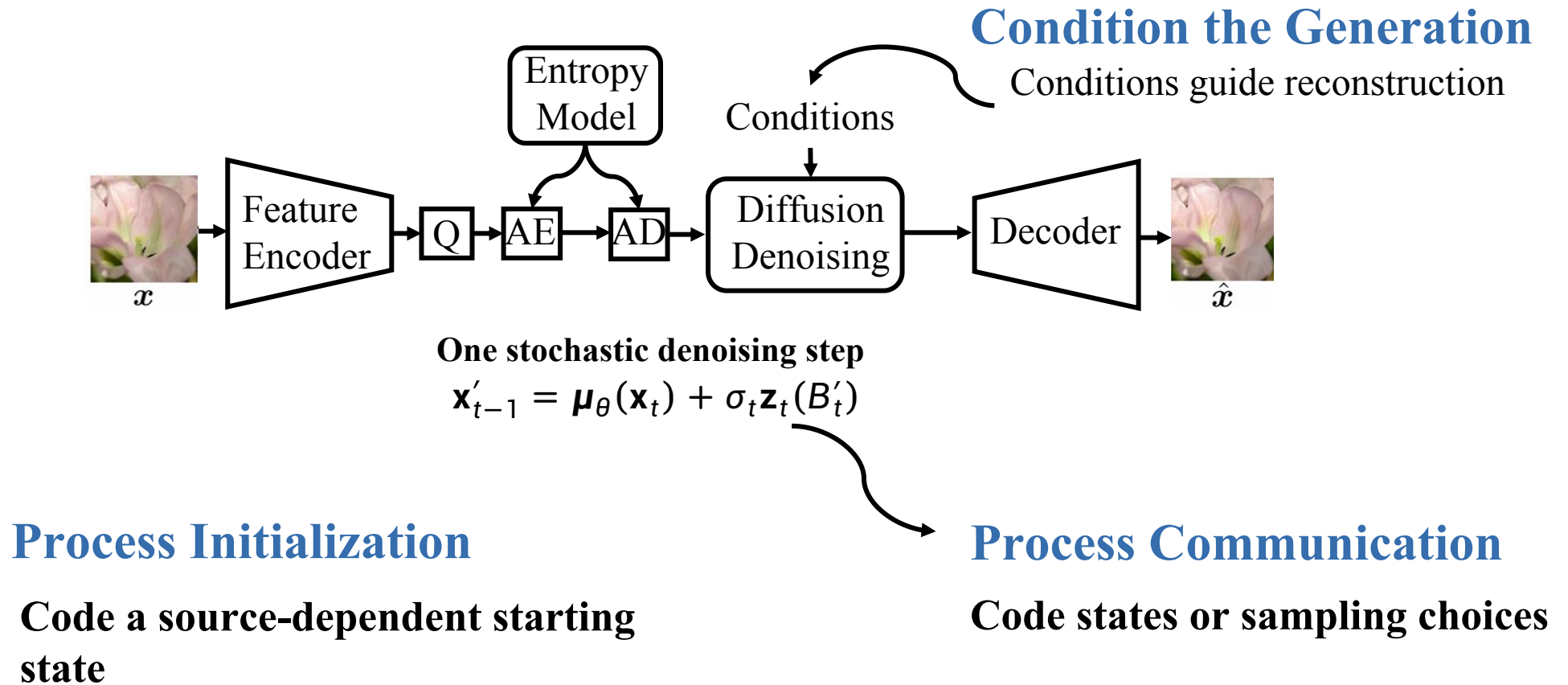
B.4 Task-Critical Evidence

C Generative Process

C.1 Process Initialization

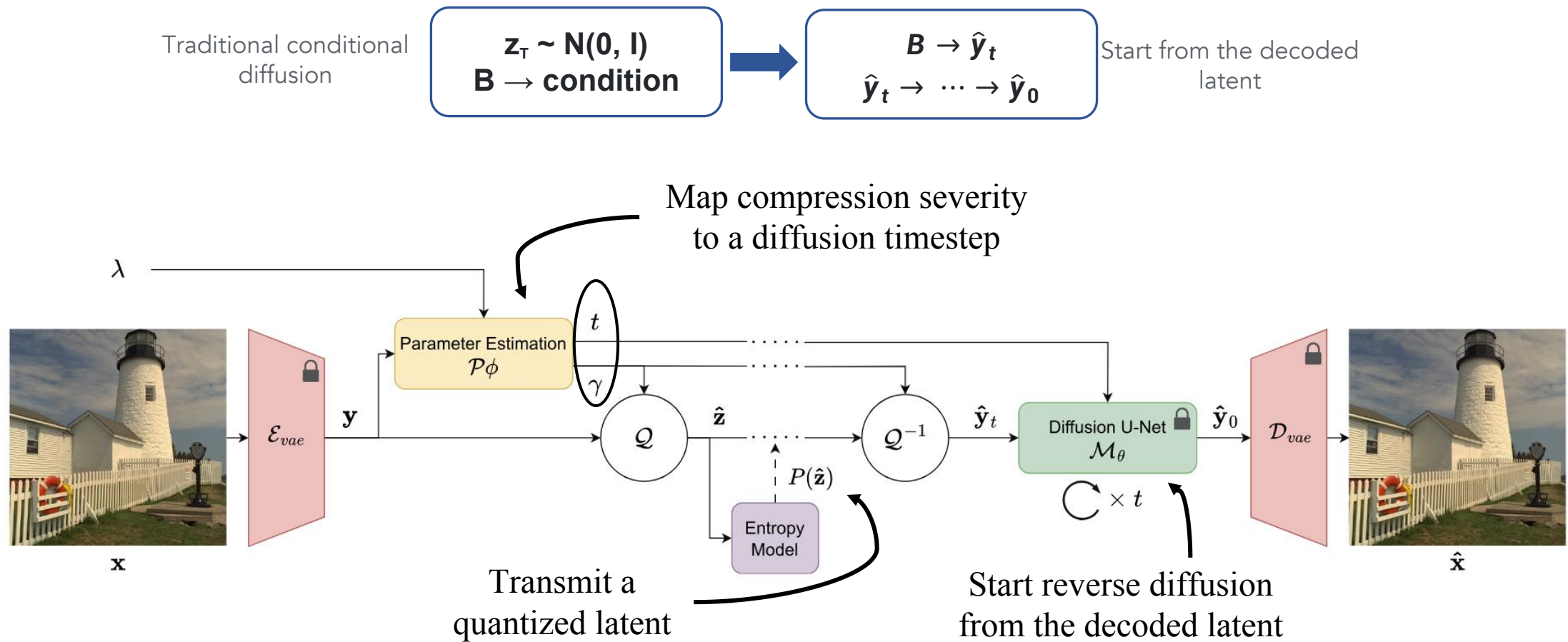
C.2 Process Communication

Beyond Conditioning: Coding the Generative Process

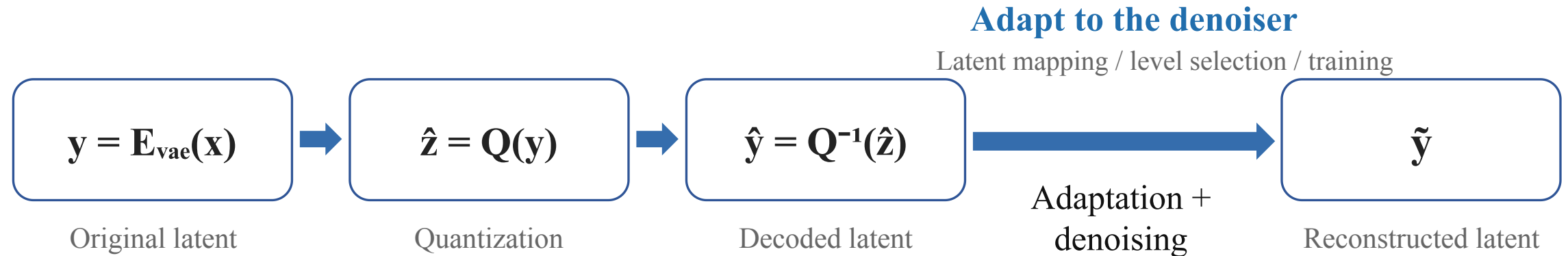


The bitstream moves from **conditioning the generative model** to becoming **part of the generative process**

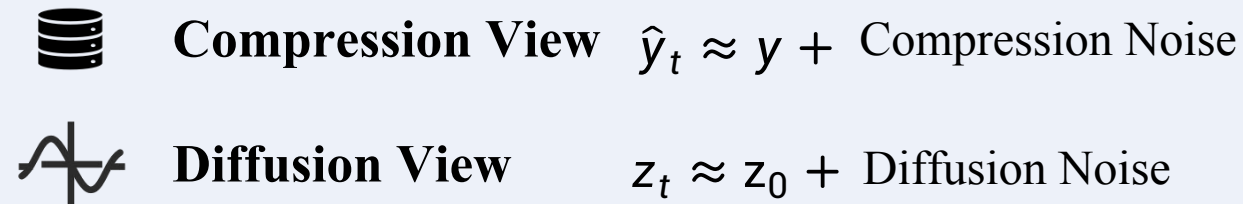
- The transmitted latent becomes the starting state of generation



Adapt the decoded latent into a diffusion starting state



Approximate Analogy



Choose the denoising setup

Optimize final reconstruction quality

Account for compression distortion

VAE error + additional quantization distortion

Compression distortion $\neq$ diffusion noise

More distortion does not imply more steps

Adaptation prepares the compressed latent for diffusion reconstruction.

Choose the starting timestep for reconstruction quality

This method predicts quantization parameters and timestep together

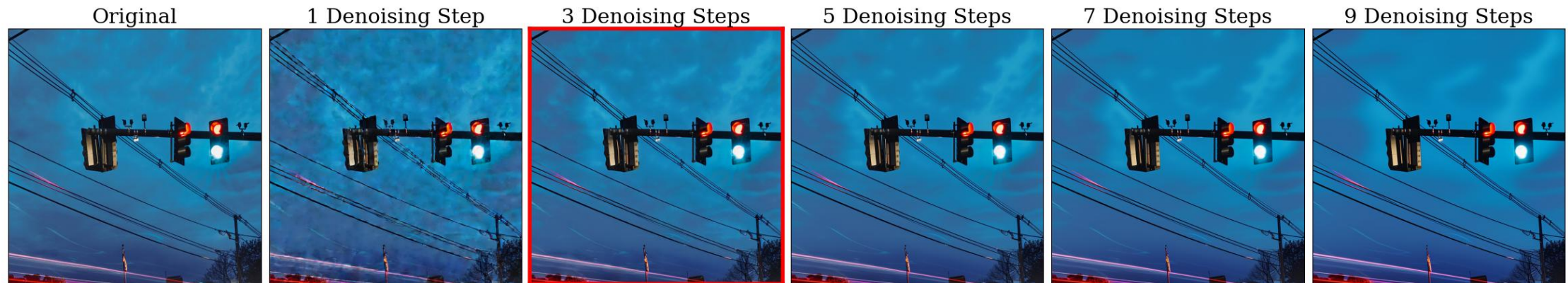
$$(\gamma, t) = \mathbf{P}\phi(\mathbf{y}, \lambda)$$

Source latent $\mathbf{y}$ + rate–distortion weight λ

$$\text{Train with } L = R + \lambda \|\mathbf{x} - \hat{\mathbf{x}}(\gamma, t)\|^2$$

Balance bitrate with squared reconstruction error

More denoising is not always better



Residual artifacts **Predicted setting**

Increasing smoothing

Learn the starting timestep to balance rate and reconstruction quality.

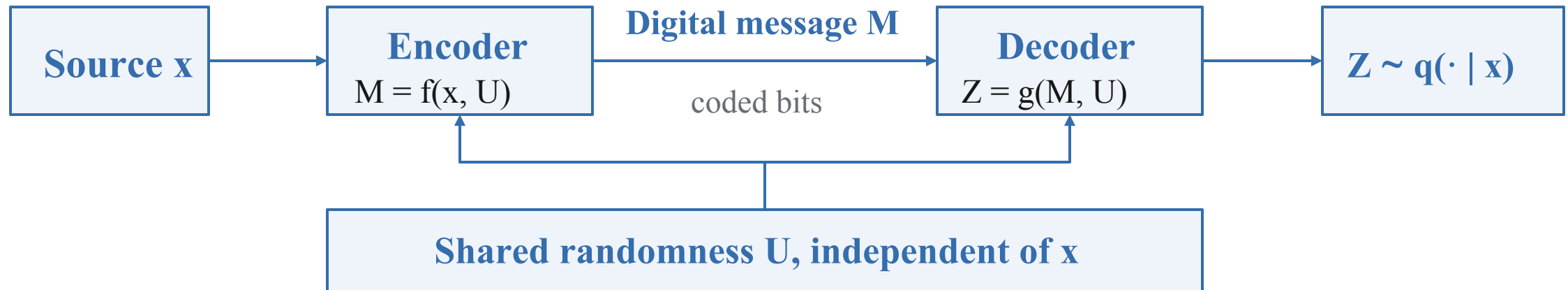
Reverse channel coding: generating the required sample

Channel coding: transmit bits through a noisy channel.

Reverse channel coding: use bits to simulate a random channel.

Sender knows x

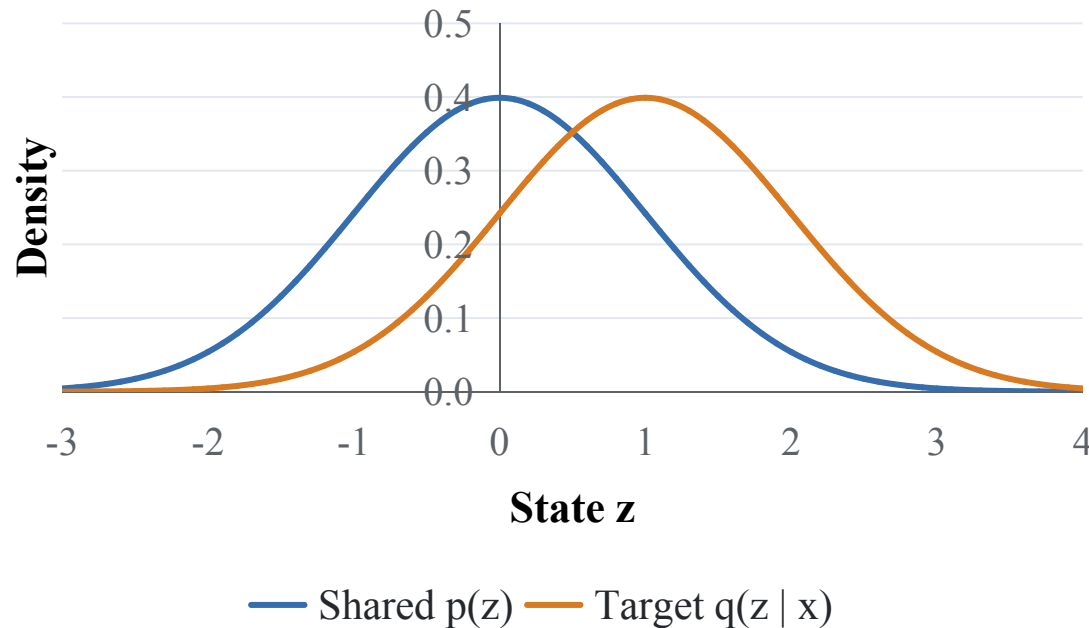
Receiver generates Z



$q(z | x)$ describes the required distribution of states for source x .

RCC reproduces the required distribution, not one fixed value.

Shared Prior and the Information Cost of a Sample



$q(z | x)$: target for this source

$p(z)$: reference shared by both sides

$$C(x) = \text{KL}(q(\cdot | x) \parallel p)$$

KL divergence measures
target–reference mismatch.

Low information cost does not imply low computation.

A well-matched shared prior reduces the communication needed.

DiffC: Encoding and Decoding the States

$$\mathbf{z}_t = \alpha_t \mathbf{x} + \sigma_t \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

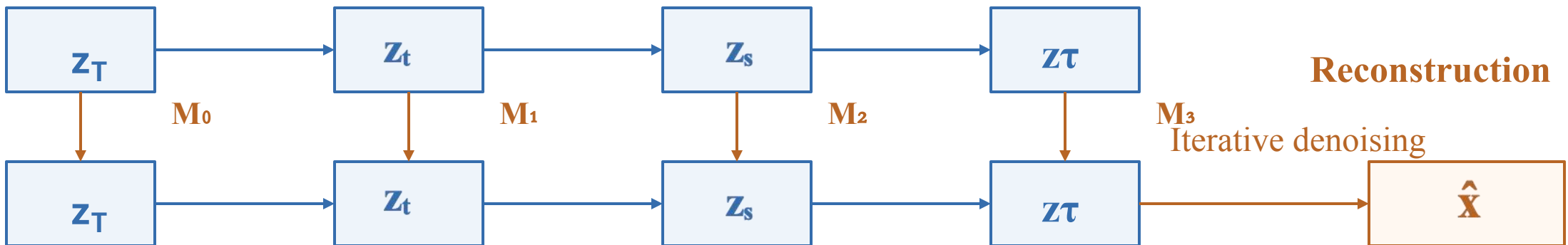
Noise level controls retained source information

Target: $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$

Shared model: $p_{\theta}(\mathbf{z}_s | \mathbf{z}_t)$

Encoder: source image defines target q

Both sides: denoiser defines reference p_{θ}



RCC messages let the decoder reproduce the states
Rates estimated using a KL-based coding upper bound

Chosen stopping level $\mathbf{z}_{\tau}$
Then continue denoising

Shared randomness + RCC message $\rightarrow$ the same next state

Practical DiffC: From Rate Estimates to Actual Coding

Reuse a pretrained diffusion model without codec-specific training.



Efficient RCC

Blockwise search, timestep scheduling and GPU acceleration

Remaining cost

Encoder searches samples; both sides run the denoiser.

The generative prior becomes part of the coding mechanism itself.

Generative Process Coding: Key Takeaways

DiffC / Practical DiffC / DDCM / Few-step RCC

01 The generative prior becomes part of the codec

Shared generative knowledge supports both sample communication and image reconstruction.

02 Bits steer generation toward the source

The bitstream specifies choices along the generative process.
For RCC, target–prior mismatch gives the information-cost benchmark.

03 Efficient communication makes the approach practical

Pretrained models provide a reusable foundation. Efficient sample search and fewer model evaluations reduce the computational cost.

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SUBTOPICS

A Where the Bitstream Acts

A.1 Model Level

A.2 Direction Level

A.3 State or Trajectory Level

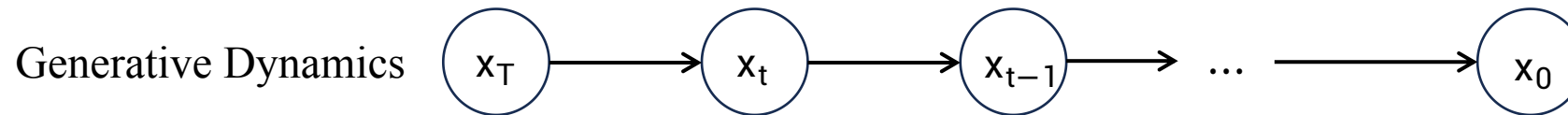
B Prior Adaptation to Compression

B.1 Full Adaptation

B.2 Lightweight Adaptation

B.3 Frozen Prior

• Three Intervention Points in Generative Decoding

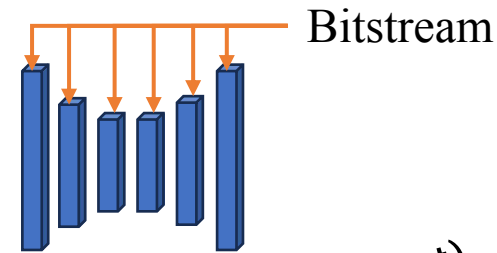


S_t : sampler step $x_{t-1} = S_t(x_t, u_\theta(x_t, t; B) + \Delta u_B(x_t, t), \xi_t)$

Model-Level Conditioning

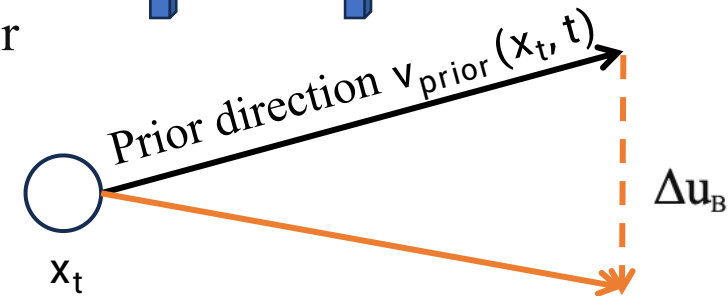
B enters the model prediction $u_\theta(x_t, t; B)$.

u_θ : score, noise prediction or velocity, as used by the sampler



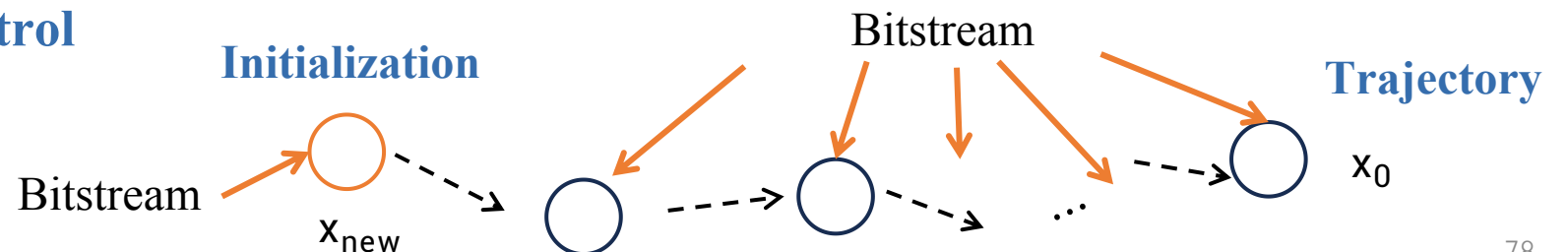
Direction-Level Control

B defines an explicit correction $\Delta u_B(x_t, t)$.



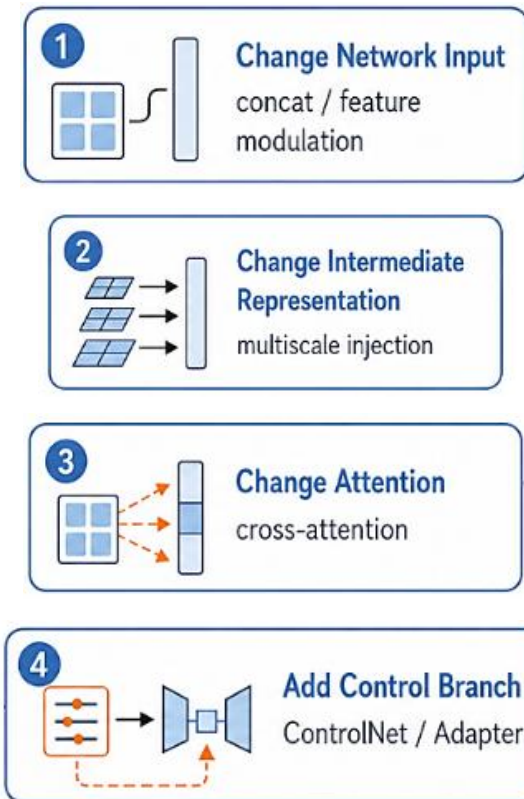
State- / Trajectory-Level Control

Bits specify the starting state or sampling choices.



Model-level: make coded evidence available inside the network

Choose an interface that matches the condition and the denoiser architecture.



Input or feature modulation

Expose the condition directly through input features or feature modulation.

Multiscale injection

Provide condition features at several resolutions in the denoiser.

Cross-attention

Let generative features query relevant information in the condition.

Auxiliary control branch

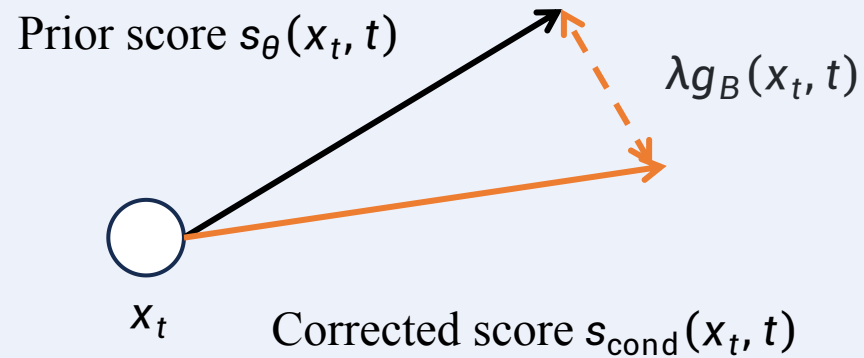
Process the condition in a separate path and inject its control features.

Match the interface to the condition. These mechanisms can be combined.

For example, a control branch can inject features at multiple scales.

- Direction-Level Control Mechanisms

Posterior / Score Guidance



$$s_{\text{cond}}(x_t, t) = s_\theta(x_t, t) + \lambda g_B(x_t, t)$$

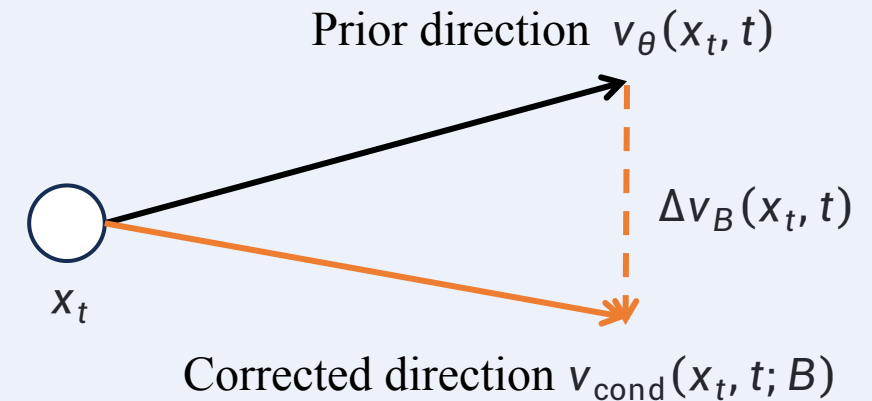
$s_\theta(x_t, t)$: score predicted by the generative prior

$g_B(x_t, t)$: condition-dependent guidance direction

λ : guidance strength $s_{\text{cond}}(x_t, t)$: conditioned score

Bayesian basis: prior score plus evidence guidance

Velocity / Update Correction



$$v_{\text{cond}}(x_t, t; B) = v_\theta(x_t, t) + \Delta v_B(x_t, t)$$

$v_\theta(x_t, t)$: prior velocity field

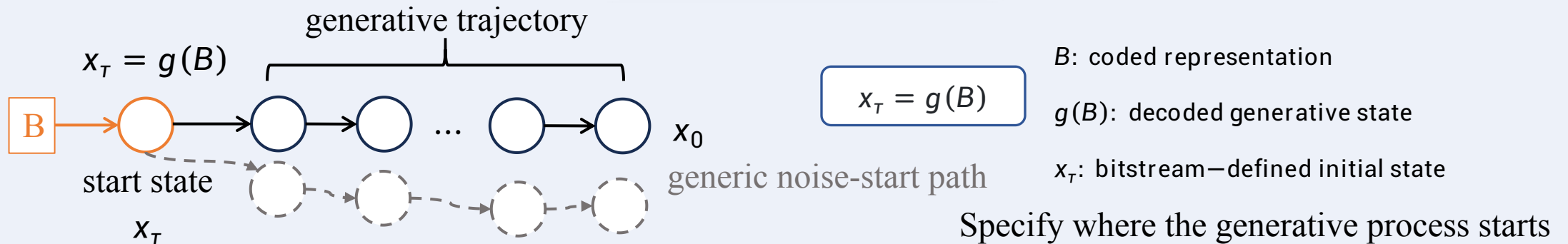
$\Delta v_B(x_t, t)$: condition-dependent correction

$v_{\text{cond}}(x_t, t; B)$: corrected generative direction

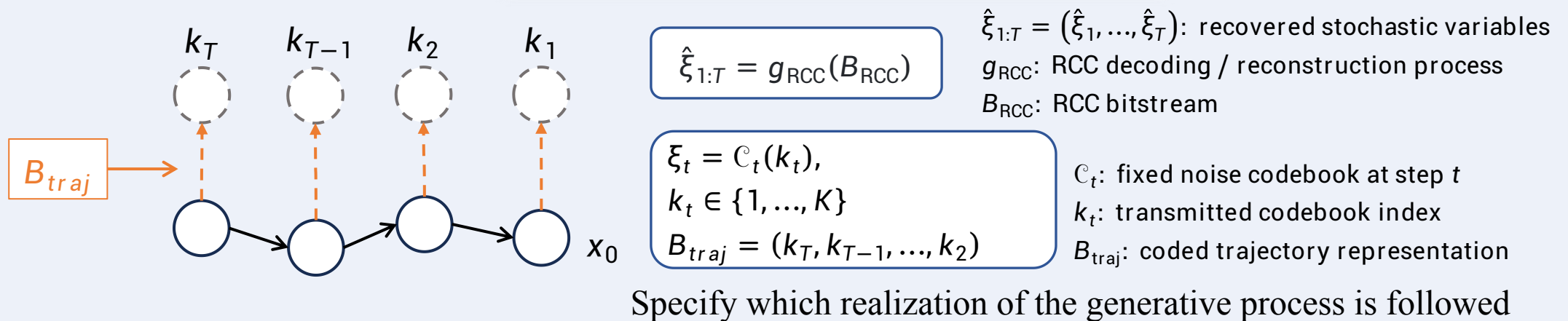
Correct the update to steer the reconstruction path

- State- / Trajectory-Level Control Mechanisms

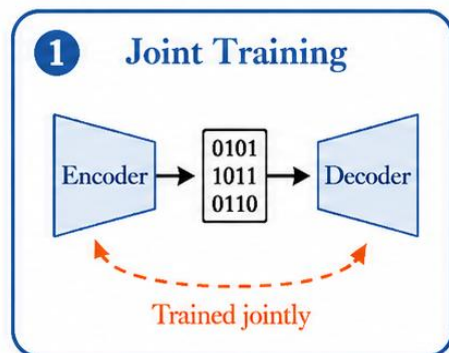
State Initialization



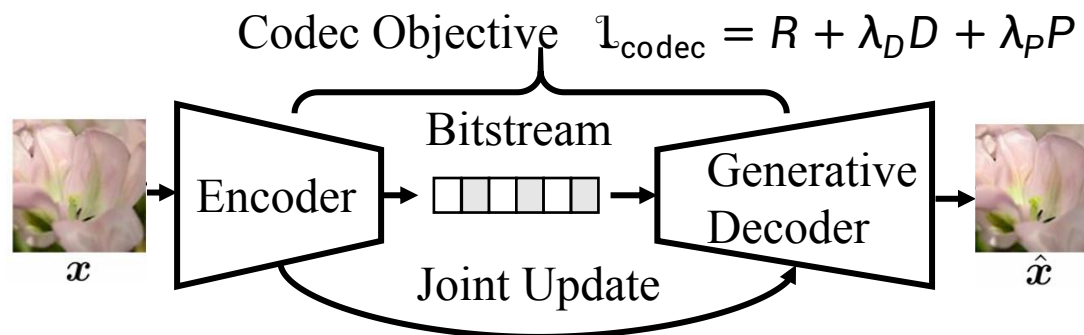
Trajectory / Randomness Coding



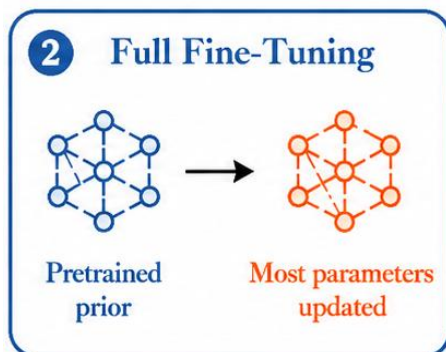
• Strong Adaptation: Learning a Compression-Specific Prior



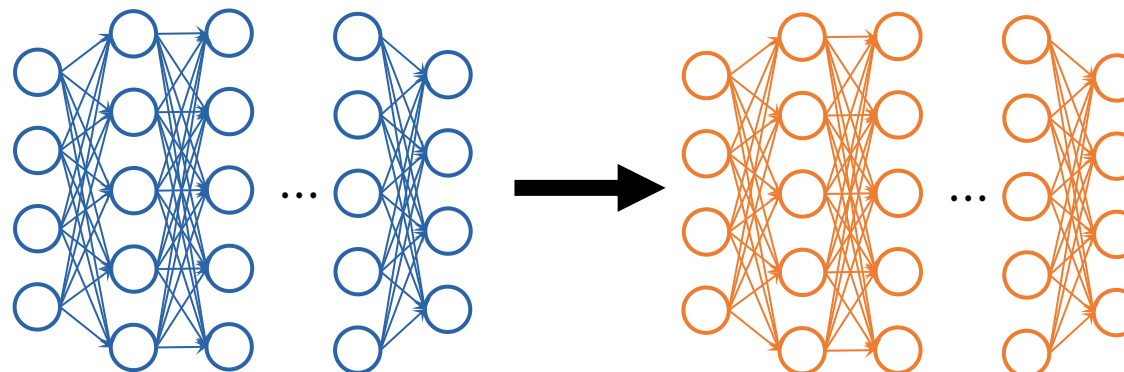
Learn the generative prior together with the codec



$$(\phi^*, \theta^*) = \arg \min_{\phi, \theta} \mathcal{L}_{\text{codec}}$$



Start from a pretrained prior, then update most parameters



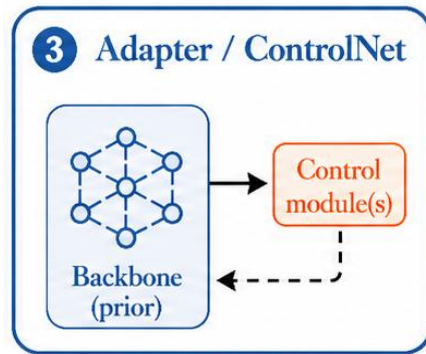
$$\theta^* = \theta_0 + \Delta\theta$$

More specialization, higher training cost

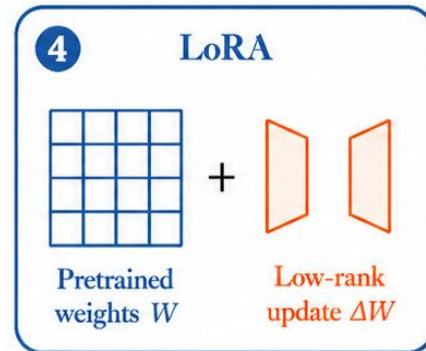
- **Initialization**
Task-specific initialization
- **Updated Parameters**
Codec + generative model
- **Role of Prior**
Learned for compression
Example: CDC¹
- **Initialization**
Pretrained prior
- **Updated Parameters**
Most / all backbone parameters
- **Role of Prior**
Initialization + transferable knowledge

¹ R. Yang and S. Mandt, “Lossy Image Compression with Conditional Diffusion Models,” Advances in Neural Information Processing Systems 36, pp. 64971–64995, 2023.

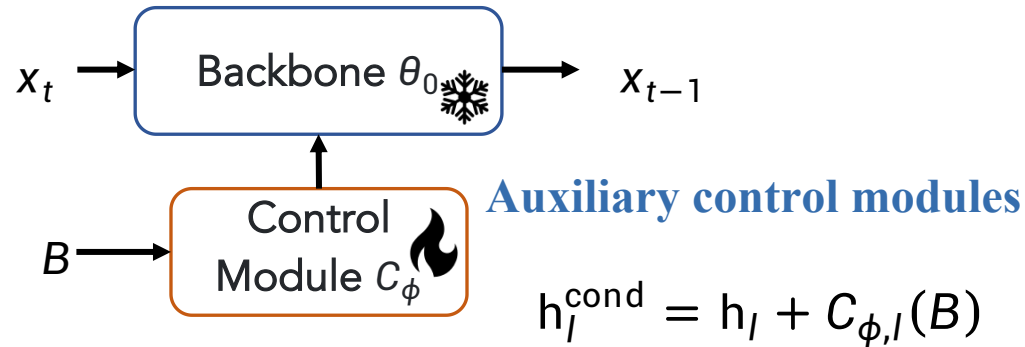
• Prior-Preserving Adaptation: Reuse More, Learn Less



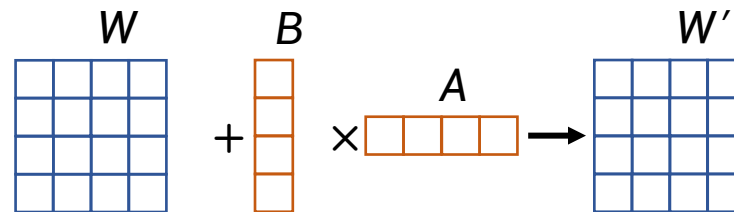
Preserve the backbone and learn auxiliary control modules



Inject low-rank updates into selected weights



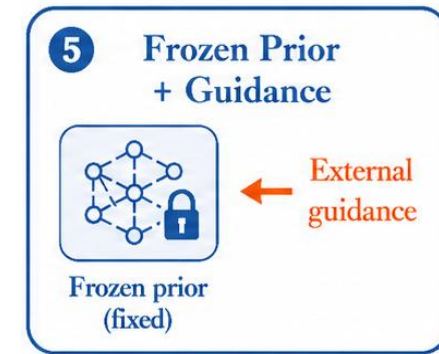
$C_{\phi,l}(B)$: control feature produced by the auxiliary module
 ϕ : trainable parameters of the control module



$$W' = W + \Delta W, \quad \Delta W = BA$$

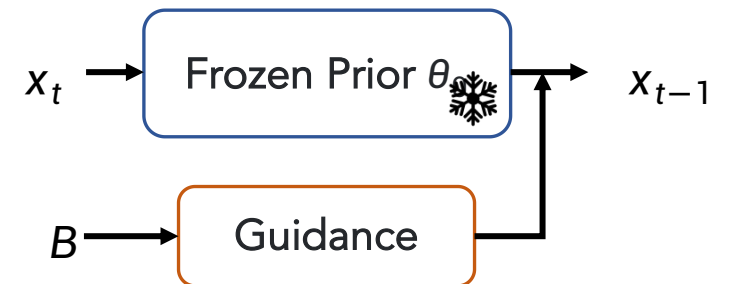
$$\text{rank}(\Delta W) = r \ll d$$

W : frozen pretrained weight matrix
 A, B : trainable low-rank matrices
 r : low-rank adaptation dimension



Keep the prior fixed and control generation externally

$v_{\theta_0}(x_t, t)$:
 generation direction from the frozen prior
 $\Delta v_B(x_t, t)$:
 condition-dependent external guidance



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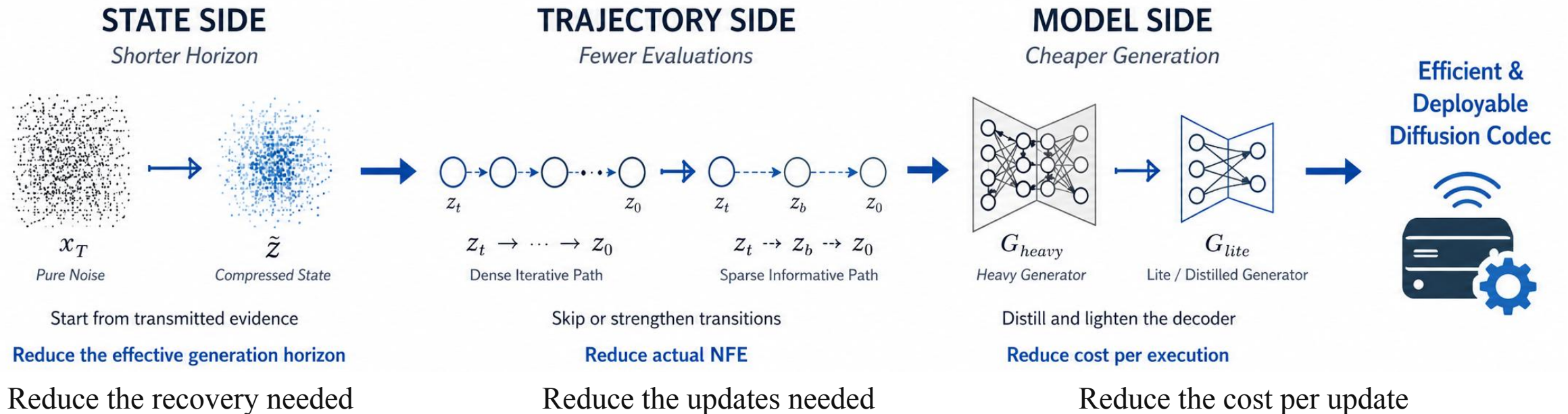
A Trajectory Side

B State Side

C Model Side

Efficient Decoding: Fewer Evaluations, Lower Cost

$$\text{Cost} \approx \text{model evaluations} \times \text{cost per evaluation}$$



Both routes can reduce steps; they act at different points.

Trajectory Side: Fewer Sampling Updates

Fast samplers

More accurate numerical updates with fewer evaluations.

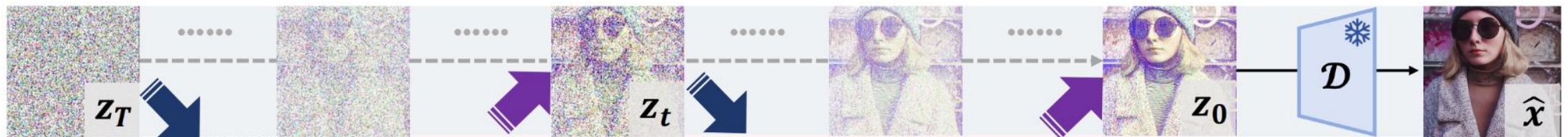
Distillation

Train a few-step student from a multi-step teacher.

Consistency training

Learn matching clean predictions across noise levels.

Example: DiffCR uses consistency-based refinement

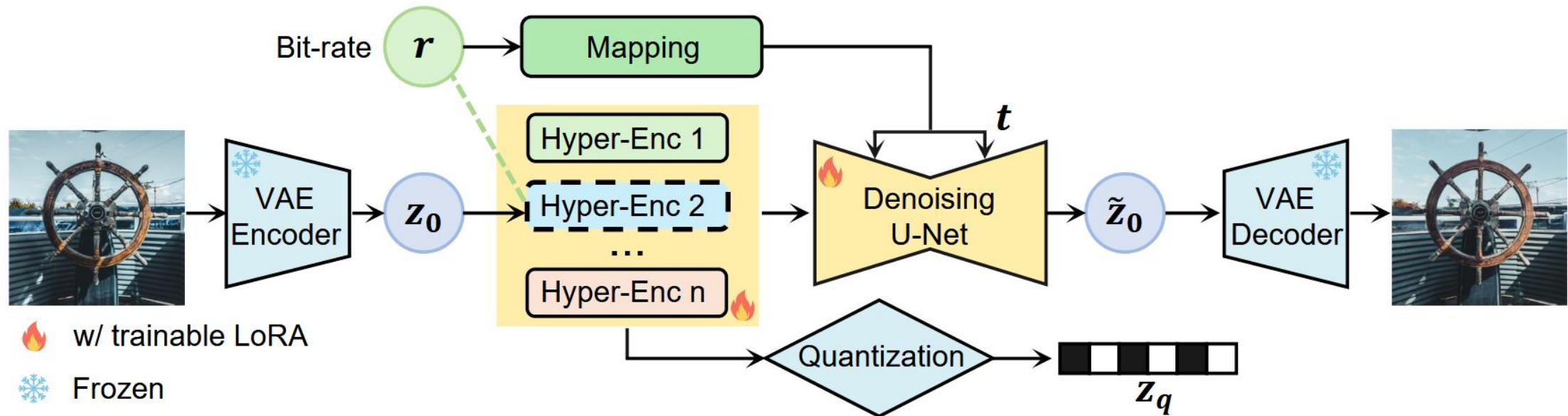


Gray: many small updates Blue / purple: two sampling updates

Consistency + compressed conditions support two-step reconstruction.

State Side: Reduce the Recovery Needed

An informative starting state reduces what remains to be recovered.



Preserve source structure

Align the compressed latent with the source image's diffusion VAE latent.

Match the recovery level

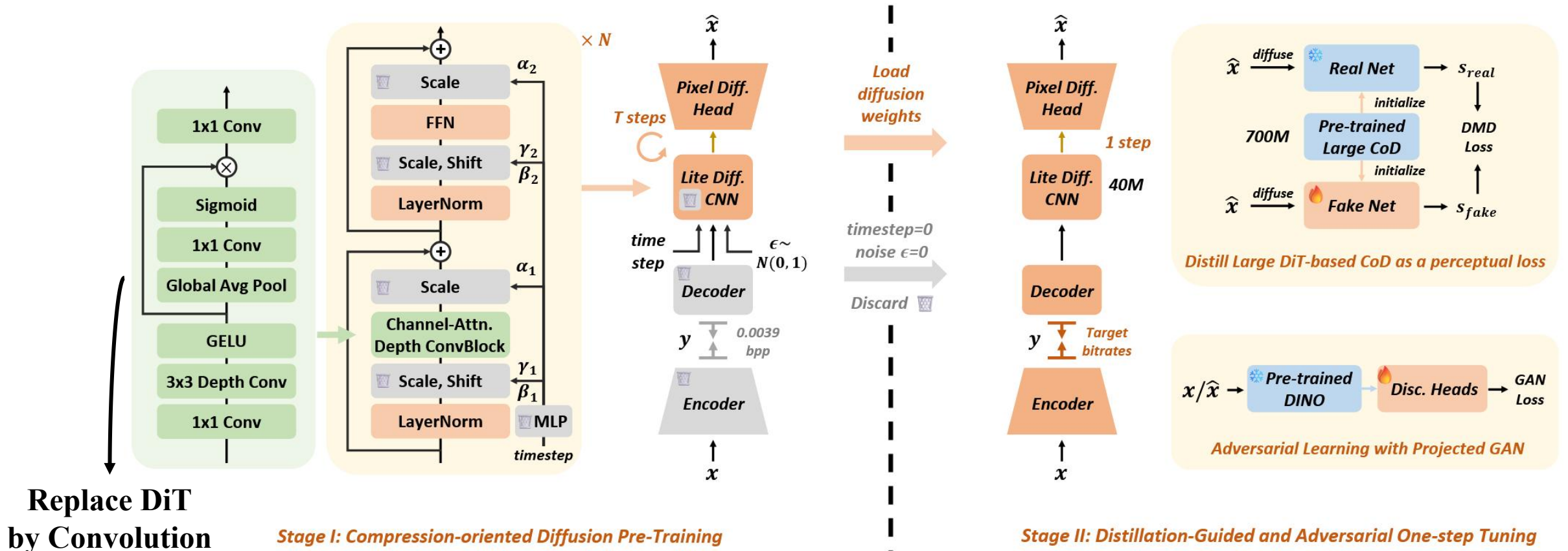
Map bitrate to a pseudo timestep.
Fine-tune for one-step recovery.

OSCAR: one denoising pass with competitive reconstruction quality.

Model Side: Lightweight Real-Time Generative Decoding

Decoding cost depends on both the number of evaluations and the model used.

$$\underbrace{T \times C_{\text{heavy}}}_{\text{Multi-Step Diffusion}} \rightarrow \underbrace{1 \times C_{\text{heavy}}}_{\text{One-Step Codec}} \rightarrow \underbrace{1 \times C_{\text{lite}}}_{\text{CoD-Lite}}$$



Compression-oriented training supports a lightweight convolutional decoder.

Selected Results: Realism and Perceptual Rate Efficiency

Ultra-low-rate realism

PerCo¹ · 2024

≈0.003 bpp

Realistic reconstructions
at ultra-low rates

A generative prior can supply
plausible visual detail.

Perceptual rate efficiency

GIC² · 2025

−69%

LPIPS BD-rate on Kodak
relative to TCM-VBR

Fewer bits at matched LPIPS.

Realistic appearance and perceptual rate efficiency.

¹ M. Careil, M. J. Muckley, J. Verbeek, and S. Lathuilière, “Towards image compression with perfect realism at ultra-low bitrates,” International Conference on Learning Representations, 2024.

² J. Chang, “Generative Image Coding with Diffusion Prior,” IEEE International Conference on Multimedia and Expo, pp. 1–6, 2025.

Selected Results: One-Step and Real-Time Decoding

One-step reconstruction

OneDC³ · 2025

1 step

0.34 s decoding

$1024 \times 1024 \cdot A100$

One-step generative reconstruction.

Real-time decoding

CoD-Lite⁴ · 2026

≈42 fps

$1920 \times 1024 \cdot A100$

~0.03 bpp · 24 ms/frame

Lightweight decoding at real-time speed.

Fewer evaluations and lighter models improve decoding speed.

Different experimental settings; these results are not a direct ranking.

³ N. Xue, Z. Jia, J. Li, B. Li, Y. Zhang, and Y. Lu, “One-Step Diffusion-Based Image Compression with Semantic Distillation,” Advances in Neural Information Processing Systems 38, pp. 41680–41716, 2025.

⁴ Z. Jia et al., “CoD-Lite: Real-Time Diffusion-Based Generative Image Compression,” arXiv preprint arXiv:2604.12525, 2026.

Diffusion Methods in Generative Image and Video Coding

OUTLINE

1 Unified Framework

2 Image Coding

2.1 What Is Transmitted

2.2 Control Mechanisms

2.3 Efficient Decoding

3 Video Coding

3.1 Frame-Level Predictive

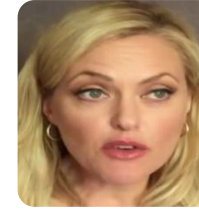
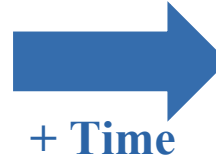
3.2 Video-Native Diffusion

From Images to Videos: Temporal Redundancy and Generation Priors



Image Generative Coding

Structure · Semantics · Perception
Recover single image from limited bits



Video Diffusion Coding

+ Temporal dimension
New opportunities + new constraints

New Opportunity · Temporal Redundancy



Shared background



Stable appearance



Continuous motion

→ Transmit new content and prediction errors

New Challenge · Temporal Consistency



Texture



Identity



Geometry

Motion

→ A single bad frame is seen as flicker

Q1 Who models temporal compensation?

Explicit Prediction

vs

Generative Prior

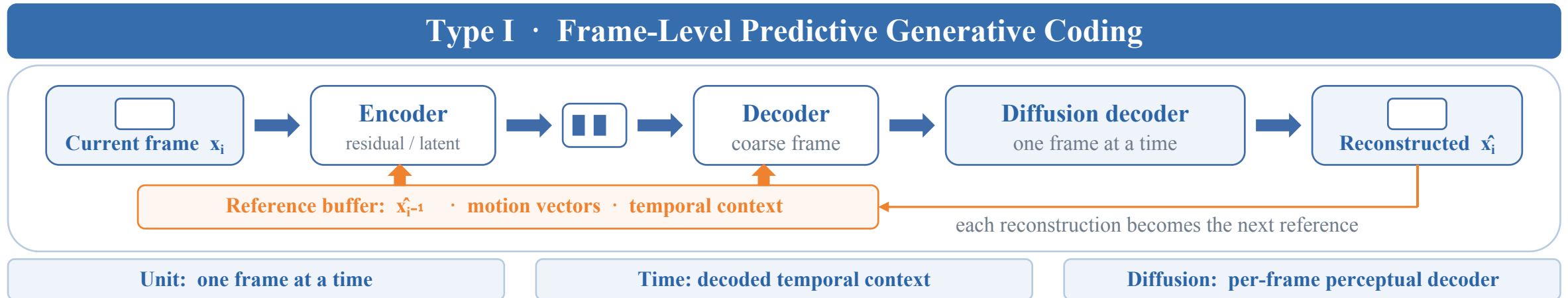
Q2 What is the generation unit?

Per-Frame

vs

Joint Spatiotemporal

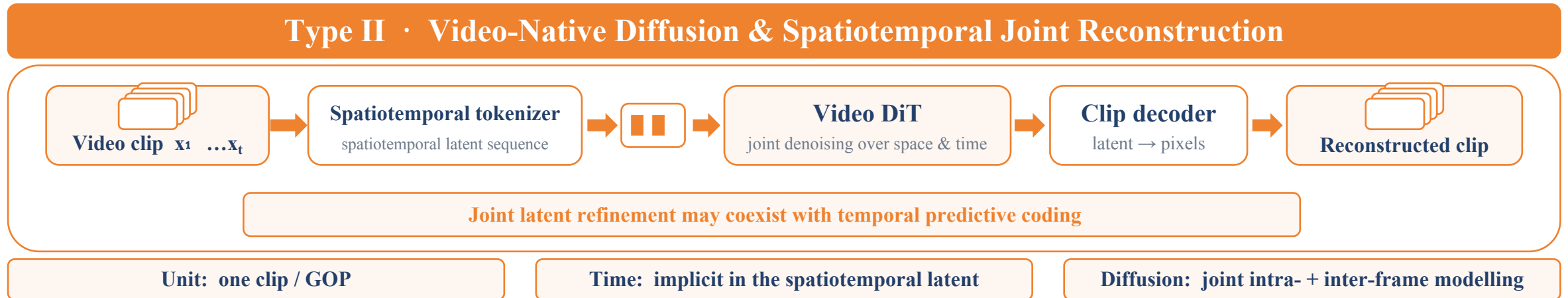
Video Coding: Frame-Level Generation



Generation unit: one frame

Decoded temporal context links successive reconstructions.

Video Coding: Joint Spatiotemporal Generation

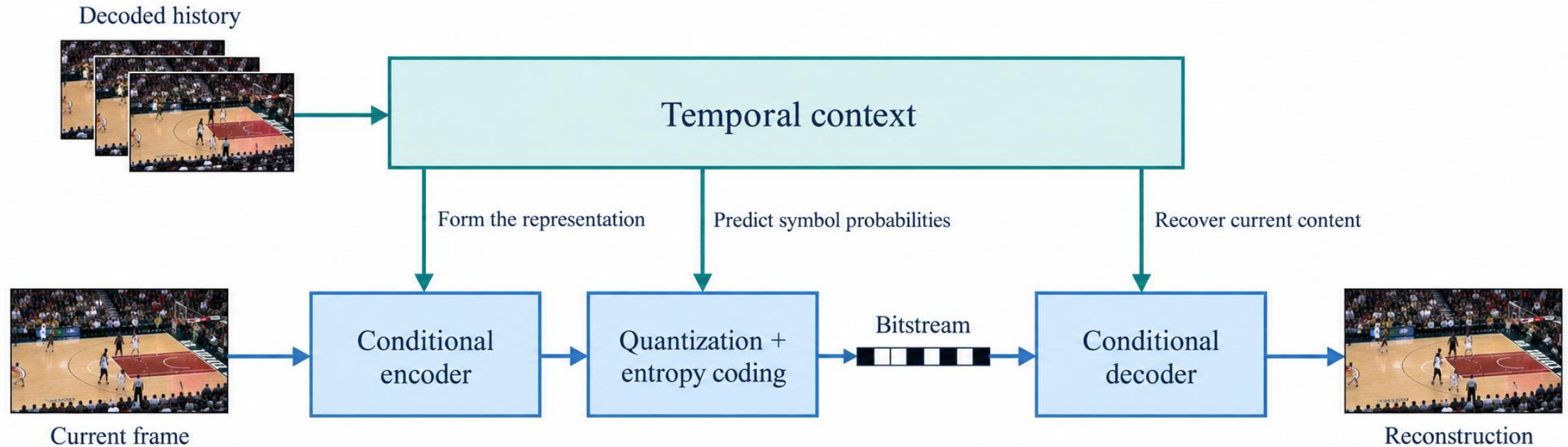


Joint generation supports temporal consistency

Multi-frame buffering adds delay.

Shorter windows and causal streaming can reduce it.

DCVC-Based Conditional Coding



Explicit motion: DCVC-DC / FM

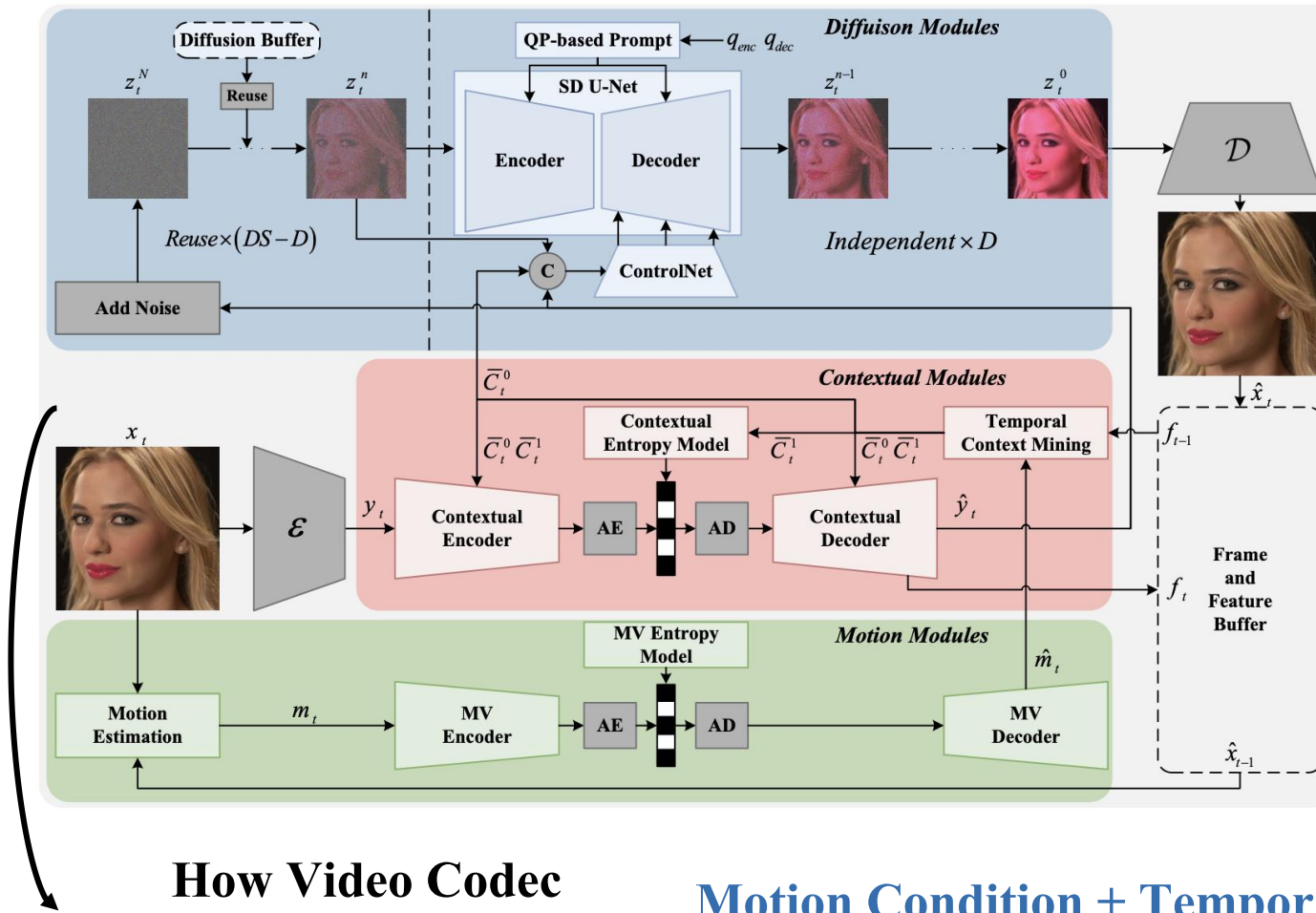
Estimate, transmit and compensate motion

Implicit temporal modeling: DCVC-RT

Use history without a separate motion stream

Temporal context conditions representation, entropy modeling and reconstruction.

DiffVC: Conditional Coding with a Multi-Step Generative Decoder



Coded content + history

Constrain reconstruction with the current latent and motion-compensated context.

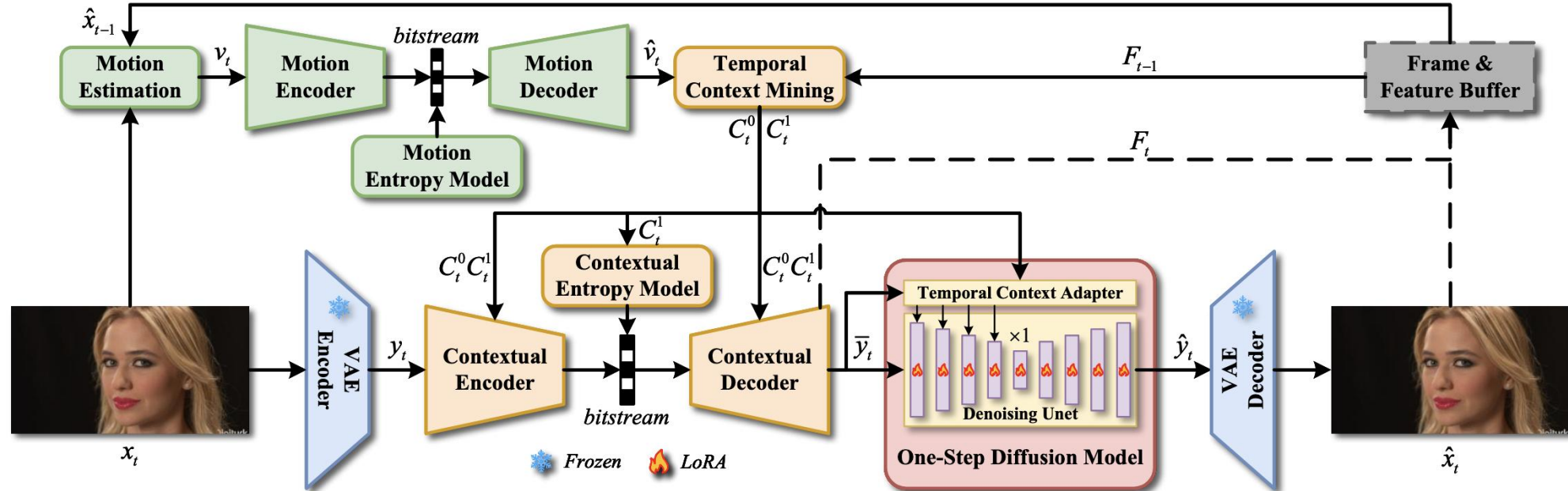
Remaining bottleneck

Multiple denoising calls for every reconstructed frame.

How Video Codec
Conditions Diffusion?

Motion Condition + Temporal Context + Current-Frame Latent

DiffVC-OSD: One-Step Recovery and Joint Adaptation



Keep the informative starting point

Decoded latent + Temporal Context Adapter (TCA)

Joint adaptation

Joint rate–distortion–perception optimization
(rate, MSE and perceptual losses)

Fewer sampling steps make joint codec–generator adaptation more practical.

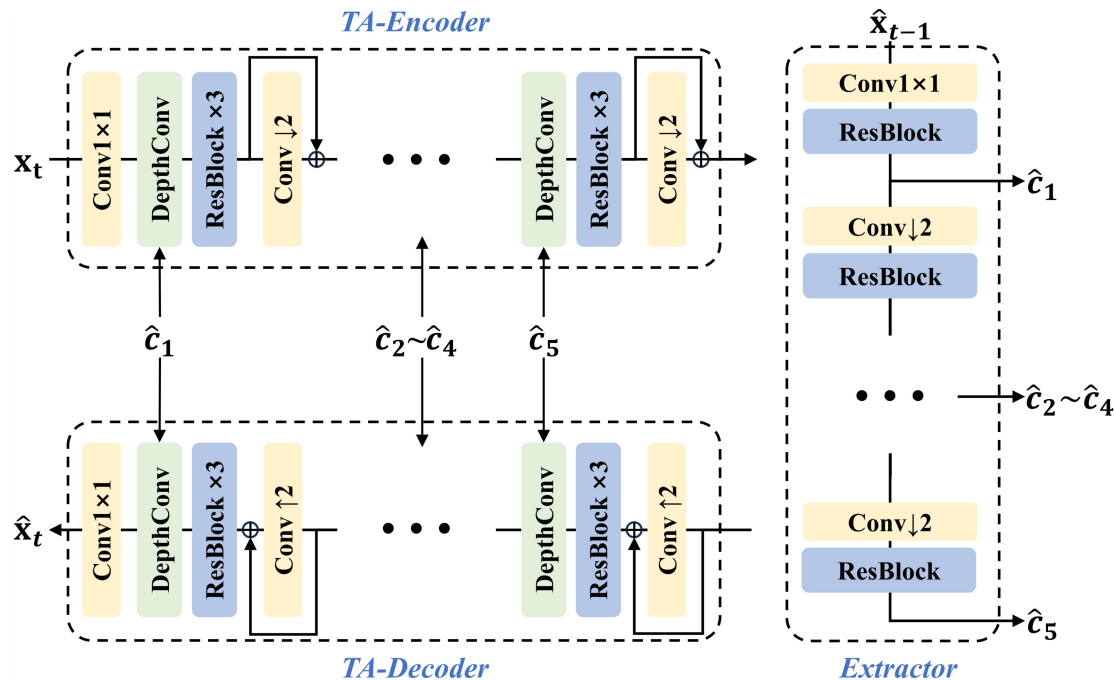
Type I. Frame-Level Paradigm

Temporal Adaptation for Frame-Level Generation

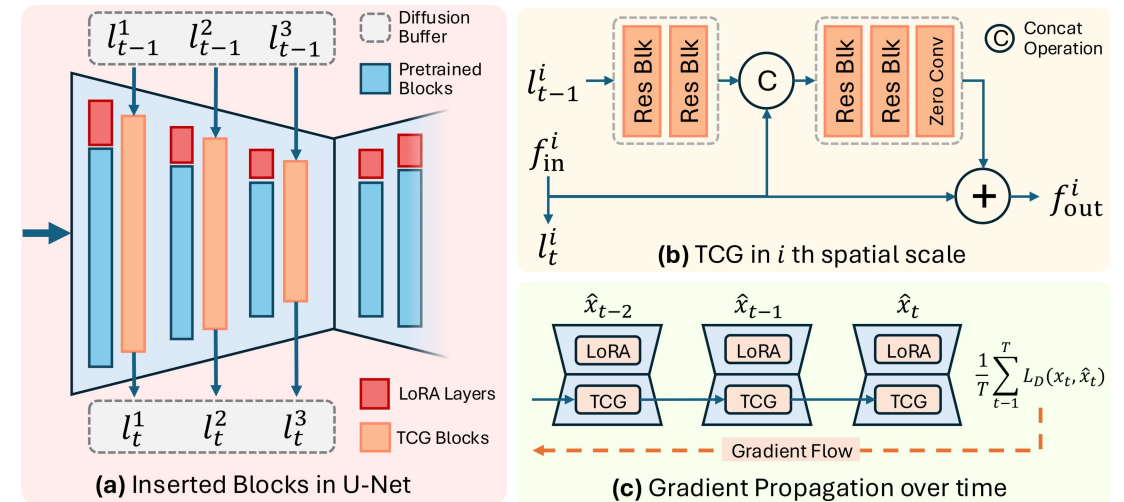
Good individual frames do not guarantee stable video.

Adapt representation and reconstruction

Stabilize generated appearance



YODA: reference features guide both the encoder and decoder.



S²VC: previous generative features help appearance persist across frames.

Two Roles of Temporal History

History for coding



History for generation

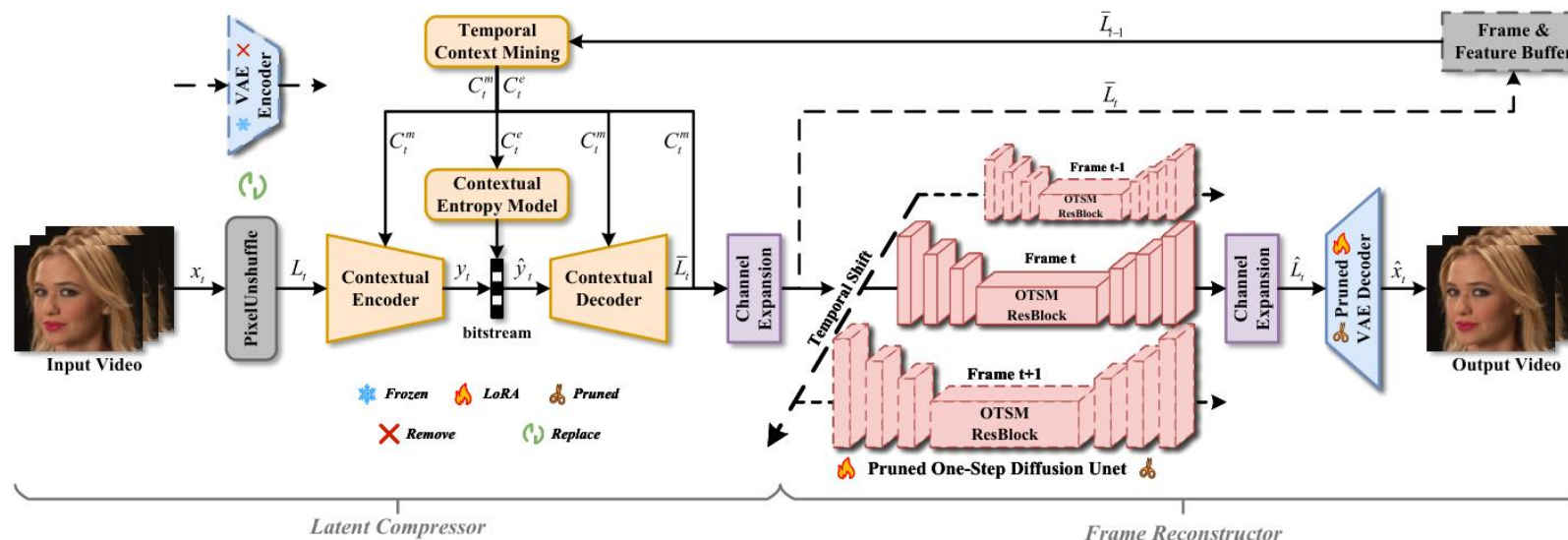


A predictable wall can still have uncertain brick texture.

Temporal history helps reduce coded redundancy and stabilize generated detail.

DiffVC-RT: Accelerating Frame-Level Generation

Even one-step generation needs acceleration for real-time video.



Reduce model cost

Prune the U-Net and VAE decoder.

Reuse temporal features to stabilize detail.

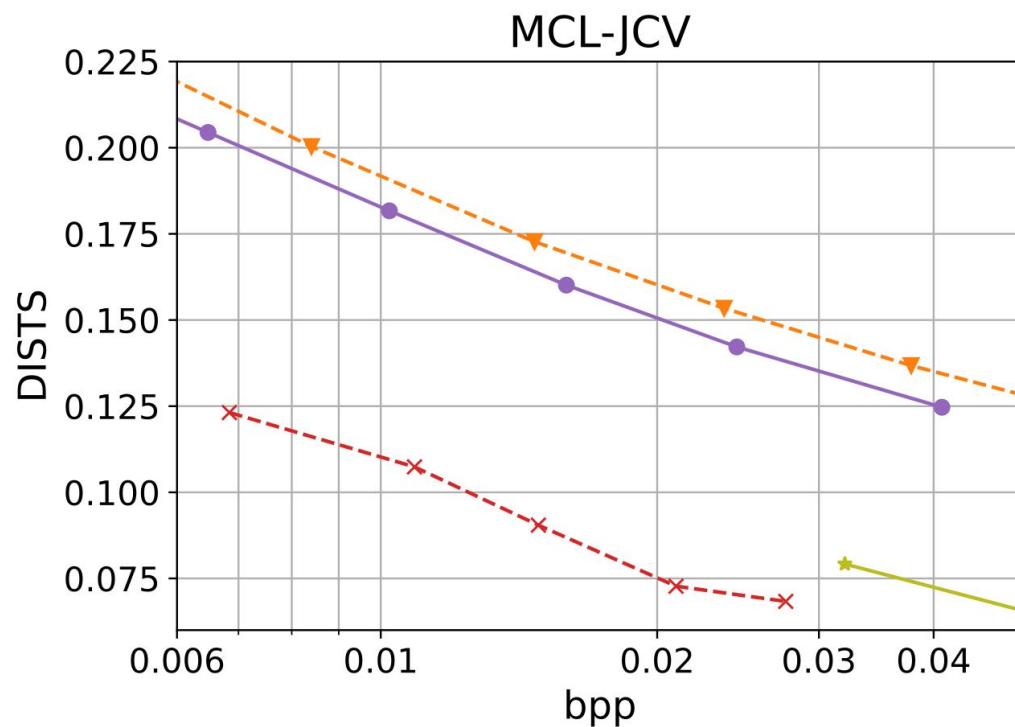
Overlap and batch reconstruction

Latent decoding need not wait for pixels.

Batch prepared latents for reconstruction.

≈ 30 fps at 720p on H800 · batch size 8

Frame-Level Generation: Perceptual Results



VTM

DCVC-RT

DiffVC

S²VC

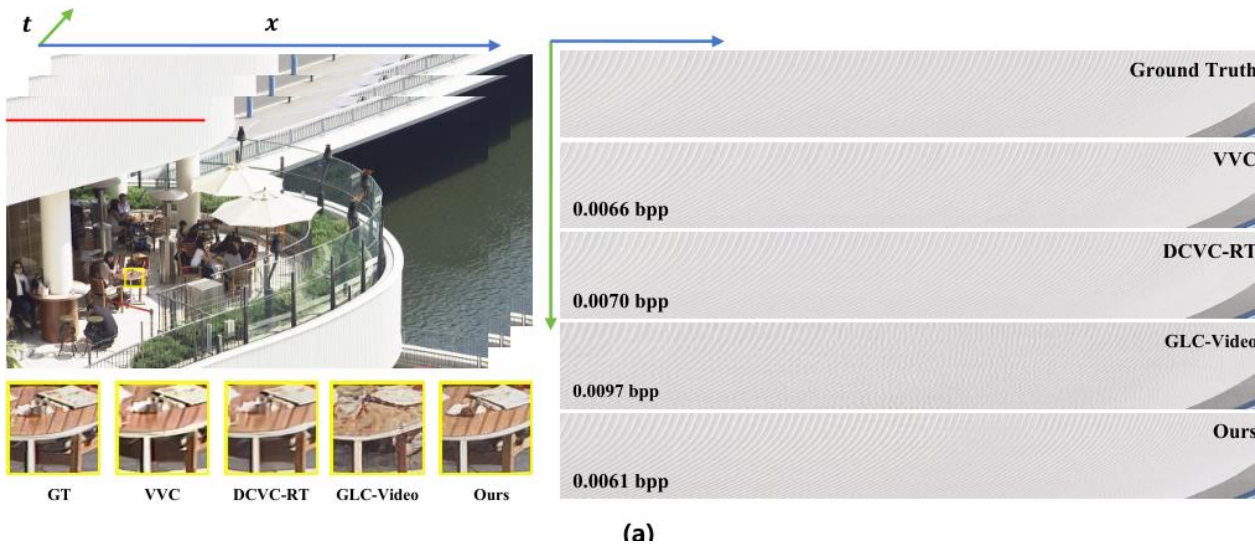
Perceptual gains

S²VC achieves lower DISTs than VTM and DCVC-RT at comparable low bitrates.

Temporal guidance

Feature propagation and training across frames help stabilize detail.

From Image Priors to Video Foundation Models



Why frame-wise generation can flicker

Missing detail admits different plausible textures. Independent noise can add variation.

A prior learned from videos

A video prior learns appearance and motion across frames, helping keep details consistent.

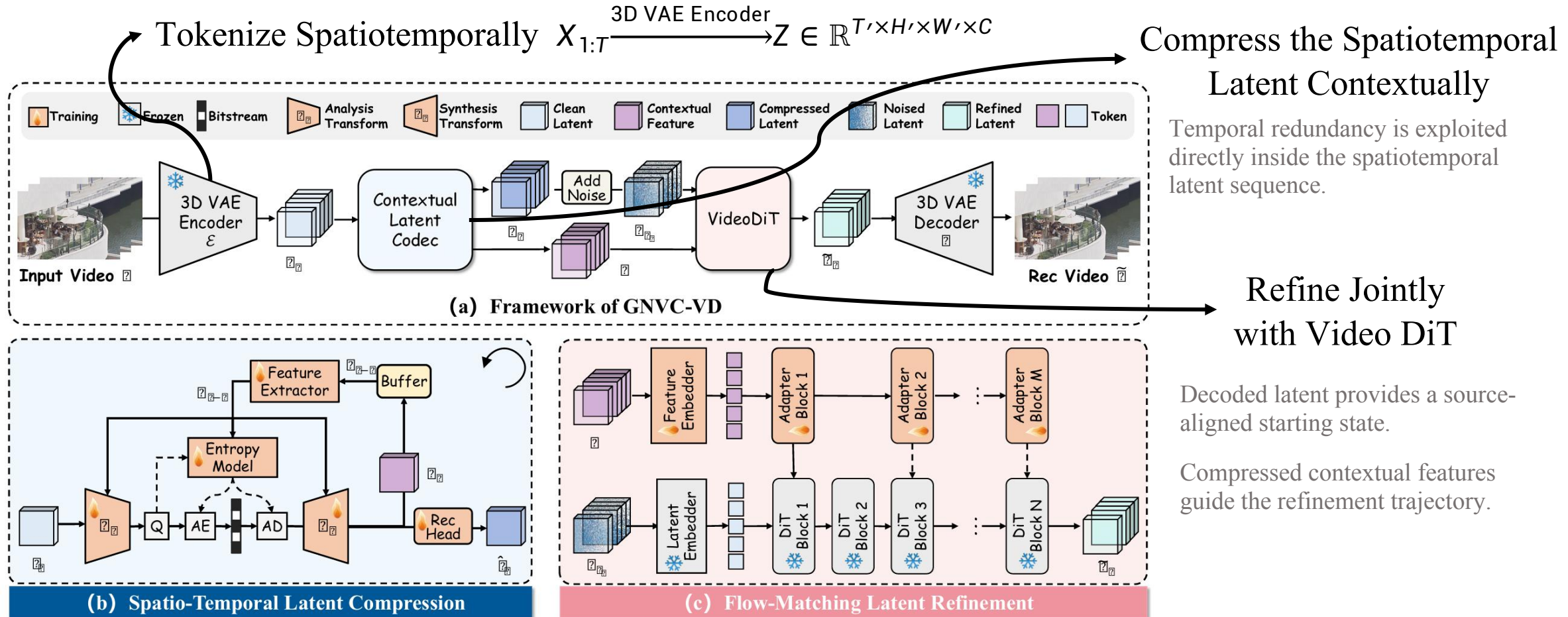
Spatiotemporal representation

The 3D VAE forms spatiotemporal latents. The video diffusion model processes them jointly.

Type II. Video-Native Diffusion Paradigm

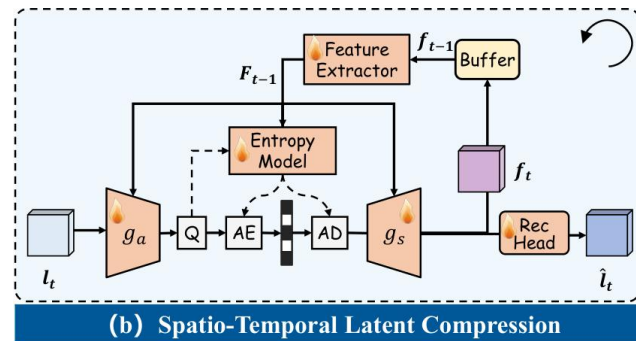
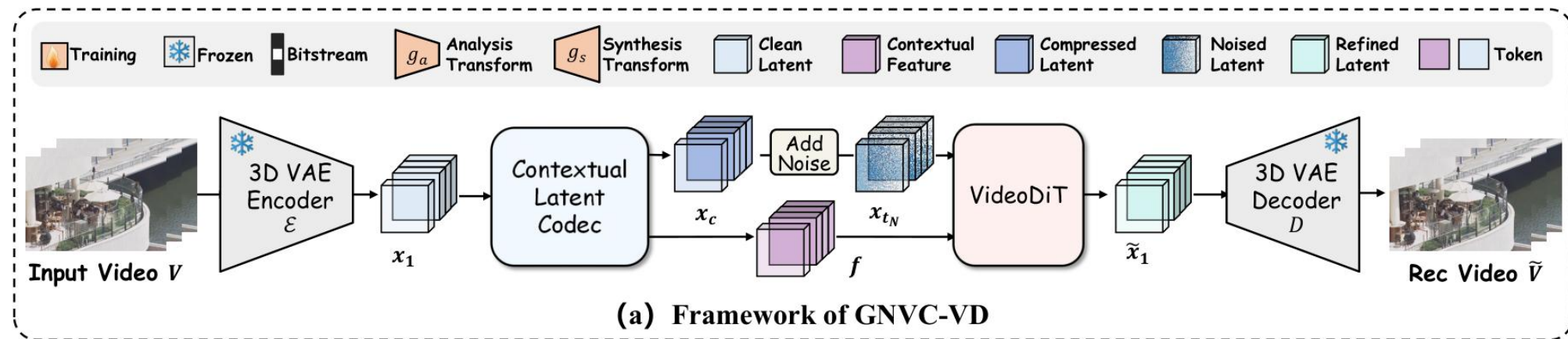
• Spatiotemporal Tokenization for Joint Video-DiT Refinement

How Does a Video-Native Codec Preserve Time throughout the Pipeline ?



Type II. Video-Native Diffusion Paradigm

- GNVC: Temporal Coding of the 3D VAE Latents



I/P coding of the latent sequence

I latents: independent coding.

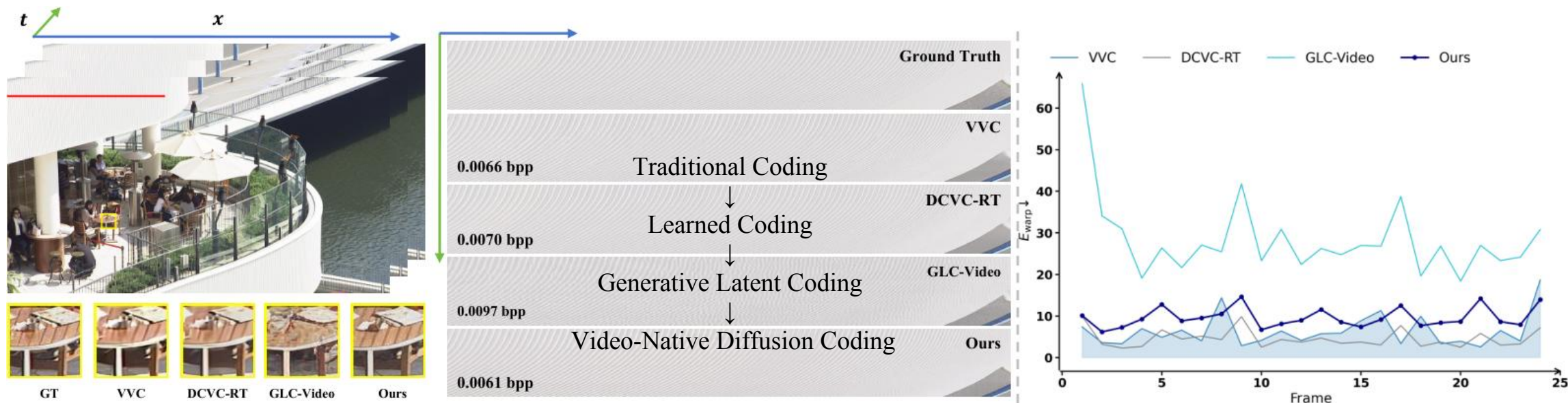
P latents: coding with decoded temporal context.

Joint video reconstruction

VideoDiT refines the recovered sequence together.

Temporal prediction for compression, joint generation for reconstruction.

• Joint Spatiotemporal Refinement



Spatial Detail vs. Temporal Stability

- VVC: Stable but Smooth
- DCVC-RT: Faithful but Detail-Limited
- GLC-Video: Sharp but Unstable
- GNVC-VD: Sharp and Stable

Stabilize Detail along Time

$$E_{\text{warp}}(t) = D(\hat{x}_t, W(\hat{x}_{t-1}, F_{t-1 \rightarrow t}))$$

Low & stable $E_{\text{warp}}(t)$

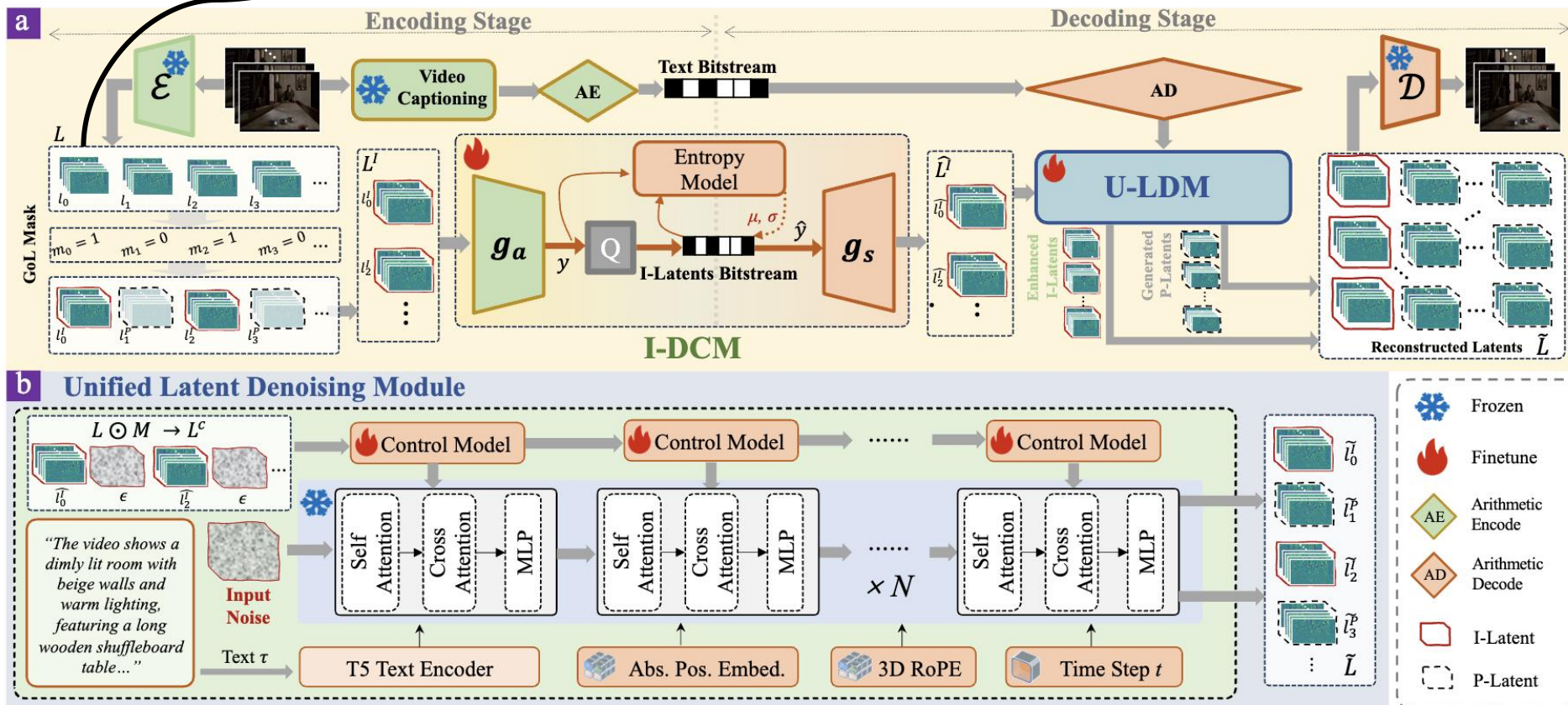
⇒ better temporal consistency

Type II. Video-Native Diffusion Paradigm

Recasting GOP in Spatiotemporal Latent Space

Organize Video as a Latent Stream

- I-Latents** Explicitly compressed and transmitted → serve as structural anchors
- P-Latents** No separate latent payload → reconstructed from prior & condition

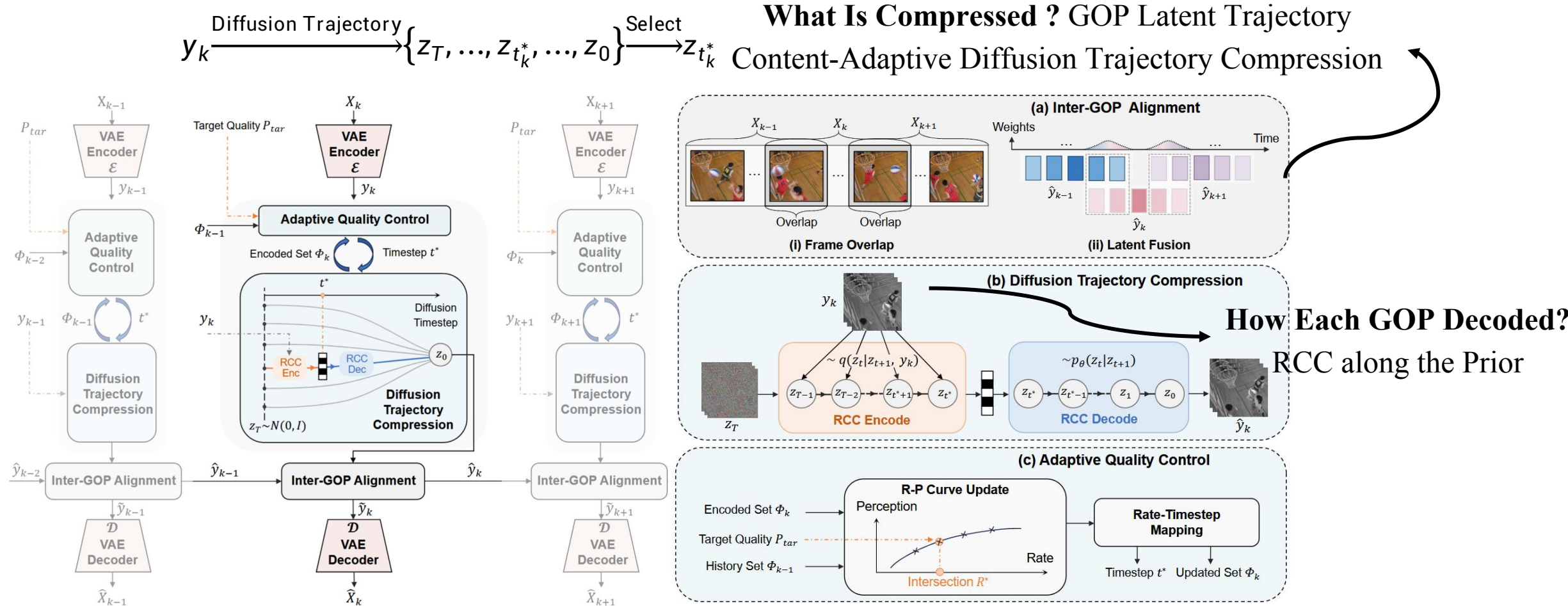


Anchor-and-Generate Latent Reconstruction

$$\tilde{L} = M \odot \hat{L}^I + (1 - M) \odot \epsilon, \\ \hat{L} = G_{\theta}(\tilde{L}, M)$$

Tokenize the video →
Select latent anchors →
Generate the missing latents

Adaptive GOP-Level Diffusion Trajectory Compression



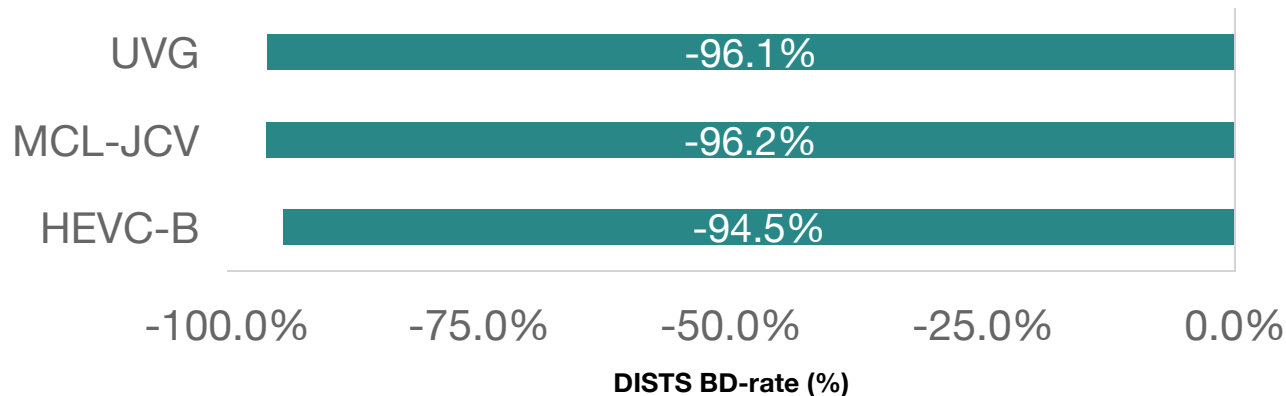
Partition into GOPs → Compress trajectories adaptively → Align across GOPs → Decode with the prior

Generative Video Coding: Perceptual Performance

Low-rate perceptual compression

GNVC-VD versus VTM-17.0

DISTS BD-rate (%)



≈95%

Bitrate savings
at matched DISTS

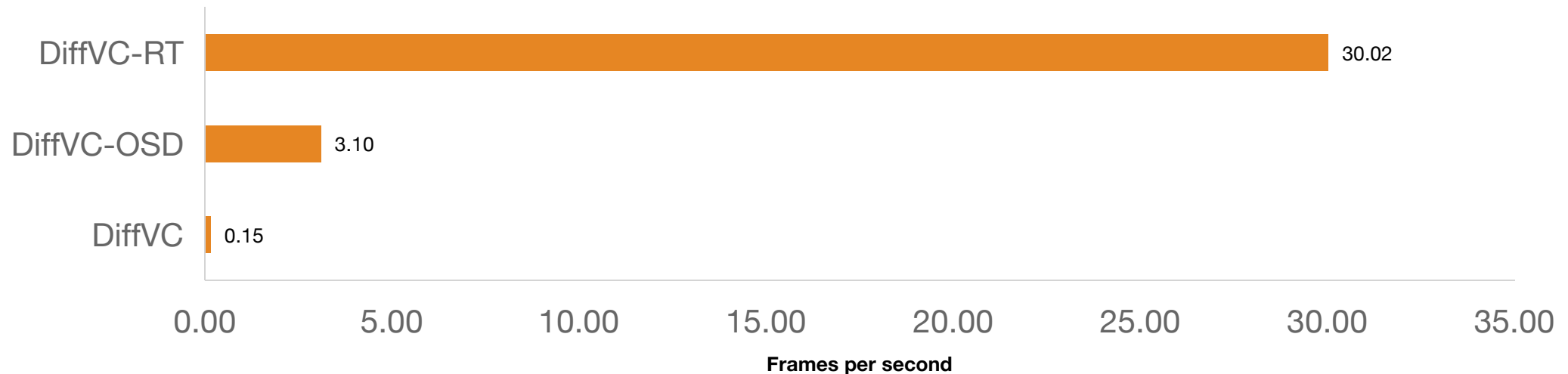
Reported operating region: below 0.03 bpp

BD-rate summarizes bitrate differences over the tested rate–quality curves.

Q. Mao, H. Cheng, T. Yang, L. Jin, and S. Ma, “Generative Neural Video Compression via Video Diffusion Prior,” Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 43239–43248, 2026.

Video Decoding Throughput

DiffVC series at 1280×720 on NVIDIA H800



Around 30 fps at 720p on a high-end GPU

Efficient decoding on everyday devices remains a challenge.

W. Ma and Z. Chen, "DiffVC-RT: Towards Practical Real-Time Diffusion-based Perceptual Neural Video Compression," arXiv preprint arXiv:2601.20564, 2026.

Core Challenges in Video Generative Coding

1 Realistic $\neq$ Faithful



Looks
natural

$\neq$

Matches
source

same bitrate budget, different reconstructions

Perception + Fidelity

Natural-looking motion can still be inconsistent with the source video.

2 Drift Accumulates



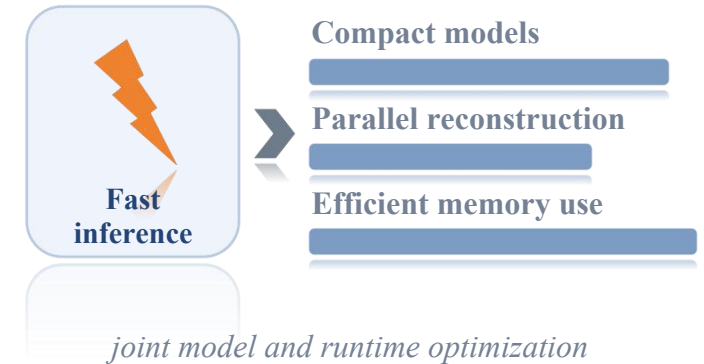
generated frames re-enter the reference chain

each reconstruction becomes the next reference

Confidence + Drift control

Small perceptual errors compound frame after frame through the feedback loop.

3 Efficient Decoding



Model + Runtime Co-design

Optimize model complexity, memory use and parallelism for target throughput and latency.

The target: one unified spatiotemporal rate–distortion–perception trade-off

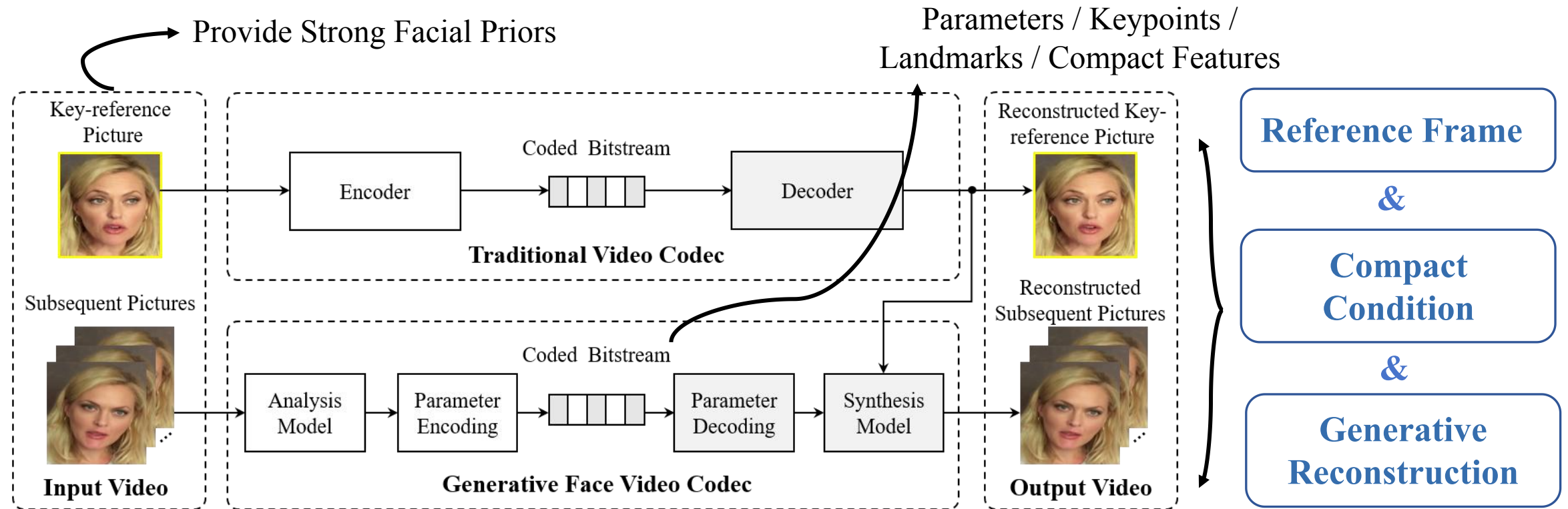
PART 03

Generative Face Video Coding and Beyond

Shiqi Wang

City University of Hong Kong
Department of Computer Science

Generative Face Video Coding: Paradigm, Evolution, and Beyond



Paradigm

What is transmitted and how to reconstructed?

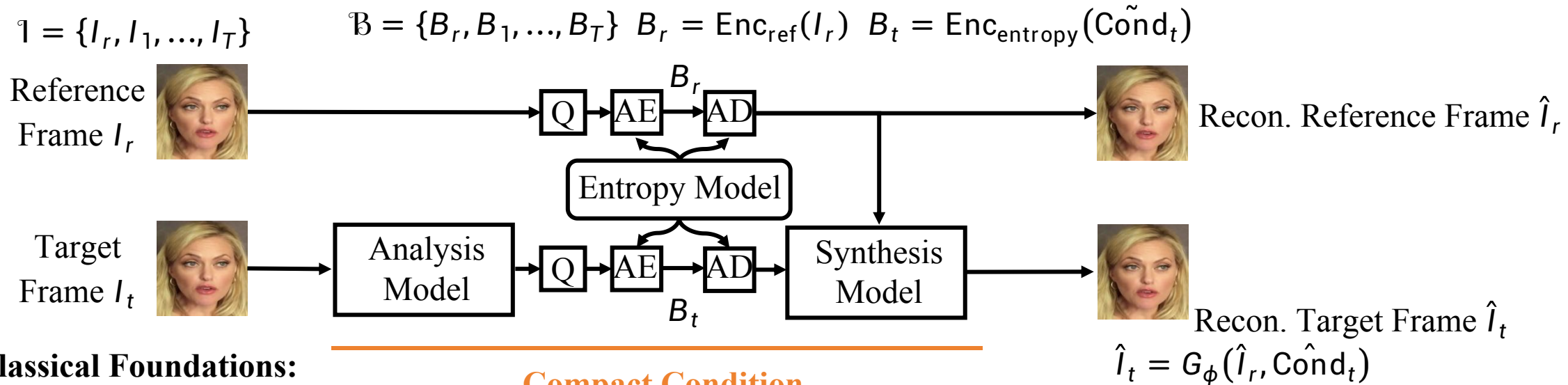
Evolution

How do compact features and capability improve?

Beyond

How does scenarios expand beyond talking-face?

GFVC: Task Formulation and Classical Foundations



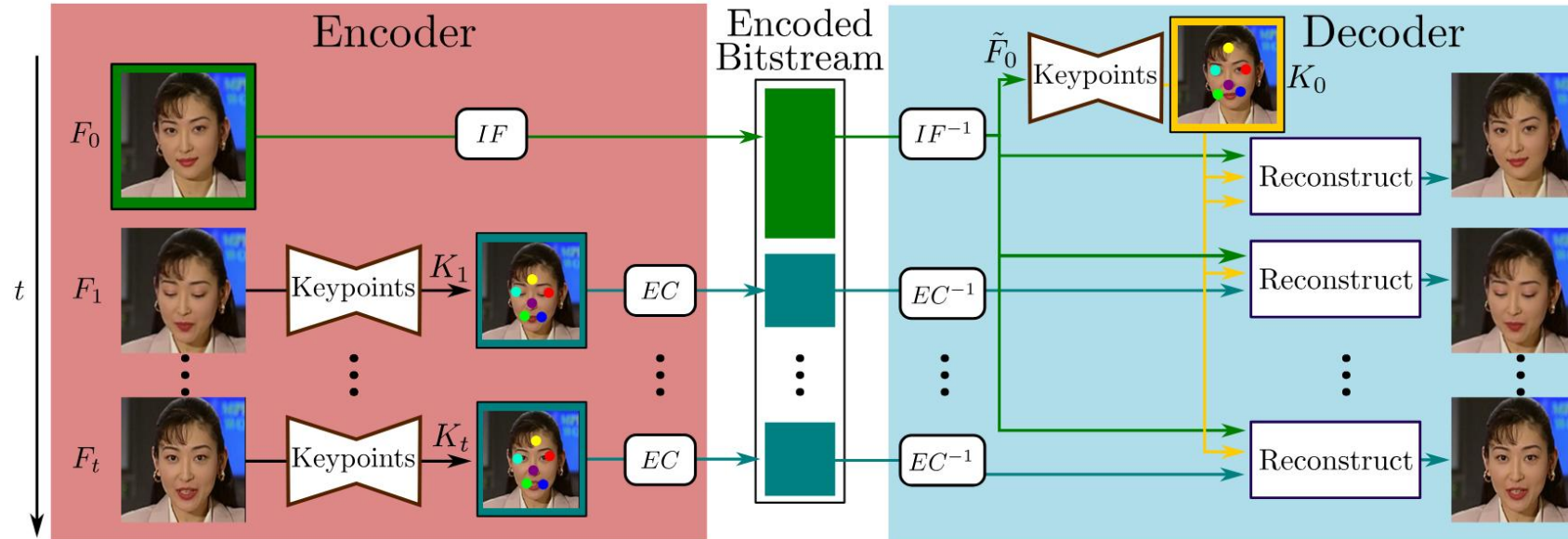
Classical Foundations:
How to instantiate the compact condition?

Compact Condition

$$\text{Cond}_t = E_\theta(I_r, I_t), \quad \dim(\text{Cond}_t) \ll \dim(I_t)$$



• DAC: Keypoint-Driven Deep Animation for Ultra-Low Bitrate Coding



Keypoint-Driven Motion Representation

$$K_t = \{k_t^m\}_{m=1}^M, K_t \in \mathbb{R}^{5 \times M}$$

Each keypoint is equipped with **2D position** and **local Jacobian Matrix**

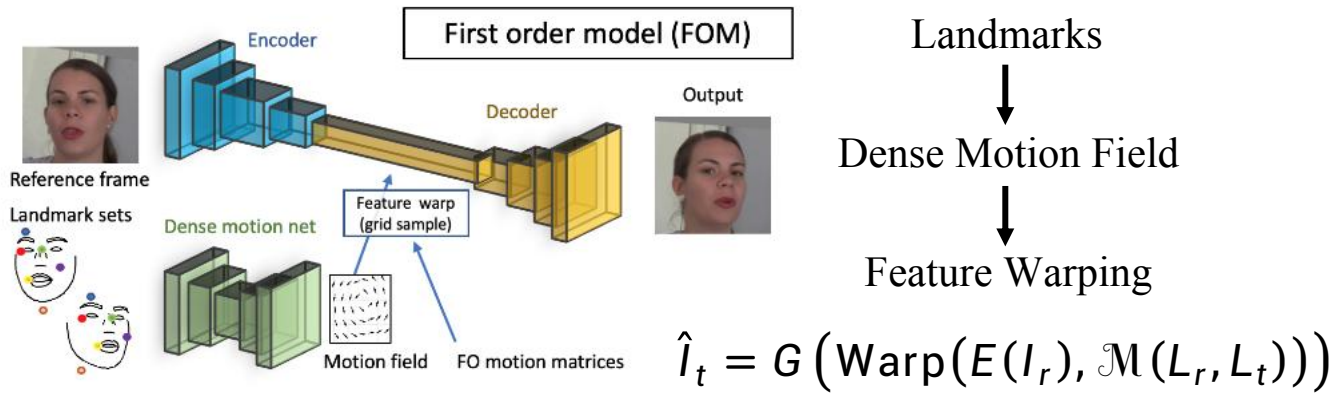
Reference $K_0 \rightarrow$ Keypoint displacement ($K_0 \rightarrow K_t$) $\rightarrow$ Dense motion $\rightarrow$ Frame reconstruction

Why Keypoint ?

- **Compact Motion Representation**
- **Geometry-Aware Correspondence**
- **Appearance-Motion Decoupling**

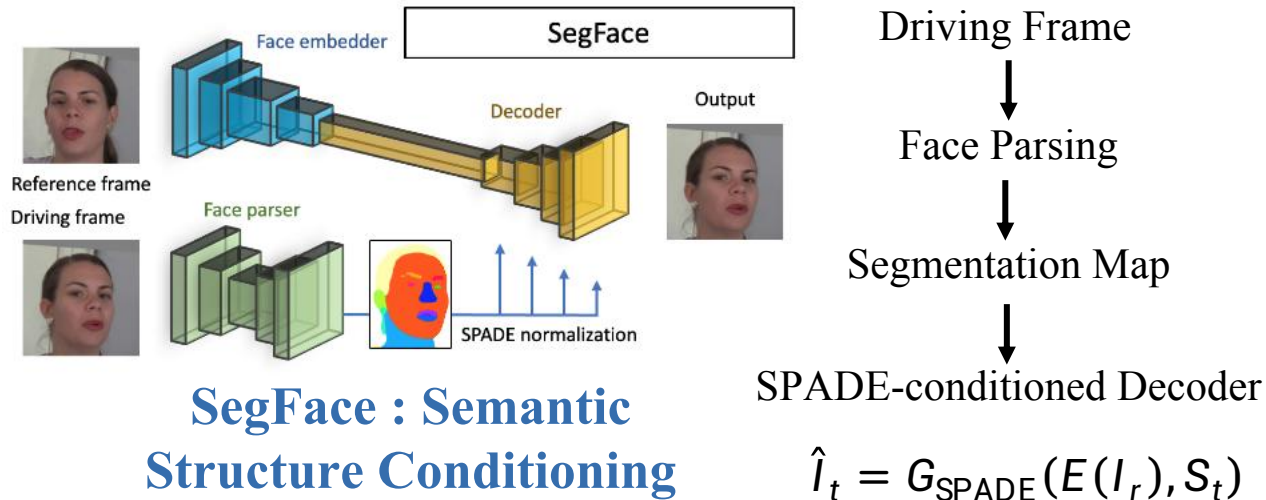
Reference Appearance $\hat{F}_0$ + Compact Motion ΔK_t $\xrightarrow{G_\phi}$ Generative Reconstruction $\hat{F}_t$

• LBVCC: Landmark-Driven Generative Reconstruction for Video Chat

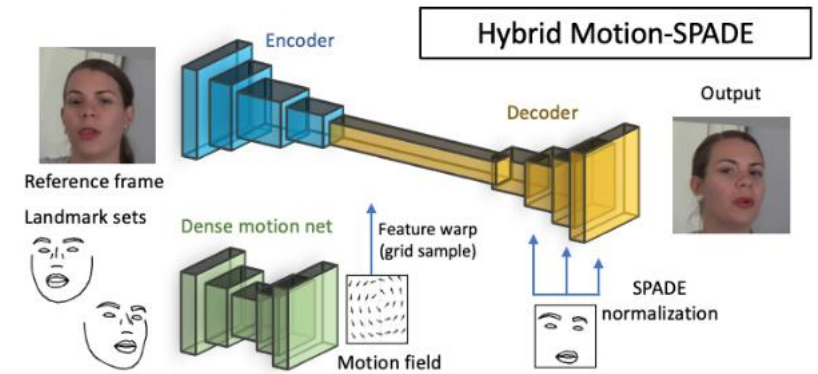


FOM : Geometric Motion Transfer

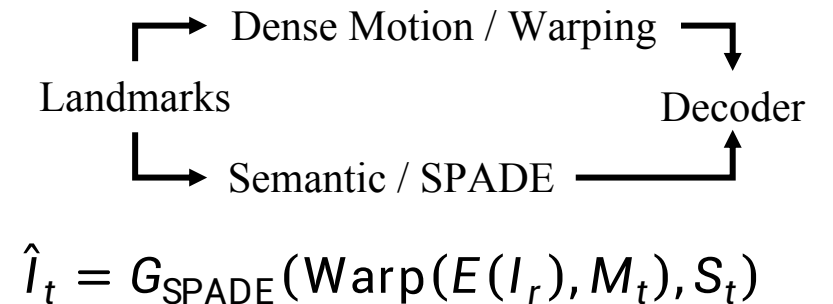
DAC = FOM-style animation turned into a codec



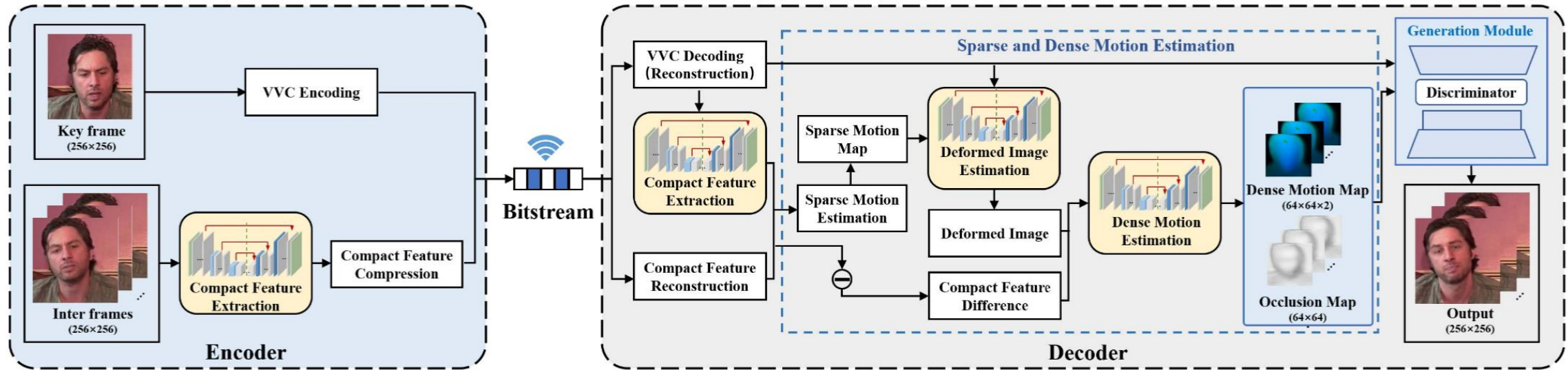
SegFace : Semantic Structure Conditioning



Hybrid Motion-SPADE : Geometric + Semantic



• CFTE: Beyond Explicit Keypoints with Learned Compact Features

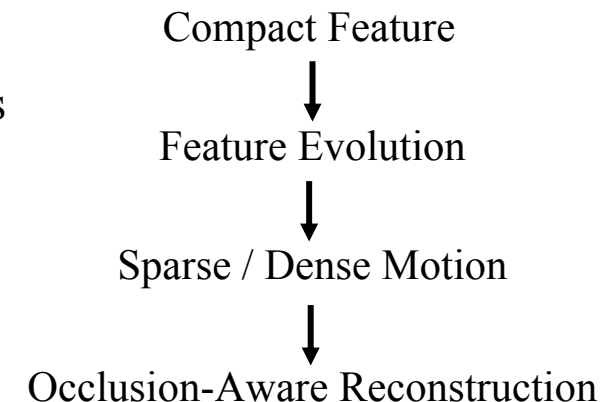


Compact Feature Extraction

Image $\rightarrow$ Learned Keypoint Detector $\rightarrow$ Explicit Geometric Symbols

$$I_t \xrightarrow{E_\theta} C_t \xrightarrow{\text{Evolution Inference}} (M_t, O_t)$$

- **End-to-End Representation Learning**
- **No Predefined Geometry**

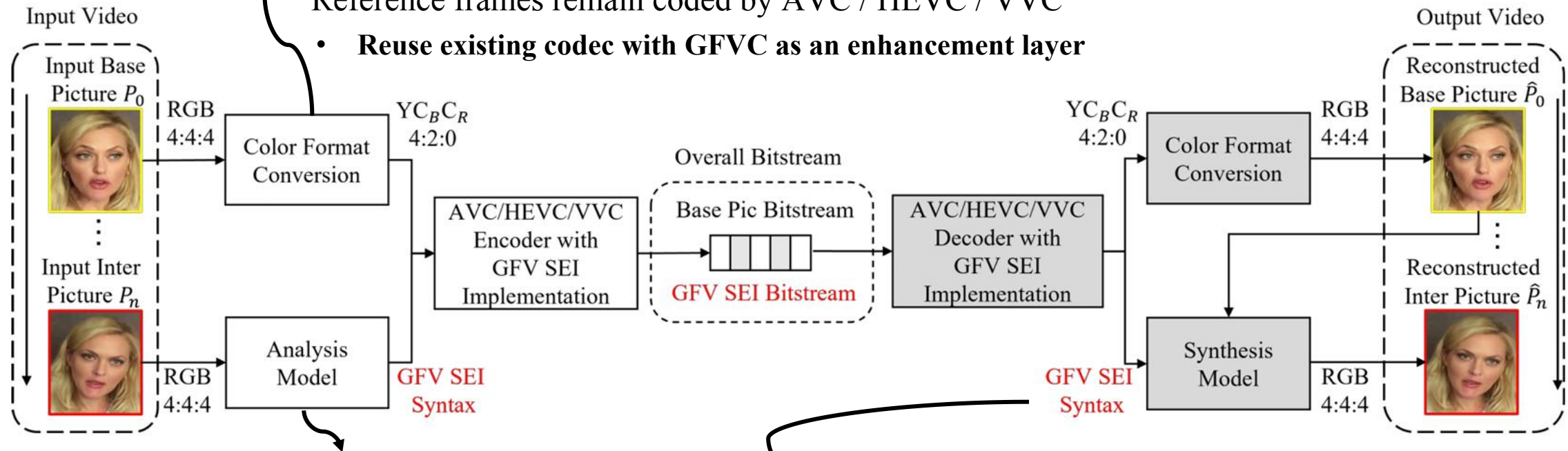


SEI-Based GFVC: Unifying Generative Representations in Standardized Bitstreams

Legacy-Compatible Reference Coding

Reference frames remain coded by AVC / HEVC / VVC

- Reuse existing codec with GFVC as an enhancement layer



Unified Signaling for Heterogeneous Conditions

Supports diverse conditions including explicit or learned ones

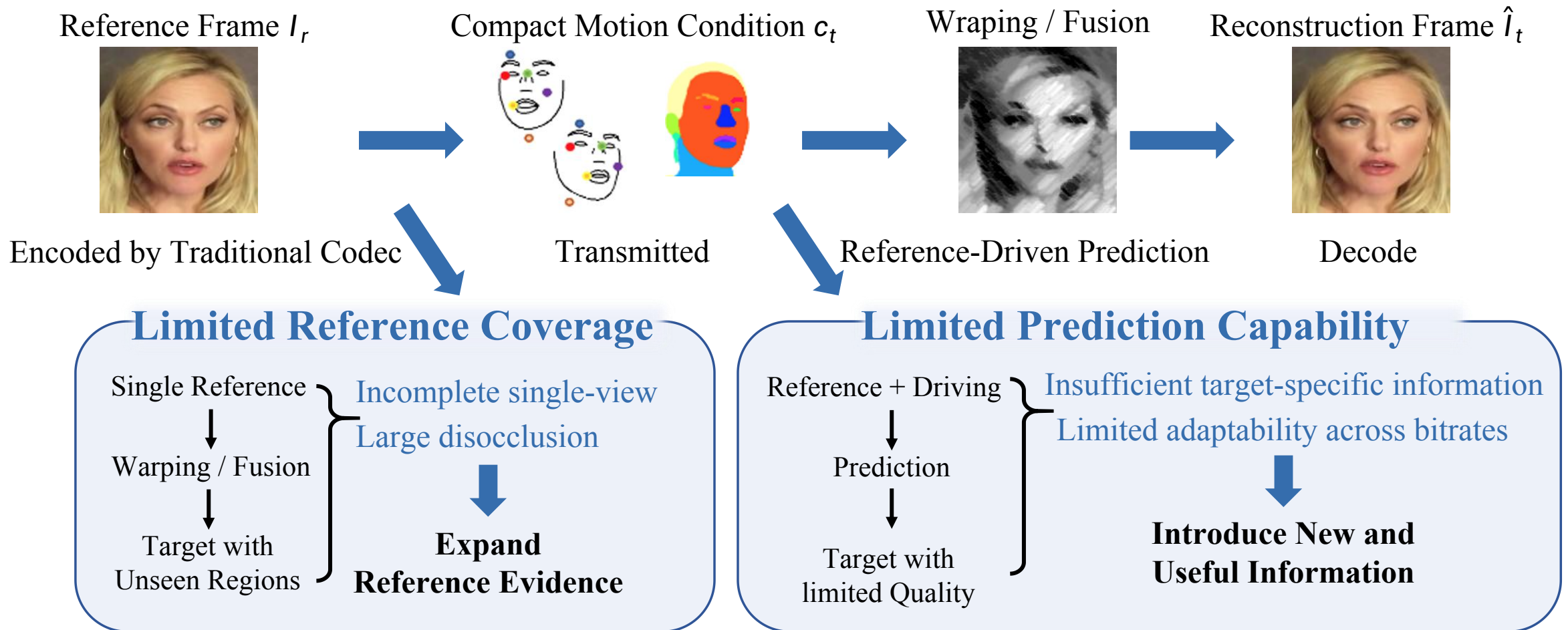
- Unified Condition Signaling within GFV SEI interface

Model-Flexible Generative Decoding

Decoupled from a specific generative model

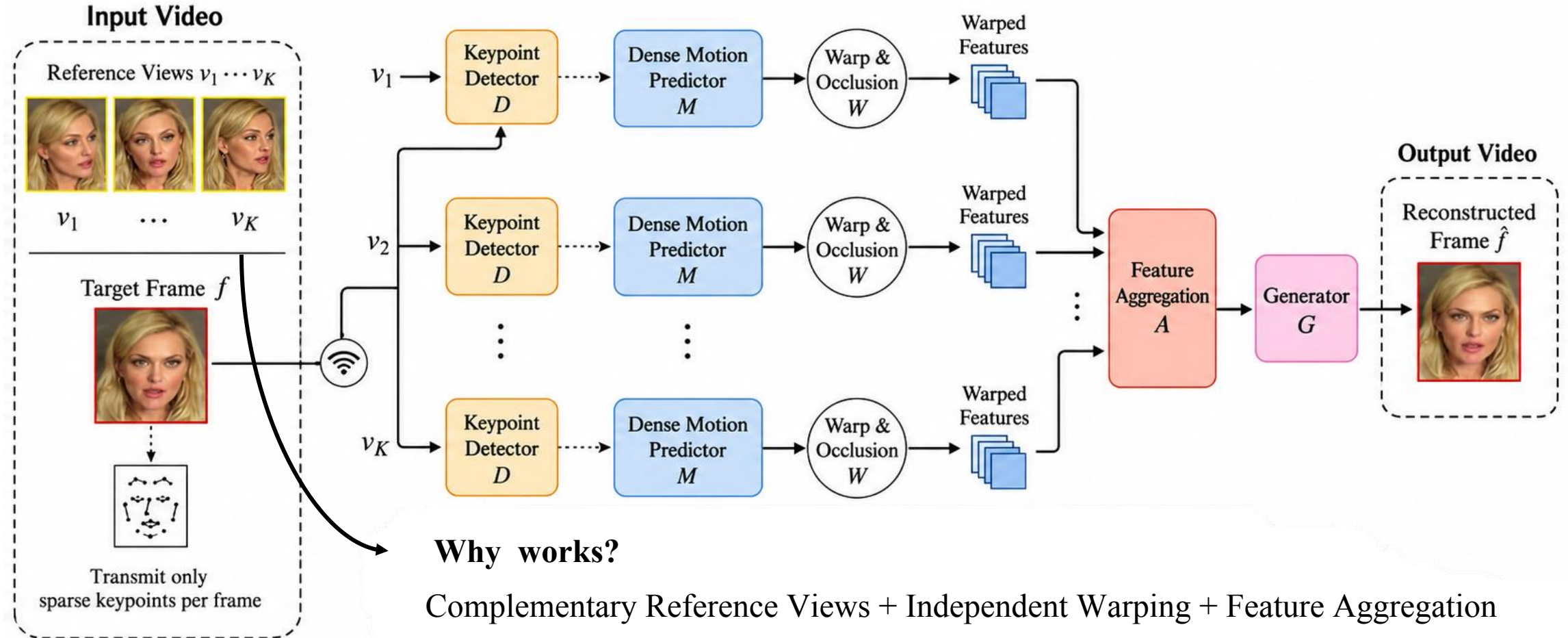
- Standardized interface with flexible generative decoder

Limits of Reference-Driven GFVC Reconstruction

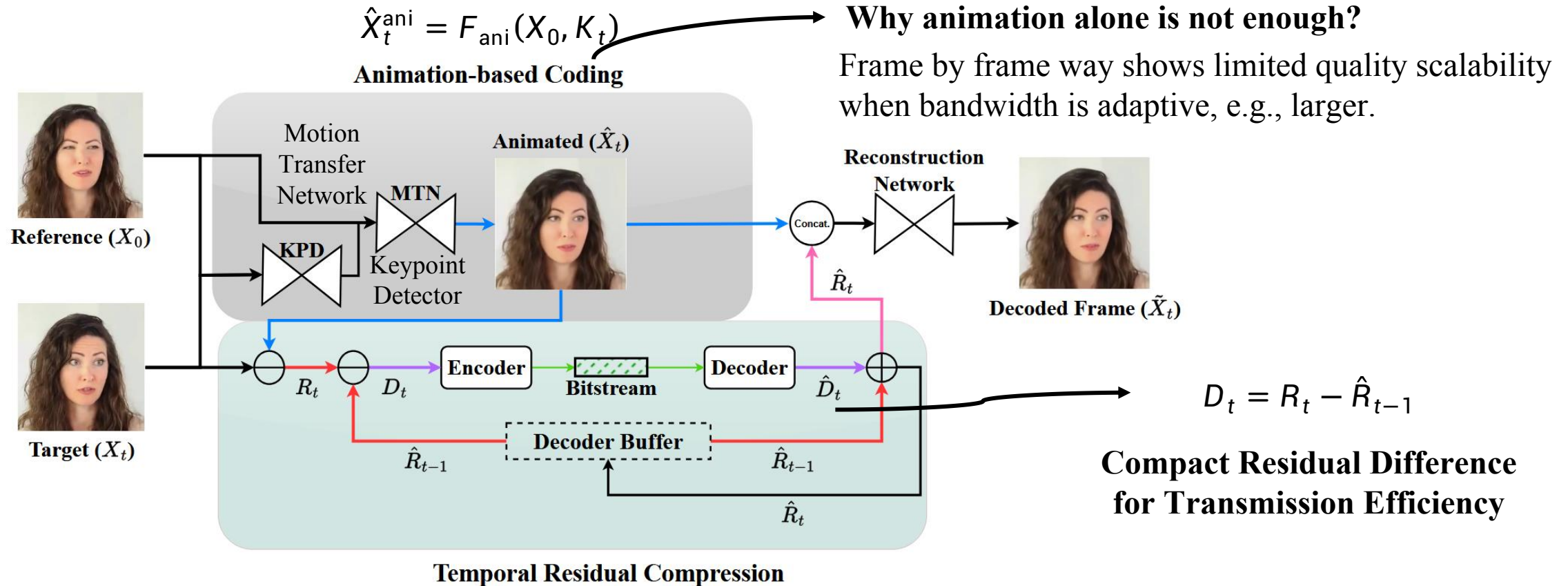


- NFVC: Expanding Reference Evidence with Multiple Views**

$$F_t^{(k)} = W(E(v_k), M_t^{(k)}, O_t^{(k)}), \quad F_t^{\text{agg}} = A(F_t^{(1)}, \dots, F_t^{(K)})$$



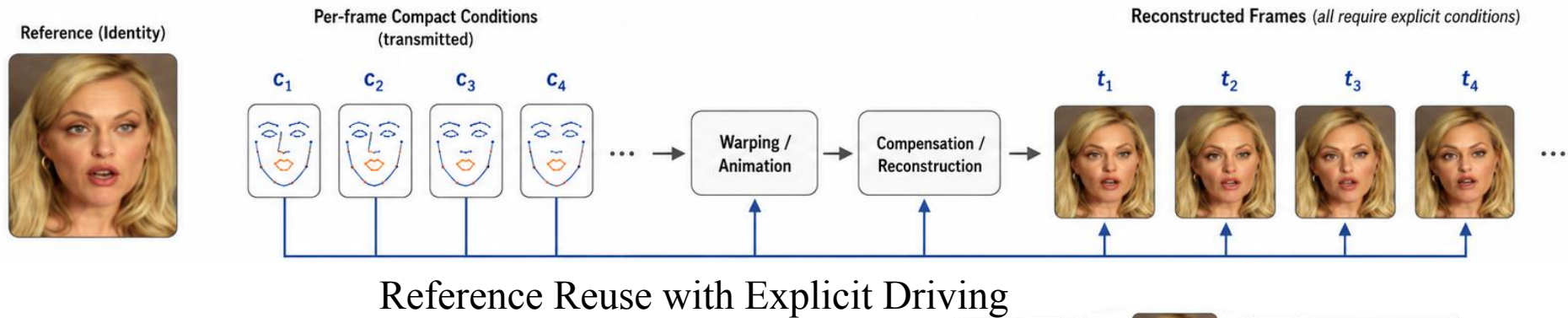
- RDAC: Supplementing Prediction with Residual Information**



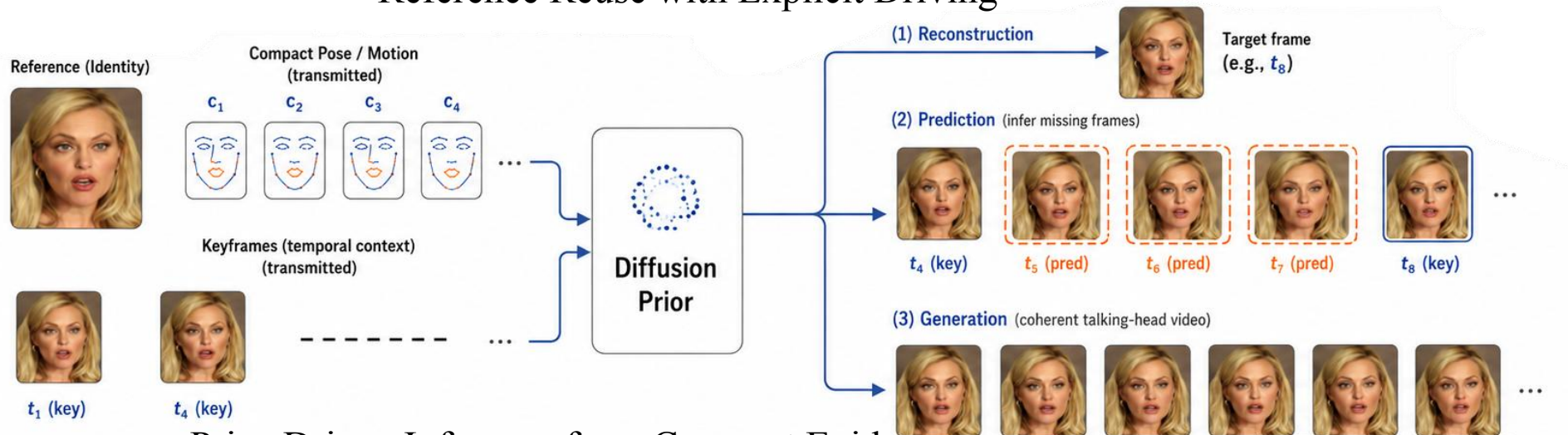
Animation-Based Prediction + Residual Supplementation + Temporal Differential Coding

Diffusion-Enhanced GFVC: Reconstruction, Prediction, and Generation

Classical GFVC

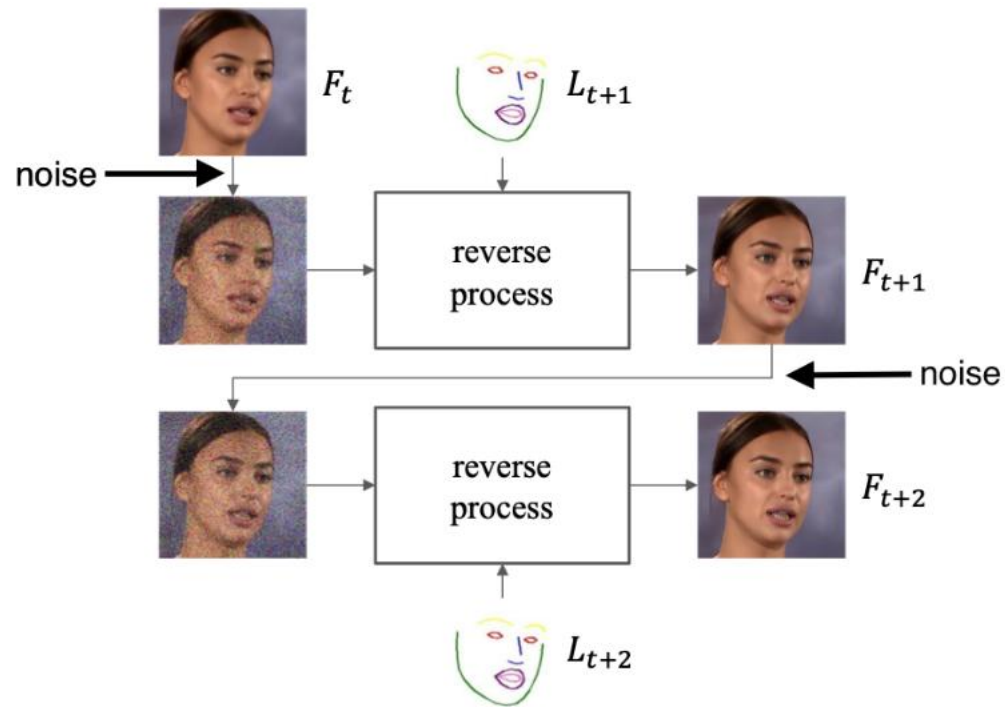


Diffusion-based GFVC



Prior-Driven Inference from Compact Evidence

- FR: Recursive Diffusion for Conditional Face Reconstruction



**Full-Frame Conditional
Reconstruction**

$$\hat{F}_{t+1} = \mathbb{D}_{\theta}(\hat{F}_t, L_{t+1}; u)$$

$$F_t \xrightarrow{\text{Forward } u \text{ steps}} F_t^{(u)} \xrightarrow[\text{Condition: } L_{t+1}]{\text{Reverse } u \text{ steps}} \hat{F}_{t+1}$$

- Partial Noising**

Retain identity and spatial information from the previous frame

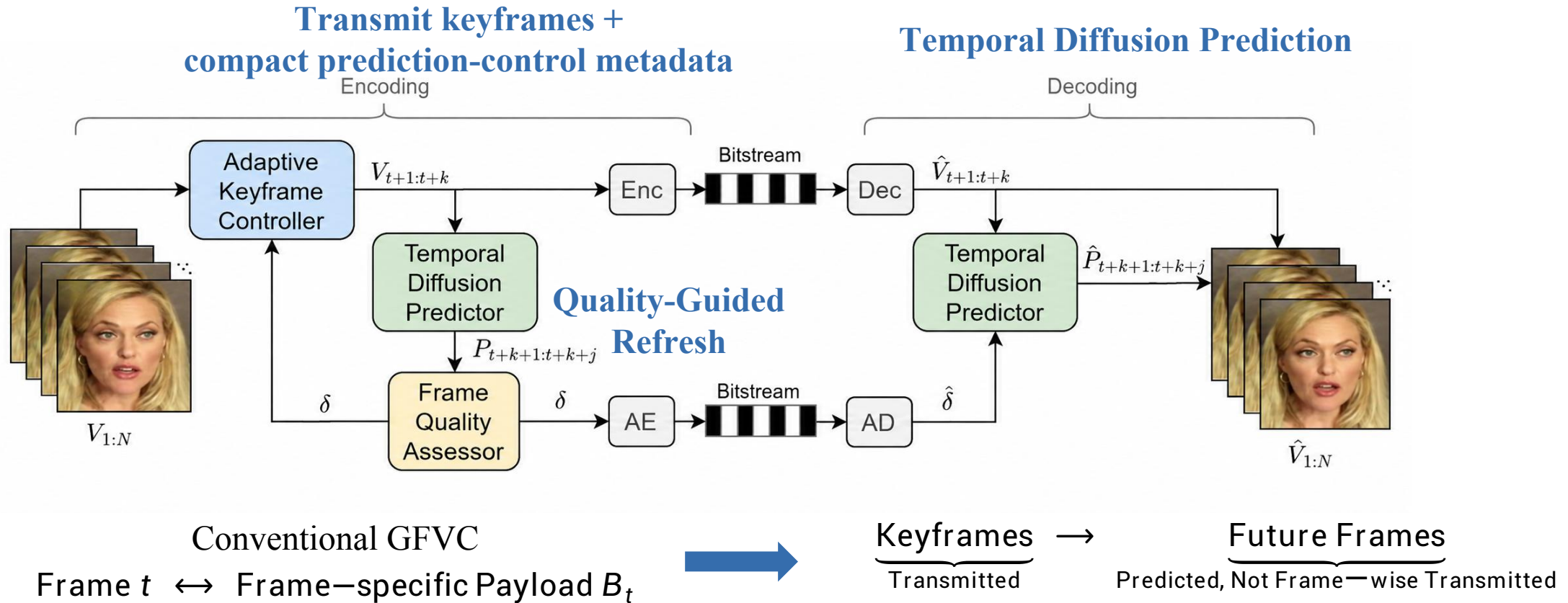
- Landmark Conditioning**

Inject the target facial expression during reverse diffusion

- Recursive Reconstruction**

Use the reconstructed frame to condition the next frame for temporal continuity

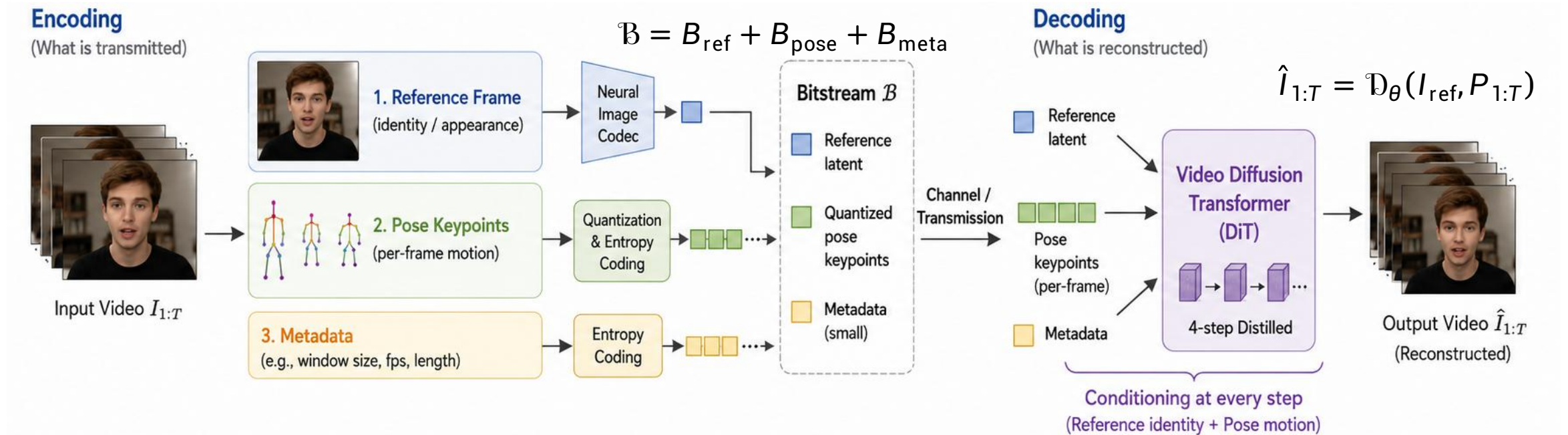
- ULB: Predicting Untransmitted Frames with Temporal Diffusion



Untransmitted Frame Prediction

From reconstructing transmitted conditions to predicting untransmitted frames.

• ReGenVC: Spatiotemporal Generative Decoding with Video Diffusion



Compact Identity–Motion Signaling

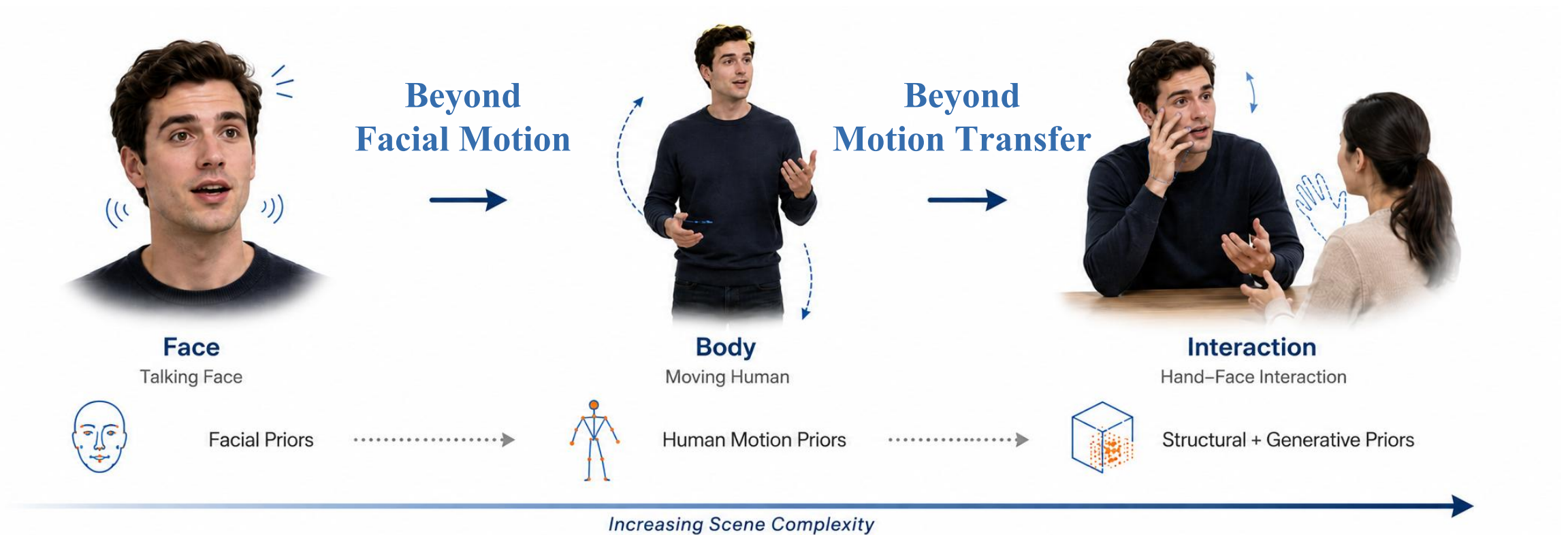
- Reference frame → identity / appearance
- Per-frame pose keypoints → motion

Windowed Spatiotemporal Generation

- Sliding-window generation for joint temporal modeling
- Anchor-frame reuse for cross-window continuity

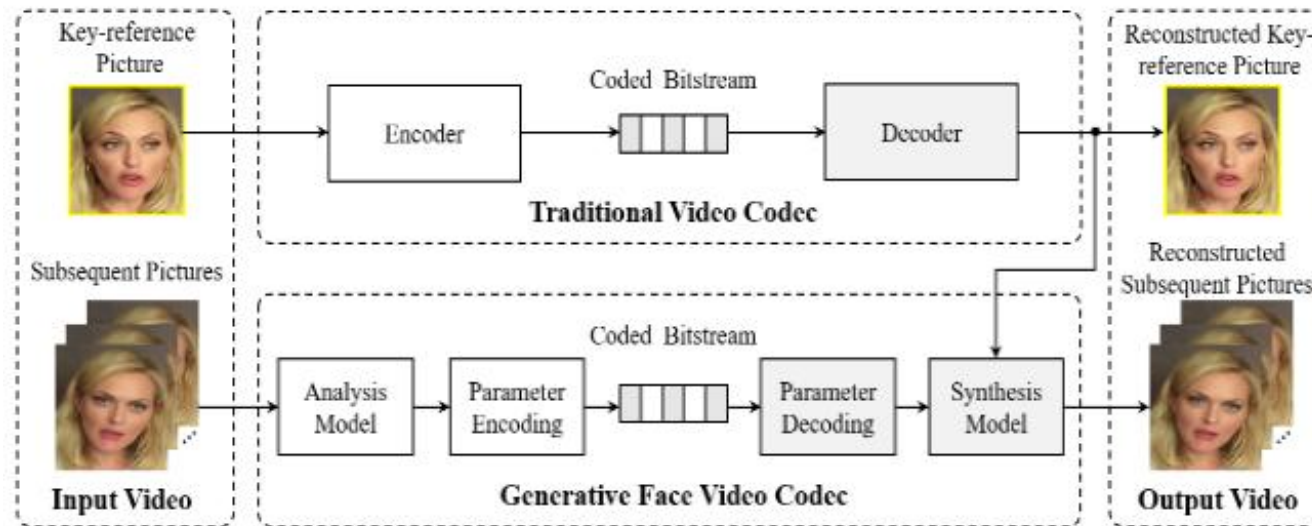
Video-Level Spatiotemporal Generative Decoding
From frame-wise generation to end-to-end video generative decoding

Beyond Face-Centric GFVC: From Talking Faces to Human Interaction

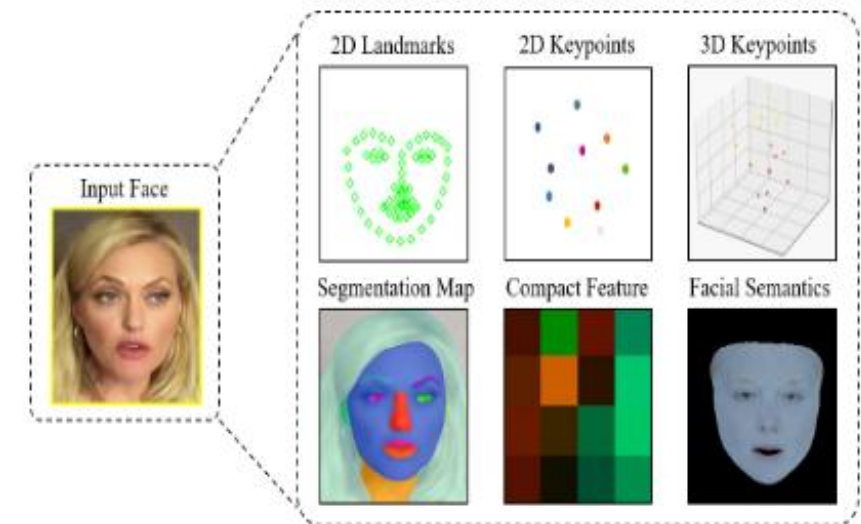


- The Face-Centric Foundations of GFVC

Reference Appearance + Compact Facial Driving + Generative Reconstruction



(a) General Framework



(b) Facial Representations

Geometric

- 2D Landmarks
- 2D/3D Keypoints

Semantic

- Segmentation
- Facial Semantics

Learned

- Compact Features

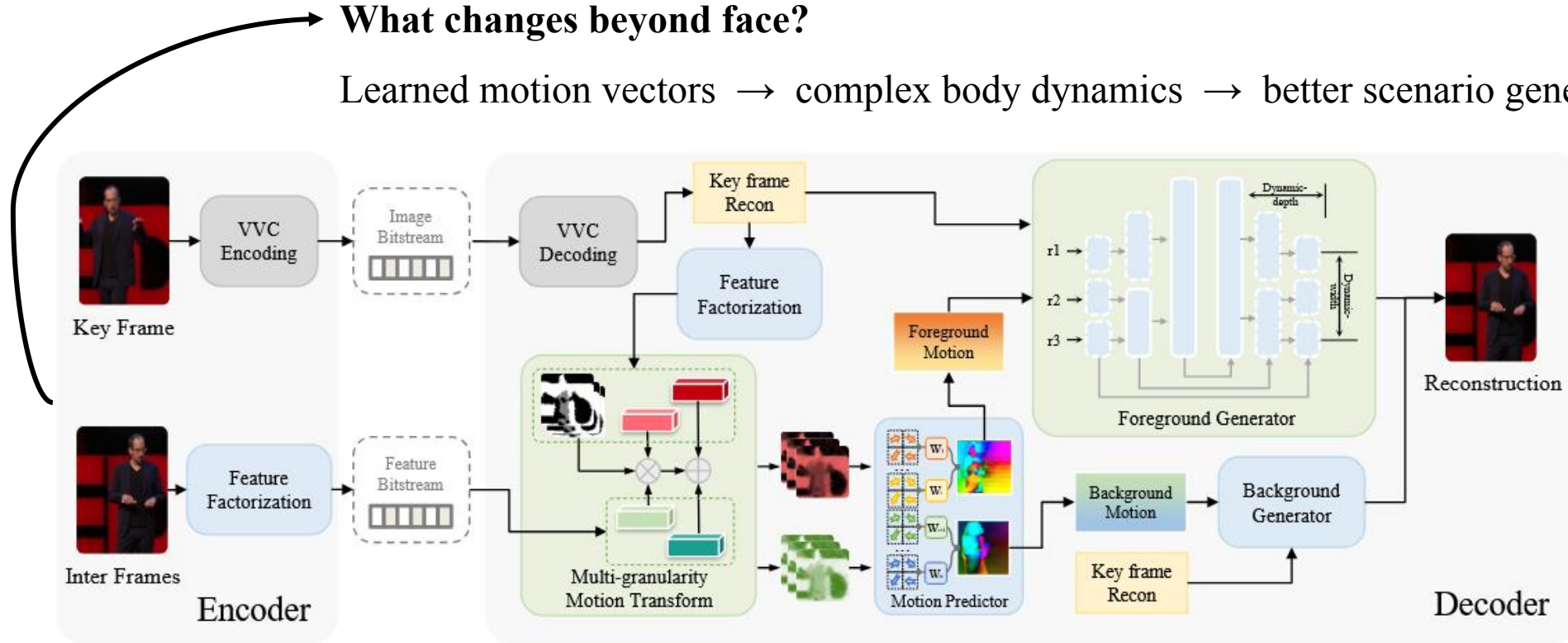
Strong Facial Priors,
Limited Scene Scope

- GHVC: From Face-Centric to Human-Centric Generative Coding**

Extend compact generative coding from facial dynamics to complex human-body motion.

What changes beyond face?

Learned motion vectors → complex body dynamics → better scenario generalization

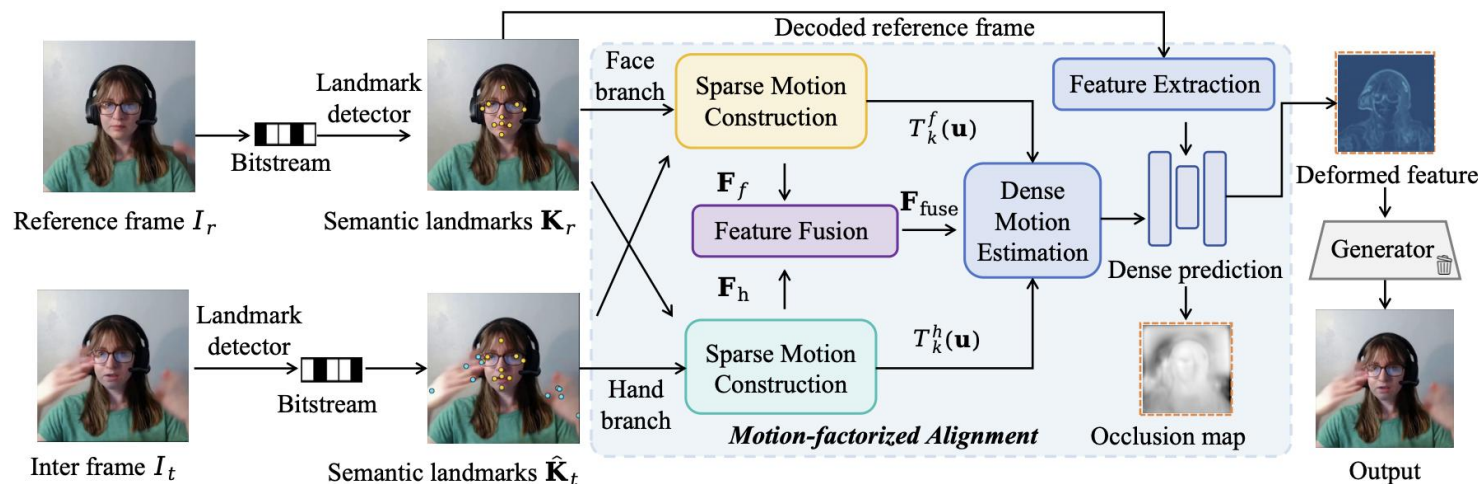


Compact Motion Factorization + Fine-Grained Motion Fields + Foreground–Background Generation

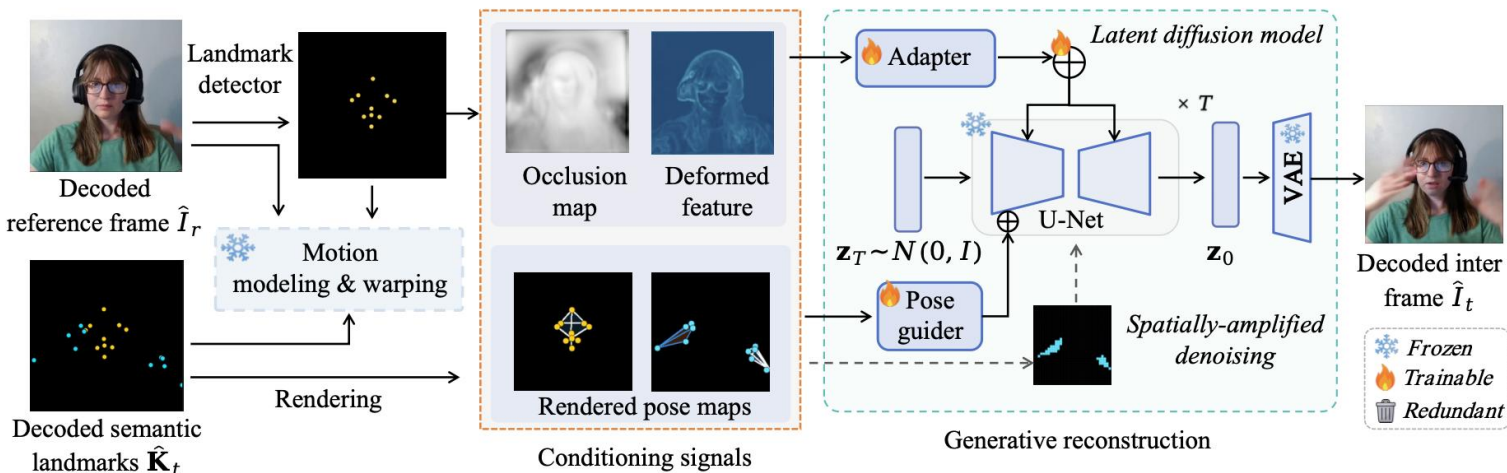
Beyond Faces: Interaction-Aware Generative Video Coding

Interactive video introduces structured motion, occlusion, and content emergence

(a)
Motion-
Factorized
Alignment



(b)
Conditional
Diffusion
Recon.



Interaction-Aware Structural Prior

Adequate face-hand semantic landmarks

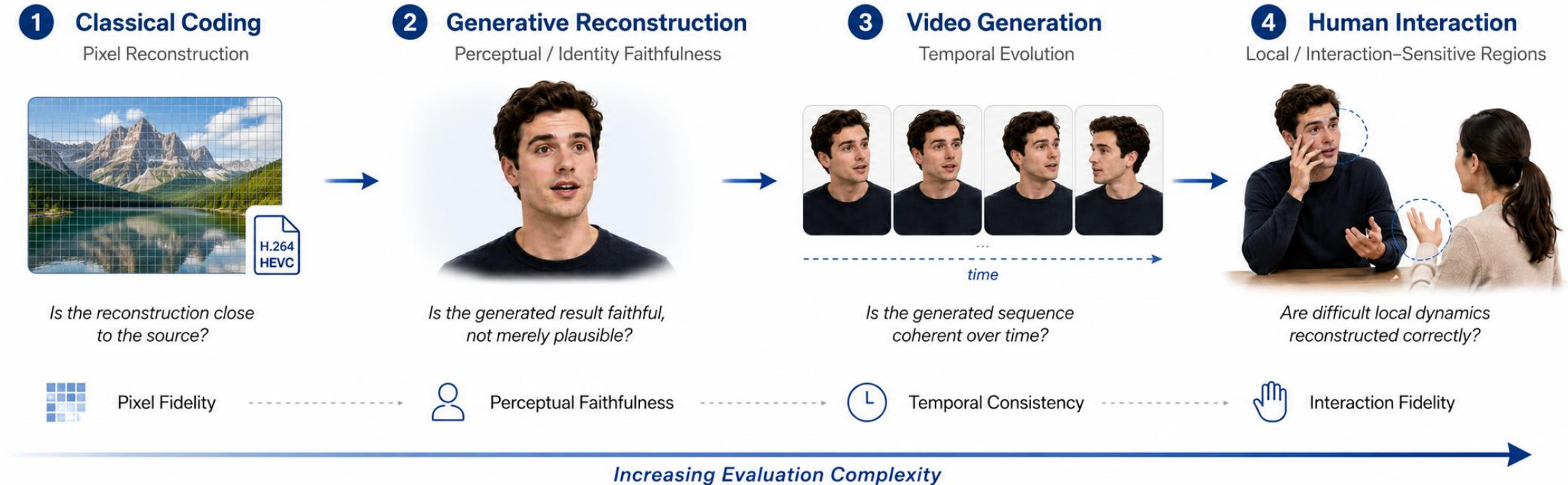
Factorized Motion Transfer

Separate face-hand motion modeling with alignment

Generation Beyond Warping

Prior-guided reconstruction for occlusion and new content

- Why must GFVC evaluation change?



As GFVC evolves from reconstruction to generation and interaction, evaluation must move beyond pixel distortion

Traditional Coding
Reconstruction closeness



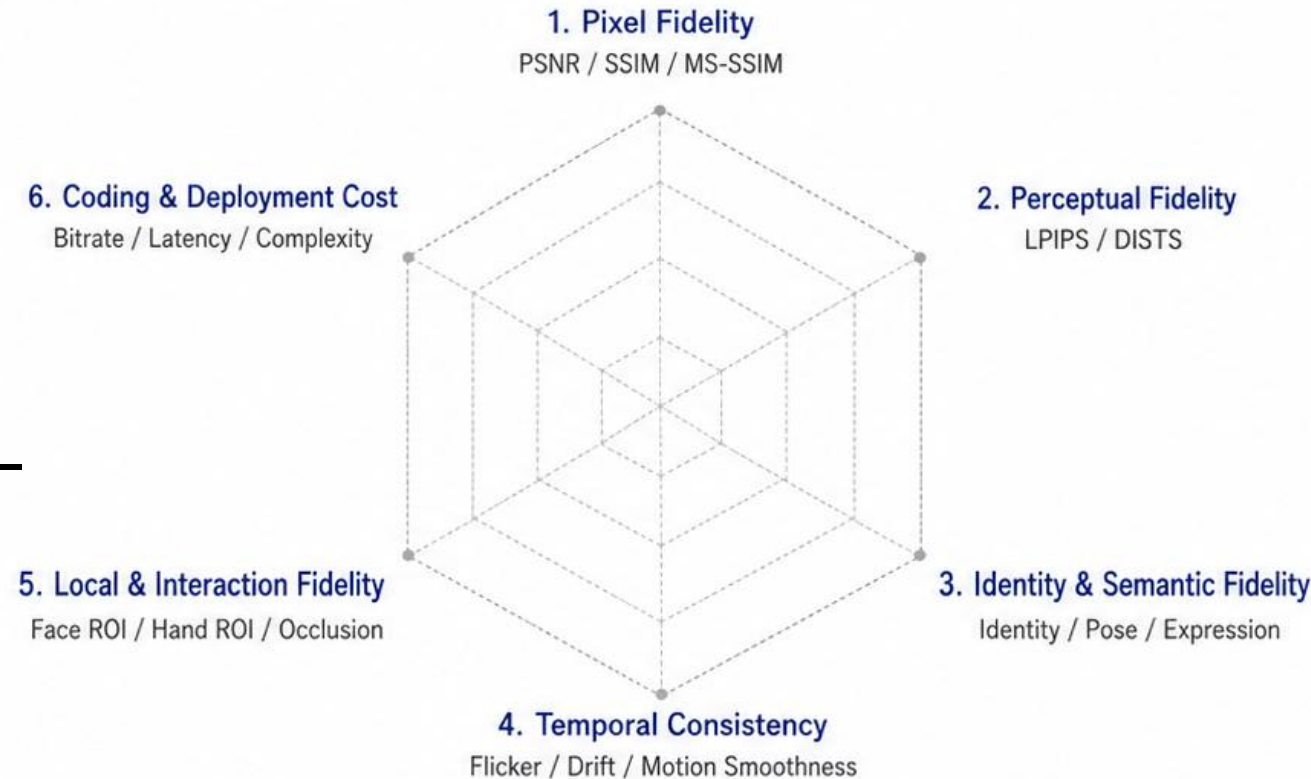
GFVC / Generative Coding
Visual faithfulness under compact evidence

- **Multi-Dimensional Evaluation for GFVC**

Video-Level Reliability

Global quality must be complemented by temporal and local evaluation.

Does it remain correct over time and locally?



Reference Faithfulness

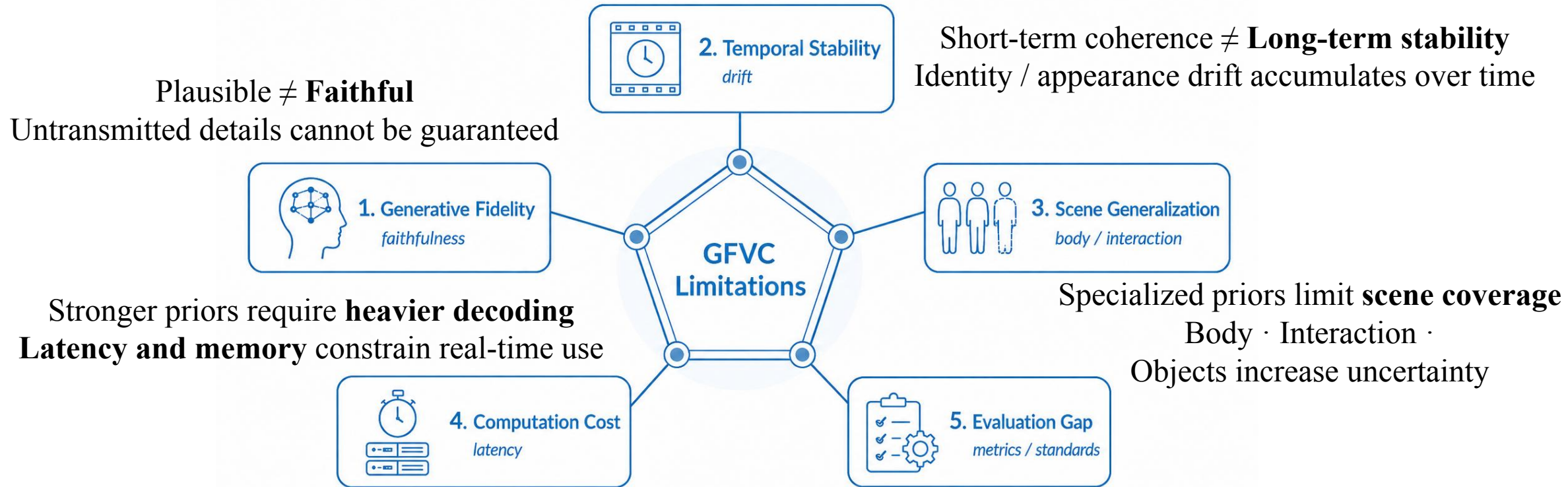
Generated content should remain faithful to the source, not merely plausible.

Does it reconstruct the right content?

Recommended Evaluation Protocol

Global Quality + Local Quality + Temporal Quality + Rate / Complexity

- **Current Limitations: What Still Holds GFVC Back ?**



Less Transmitted Evidence → Higher Reliance on Generative Priors → Greater Uncertainty & System Cost

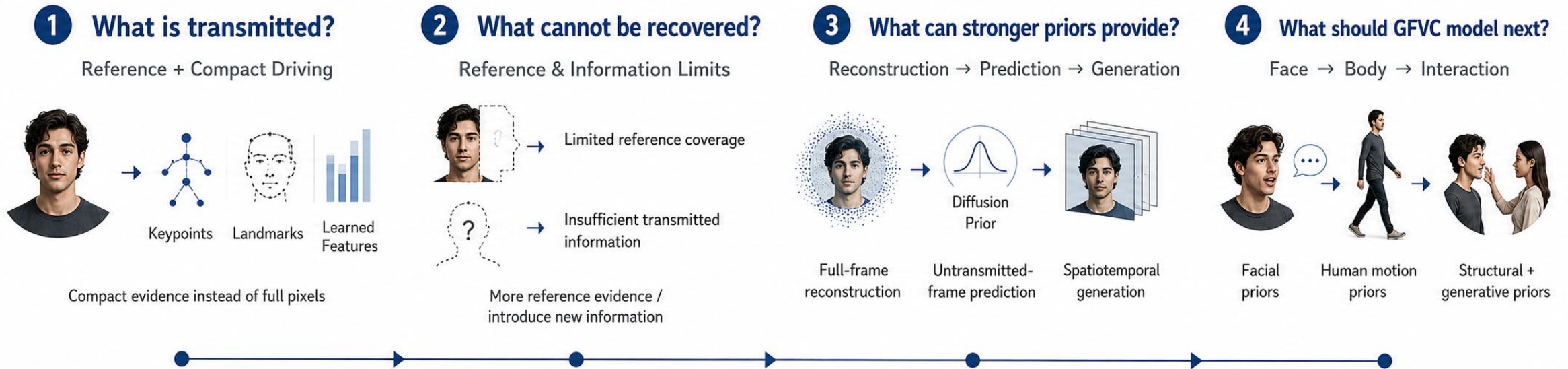
- **Future Directions: From Limitations to Practical GFVC**

Future should go toward reliable, stable, generalizable, efficient, and standardized GFVC

Current Limitation	Future Direction	Key Goal
Generative Fidelity	Reliable & Controllable Generation	Faithful Reconstruction
Temporal Stability	Long-Horizon Spatiotemporal Modeling	Long-Term Consistency
Scene Generalization	Richer Human Interaction Modeling	Broader Scene Coverage
Computation Cost	Efficient Generative Decoding	Real-Time Efficiency
Evaluation Gap	Unified Evaluation & Standardization	Unified Metrics & Interfaces

Faithful → Stable → Generalizable → Efficient → Standardized

From Compact Evidence to Generative Communication



Transmit Less → Infer More → Model Richer Interactions



Toward Practical & Reliable GFVC

Faithful · Stable · Generalizable · Efficient · Standardized

GFVC is evolving from coding motion to generating communication

References (1/3)

- [1] J. Ballé, V. Laparra, and E. P. Simoncelli, “End-to-end Optimized Image Compression,” International Conference on Learning Representations, 2017.
- [2] J. Ballé, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, “Variational image compression with a scale hyperprior,” International Conference on Learning Representations, 2018.
- [3] G. Lu, W. Ouyang, D. Xu, X. Zhang, C. Cai, and Z. Gao, “DVC: An End-To-End Deep Video Compression Framework,” IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10998–11007, 2019.
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